



BSI Standards Publication

**Explosive and toxic atmospheres —
Hazard detection mapping — Guidance
on the placement of permanently
installed flame and gas detection
devices using software tools and
other techniques**

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Foreword

Publishing information

This British Standard is published by BSI Standards Limited, under licence from The British Standards Institution, and came into effect on 30 September 2020. It was prepared by subcommittee EXL/31/1, *Gas detectors*, under the authority of Technical Committee EXL/31, *Equipment for explosive atmospheres*. A list of organizations represented on these committees can be obtained on request to their committee managers.

Information about this document

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Where websites and webpages have been cited, they are provided for ease of reference and are correct at the time of publication. The location of a webpage or website, or its contents, cannot be guaranteed.

Use of this document

As a guide, this British Standard takes the form of guidance and recommendations. It should not be quoted as if it were a specification or a code of practice and claims of compliance cannot be made to it.

Presentational conventions

The provisions of this standard are presented in roman (i.e. upright) type. Its requirements are expressed in sentences in which the principal auxiliary verb is “shall”.

Commentary, explanation and general informative material is presented in smaller italic type, and does not constitute a normative element.

Where words have alternative spellings, the preferred spelling of the Shorter Oxford English Dictionary is used (e.g. “organization” rather than “organisation”).

Contractual and legal considerations

This publication does not purport to include all the necessary provisions of a contract. Users are responsible for its correct application.

Compliance with a British Standard cannot confer immunity from legal obligations.

0 Introduction

0.1 General

This British Standard is written in the form of guidance and supports flame and gas (F&G) detection standards across the industry.

One of the most challenging activities faced by a design engineer is deciding the quantity and location of gas detectors and/or flame detectors.

This British Standard refers to F&G coverage factors but does not specify target coverage factors for different applications. Coverage factors are only broad targets and are easily manipulated by changing device sensitivities, alarm trip levels, voting configurations, target gas concentrations, target flame size and other factors.

This British Standard also provides guidance on sensing technologies and the physical format of detectors which could greatly affect coverage (quantity and position). Irrespective of the quantity and positioning of detectors by a manual or software-related activity, the selection of incorrect sensing technologies leads to incidents not being detected or an increase in spurious trips.

This British Standard provides guidance on the full life cycle of a gas detection or flame detection system, emphasizing that mapping and/or modelling is an ongoing activity and not simply an activity that is carried out once in the design of a new facility. Routine surveillance of detector coverage during the full life cycle of a facility ensures that facility modification, changes in hazards, etc., are all addressed, and the management of change highlights any deficiencies or opportunities for improvement.

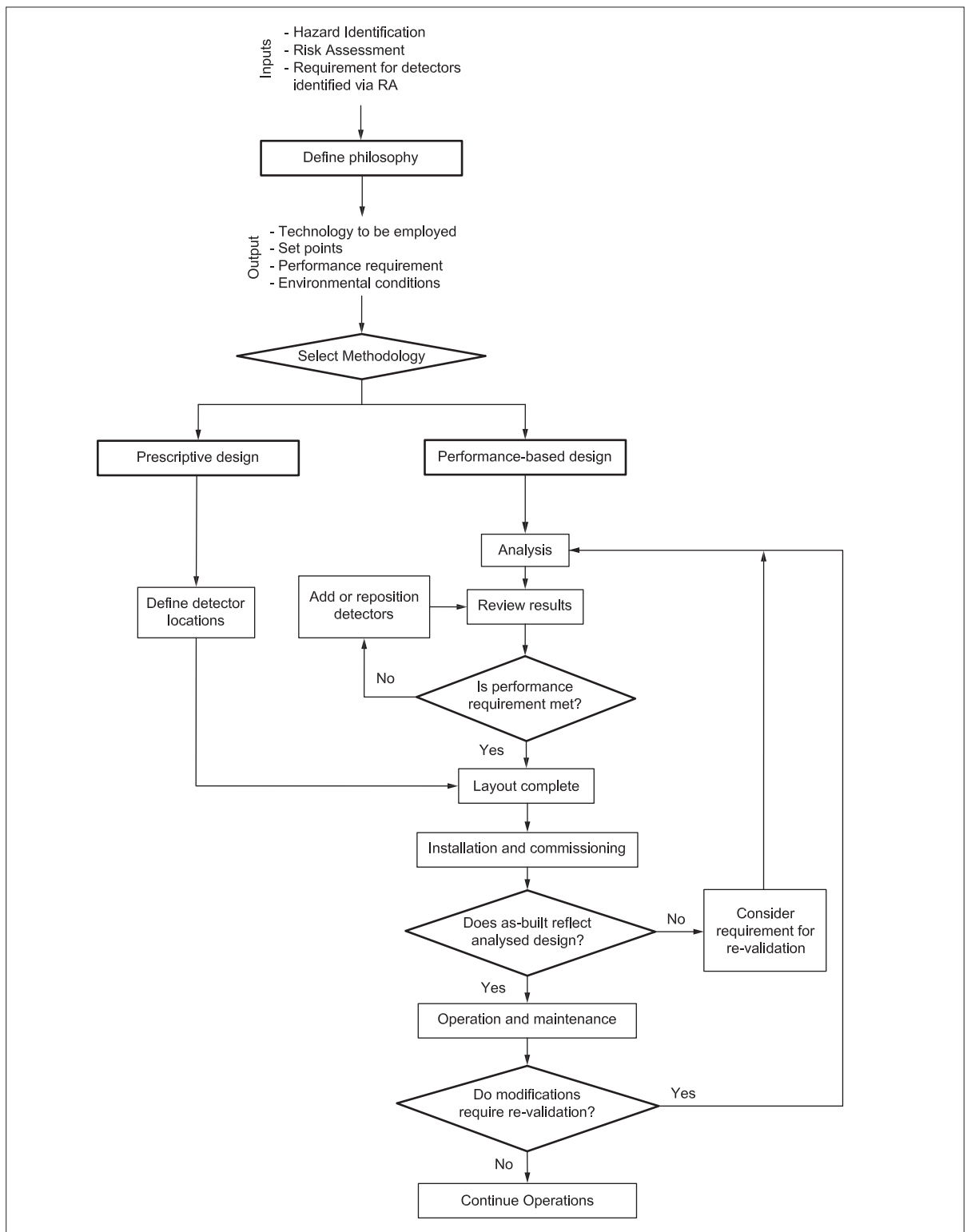
Detector coverage within this British Standard means permanently-installed devices, which are the only devices for which guidance is provided. The use of additional personal monitors, e.g. portable gas detectors, does theoretically increase the density of “detecting devices” when persons are actively working within a plant area or location.

This British Standard is based on existing and established sensing and detector technologies and configurations. It is also based on many years of industrial experiences and lessons learned. This British Standard does not exclude emerging technologies or innovative ideas; however unless there is reasonable and practical evidence that these technologies or methodologies offer equal benefits, caution is advised.

This British Standard provides informative guidance on detection symbols (see [Annex E](#)), however this guidance does not preclude continued application of any established symbols applied by an operator or designer.

0.2 Document structure

The structure of the document and the process for F&G mapping is illustrated in [Figure 1](#), with detail provided in [Clause 4](#) to [Clause 12](#).

Figure 1 — Document structure

1 Scope

This British Standard provides guidance on the placement of permanently-installed F&G detectors, including coverage and technology selection.

NOTE 1 "Permanently-installed detection systems" include optical flame detection (including ultraviolet, infrared and imaging/visual), flammable gas/vapour detection and toxic gas detection.

This British Standard provides guidance on the most commonly used methods of mapping and/or modelling: prescriptive, volumetric-based and scenario-based.

This British Standard is applicable to applications where:

- a) optical flame detection is used as a means of detecting flaming fires in internal and external environments;
- b) a gas detection system is used as a method of explosion protection;
- c) the hazard arises from the release or accumulation of explosive gases and vapours; and
- d) the hazard arises from the release or accumulation of toxic gases.

This British Standard also provides guidance on operations, maintenance and the availability/reliability of the system to complement relevant standards.

This British Standard does not provide guidance for:

- 1) addressing hazards that are a product of vehicle exhaust fumes;
- 2) applications in the domestic and general public environment;
- 3) applications in mining and tunnelling, heating ventilation and air conditioning (HVAC), medical applications or environments;
- 4) smoke detection; or
- 5) toxic gas detection with regard to business continuity, e.g. food tainting.

Mitigation effectiveness is an important aspect of F&G detection design but is out of scope of this British Standard.

NOTE 2 Oxygen detection is typically biased towards prescriptive methods for detector coverage and is generally excluded from this British Standard, considerations regarding typical oxygen detection are described in [Annex A](#).

2 Normative references

There are no normative references in this British Standard.

3 Terms and definitions

For the purposes of this British Standard, the following terms and definitions apply.

3.1 baseline sensitivity

sensitivity or performance capability of a device in controlled test conditions, i.e. before site specific performance influencing factors are applied

3.2 beam path absorption

absorption of the detection beam in the presence of gas (IR, UV and laser)

3.3 consequence

qualification/quantification of the harm caused by a hazard

NOTE Examples include death, injury, escalation.

3.4 coverage

criteria by which the volume/scenario is monitored or unmonitored with respect to detection success

NOTE Coverage can be represented as a percentage calculation, visual representation or both.

3.5 coverage factor

percentage coverage of the mapping zone

3.6 detection criticality factors

individual importance of a single detector, if known

3.7 escalation potential

potential consequence where no mitigating measures are implemented

3.8 emerging technology

devices with little proven in-use justification

NOTE The method by which detection occurs or how detection coverage is analysed can differ dependent upon the nature of the technology.

3.9 fire size

representation of fire as a function of geometric volume or radiant heat output

3.10 flame and gas detector modelling

application of software tools to generate a quantitative coverage of F&G detection in a specific space

NOTE Results of this modelling are often a valuable input to the F&G mapping process when applying volumetric-based and scenario-based methods.

3.11 flame and gas (F&G) mapping

process by which flame or gas detection layout is determined

NOTE This can include a combination of prescriptive adherence to facility philosophies, detection engineering knowledge, experience and software modelling tools.

3.12 fire and gas (F&G) philosophy

project document covering philosophy/strategy of the fixed F&G system

NOTE For the purposes of this British Standard, this is assumed to be different to a particular company's F&G philosophy/strategy that provides generic guidance for all installations/sites.

3.13 graded volume

volume within the mapping zone where specific detection performance targets are specified

3.14 hazard

source that has the potential to cause harm to someone or something

3.15 inverse square law

specified intensity inversely proportional to the square of the distance from the source of that physical quantity

NOTE Inverse square law is the determining factor in flame detection performance.

3.16 judgement-based placement

consensus between competent persons from different disciplines who holistically apply best judgement and experience to locate a detector

3.17 lower explosive limit (LEL)

volume fraction of flammable gas or vapour in air below which an explosive gas atmosphere does not form, expressed as a percentage

NOTE LEL can be used interchangeably with lower flammable limit (LFL) in this context.

3.18 mapping zone

boundary of the volume to be assessed as a single voting grouping

3.19 open-path detector (gas)

detector which measures the concentration of gas in a straight line between two points

NOTE 1 Concentration measured is displayed as metres (m), e.g. ppm/m or LEL/m.

NOTE 2 The concentration of gas is typically from >0 m to 250 m.

3.20 performance target

set of detection goals against which success of detection can be measured

NOTE Examples include a target maximum fire size or gas cloud volume.

3.21 point detector (gas)

single detector to measure a gas concentration at its location

3.22 prescriptive method

method of detector placement based on preset rules which can be followed by applying “pass/fail” criteria

3.23 radiant heat output (RHO)

energy (in kW) emitted from a flame in the radiant region of a fire

NOTE This is not to be confused with radiant heat flux (kW/m²).

3.24 risk

combination of the frequency/likelihood of harm occurring and the consequence severity

3.25 scenario-based method

method of detector placement taking into account a number of probabilistic scenarios which might result in detection

3.26 volumetric-based method

method of detector placement which measures a geographic volume where a hazard can occur which might result in detection

NOTE This is often referred to as geographic.

4 Hazard identification and risk assessment

Once a hazard has been identified employers need to conduct a risk assessment.

NOTE 1 Attention is drawn to The Health and Safety at Work Act 1974 [1].

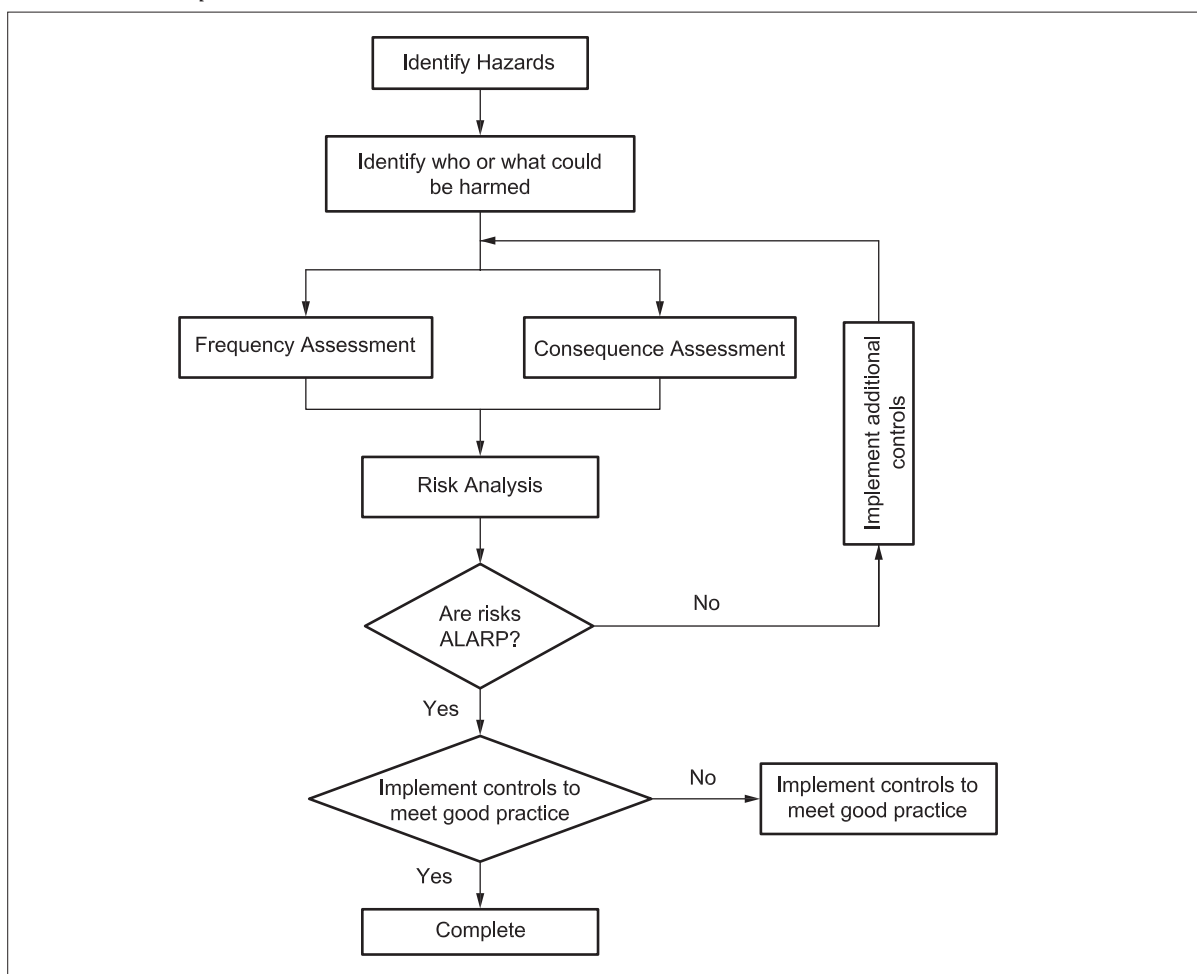
A typical risk assessment process is shown in [Figure 2](#) with the objective of determining if adequate control measures are in place so that risks can be deemed As Low As Reasonably Practicable (ALARP). For flammable gases and vapours, this process is typically conducted as part of a Dangerous Substances and Explosive Atmospheres Regulations 2002 (DSEAR) [2] assessment. For toxic gases

and vapours this is often carried out as part of a the Control of Substances Hazardous to Health Regulations 2002 (COSHH) [3] assessment.

One control measure is flame and/or gas (flammable and toxic) detection. In most cases F&G detection only detects the hazard rather than preventing the hazard and offers limited benefit in terms of reducing the potential for immediate fatalities. It is when combined with additional risk reduction measures (e.g. blowdown, deluge, alarms) that consequence and hence risk is lowered by reducing the potential for escalation. The F&G detection system is classed as mitigation when the centre of the bowtie is the release of flammable/toxic materials, or a fire.

NOTE 2 For further information on bowtie, see <https://www.gov.uk/government/news/bowtie-a-visual-tool-to-keep-an-overview-of-risk-management-practices>.

Figure 2 — Risk assessment process



The risk assessment process can range from qualitative to fully quantitative and the level of effort should be proportional to the risk; a concept known as proportionality. For example, it is not necessary to conduct a fully quantitative study if determining the flammable risk arising from an isolated single flammable source (e.g. propane bottle) in an unmanned room. This British Standard assumes that a risk assessment has been conducted with the output being that flame and/or gas detection is required to demonstrate that risks are ALARP. This British Standard provides guidance on the process to justify the number of detectors to employ and where to put them.

Toxic gas detection introduces different layers of complexity compared to flame and flammable gas. For flame and flammable gas, the risk assessment process identifies any requirement for fixed flame/gas detection and the detector location requirement is based on the required control measures in place to mitigate the event of a fire/flammable gas release. For toxic gas, the risk assessment

identifies the requirement for fixed detection and informs the decision on how many detectors are required and their location. It is therefore necessary to take account of the other control measures that are in place, along with the desired outcome of detection with respect to toxicity, when making the decision on positioning requirements for toxics. Further guidance on other control measures and how they affect decisions on the location and number of detectors is given in [7.6](#).

5 Fire and gas (F&G) philosophy

5.1 General

The fixed detection philosophy or strategy with respect to F&G detection should be defined at the outset of the project to allow for a consistent approach to be taken for the long-term maintenance and upgrading of the system, i.e. if new equipment is installed or changes are made to the plant, additional detectors can be installed or be relocated in a manner consistent with the philosophy (see [Clause 11](#) for further guidance on operation and maintenance of F&G detection systems).

In some instances, larger companies might have internal guidance (also sometimes referred to as a philosophy) on F&G detection that outlines preferred methodologies and metrics to ensure consistency across sites/installations. Where this is the case, a philosophy document should still be developed for the site in question, but the internal guidance can be referenced within the philosophy to avoid duplication.

The philosophy should include, as a minimum:

- a) the metrics to use to assess a system, e.g. percentage coverage, target fire size, gas cloud of a certain size, gas cloud/fire that could lead to escalation of a certain frequency of occurrence (see [7.3.6](#), [7.5.6](#), [7.6.6](#));
- b) whether separate criteria are set for the environment, personnel, asset and business continuity. If multiple criteria are applied, the system should be designed based on the most onerous criteria. Further, toxic gas detection is typically required only to protect personnel;

NOTE Toxic gas detection can be applied solely for business continuity protection (i.e. food tainting by ammonia, use in the semi-conductor industry), however, this is outside the scope of this British Standard.

- c) the mapping zones, including the methodology used to define them, if not included within the company's internal guidance document;
- d) area grading rules (if applicable) that define the performance criteria based on different assessed risk categories (see [7.3.2](#), [7.5.2](#), [7.6.2](#));
- e) what technologies are appropriate for the environment and hazard;
- f) set points and voting requirements for detectors; and
- g) whether there are multiple design authorities/certification bodies/stakeholders for a particular installation/site, e.g. for a floating production storage and offloading (FPSO) installation the certification body would differ between the hull and topsides. Ideally a single consistent philosophy should be developed for the installation but if this is not possible it might be necessary to include details of how the philosophy differs across the installation/site and how the interfaces are managed.

The mapping methods to be applied could differ as the design progresses. More detailed methods might not be appropriate early on in a design as sufficient information might not be available.

5.2 Detector location – practical considerations

5.2.1 General

An important element of mapping involves taking account of engineering practicalities when deciding where to locate detectors. The positioning of detectors should be determined during the mapping exercise and not left until detailed engineering.

5.2.2 Mounting

Once positioning has been determined by a mapping study, correct mounting becomes important for the correct operation of devices. Detector mounts should be permanently secured to structural steelwork, where practicable. Where mounting posts are to be used, consultation with the civil engineer might be necessary to determine the suitability of locations, e.g. taking account of drainage channels where mounting foundations might not be possible.

5.2.3 Maintenance access

Detectors should be accessible from ground or a platform without the need for specialized equipment or temporary construction, such as scaffolding, or lifting devices, or fall protection, where practicable. This includes the orientation of mountings so that displays and important written information are visible from an accessible location.

For line of sight (LOS) detectors, this includes the ability to align the optical beam from an accessible location.

Access for testing and maintenance should be confirmed during the design.

In some cases, it might be necessary to mount point gas detectors at locations inaccessible without scaffolding; in such cases test gas tubing up to the detectors should be specified to enable function testing. For locations where this is not possible, e.g. on brownfield sites, rope access can be provided and used by fully trained and properly equipped personnel.

5.2.4 Impact of normal operational activities

Detectors should be located so that the potential for damage during normal plant maintenance is minimal.

The location of LOS-type gas detectors or optical flame detectors should allow for normal activities in the region of the detection that could block the sensor. This can include regular maintenance activities such as those requiring scaffold.

5.3 Brownfield versus greenfield considerations

The mapping method for a new greenfield project might be different from an existing operational (brownfield) site.

For a greenfield project it should be possible to optimize the detection performance close to the theoretical limit of the modelling, as there is usually a lot of flexibility in mounting locations and system configuration. A typical optimization aim is to achieve a detection target with the minimum number of detectors.

For a brownfield site which is already operating with an existing F&G system, there are likely to be other significant constraints. For example, to relocate a detector could be disruptive and require significant engineering effort. The cost of installing new cabling, cable trenches, mounts, and modifying drawings typically outweighs the purchase cost of a detector. The relative cost of moving a detector versus installing a new one should be determined (this varies from site to site).

NOTE The cost of a "small" move (e.g. within the cable slack already provided) or re-orientation might be trivial.

An apparent “optimum” mapping result which requires multiple detectors in a zone to be moved is typically more expensive to engineer than purchase and installation of an additional detector, so an alternative solution involving the addition of a new detector to improve coverage might be more cost-efficient and practical.

Where this could present an issue, it is beneficial to agree an optimization policy prior to starting the mapping exercise, typically based on overall cost efficiency and effectiveness rather than a simple metric of total detector numbers, to achieve the desired coverage or risk reduction target.

An example would be a brownfield policy for flame detection optimization making changes to the layout, where necessary to close gaps or meet targets, in the following order of preference.

- a) Change angles/orientations of flame detectors.
- b) Consider making one or two small moves (preferably within the cable slack).
- c) If target still not met, consider adding new detectors in new locations rather than multiple relocations of existing detectors.

Consideration of risk reduction to ALARP is particularly relevant when justifying addition of new detection in brownfield studies, as practicability of implementation is a significant issue.

6 Detection technology

6.1 General

For all types of detection, the detector technology selected needs to be suitable for the hazard types which might be present in the particular location, e.g.:

- a) if there is a risk of a unique fuel (i.e. hydrogen), it is important to be aware of the fire detector types and capability; and
- b) certain flammable vapours, such as benzene, are not detected by standard infrared line of sight (IR LOS) detection.

Interoperability between device and system could have an impact on device selection. This should be taken into account as early as possible in the design philosophy. If this information is not available, conservative assumptions should be made with respect to coverage and updated as the project evolves.

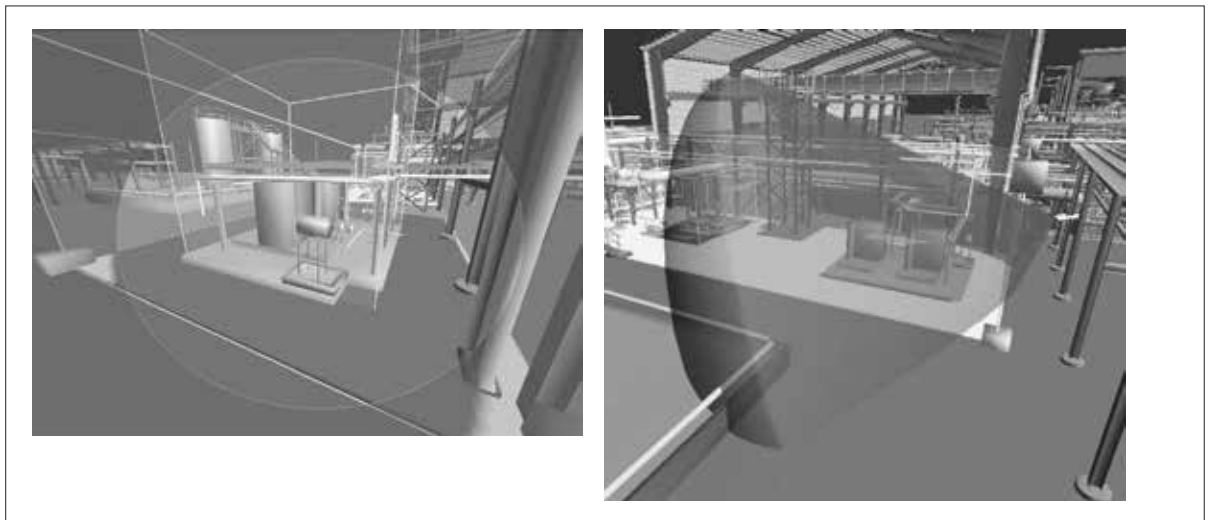
6.2 Flame detection

6.2.1 General

For all detector types the manufacturer's and third-party performance testing of detector performance for different fire types should be taken into account and guidance on installation followed.

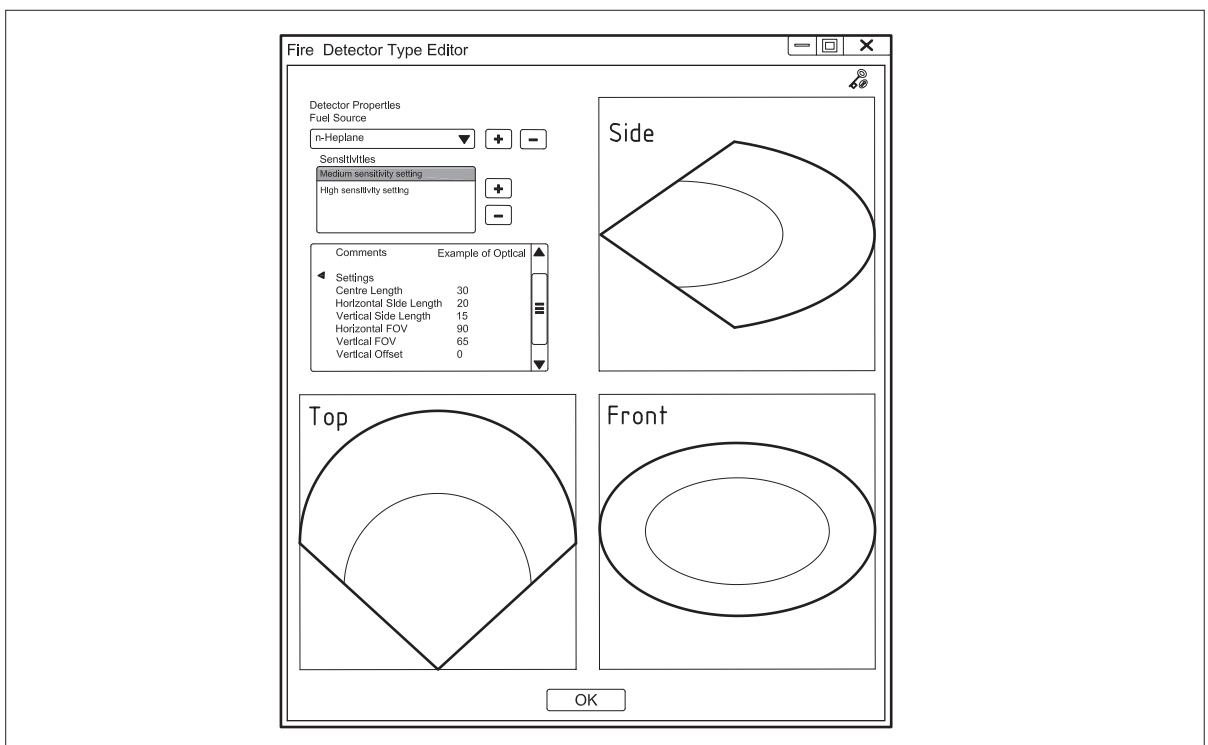
6.2.2 Mapping of optical flame detection

Optical flame detectors operate on a “detection envelope” principle which is characteristic of the model type. The detection envelope depends on the “field of view” (detection distance and angles defining the cone of vision), the centre length distance (distance of detection for the specified flame), and the fall-off in sensitivity around the periphery of the cone of vision (see [Figure 3](#) for an example from one software package).

Figure 3 — Typical point of view (POV) alongside external view of the same envelope

This information is typically included in the detector technical manual and should be used, if possible, to define the volume of coverage in the mapping software.

An example of how this is characterized in one software package is shown in [Figure 4](#). In this case, the inner and outer envelopes represent the detection ranges for two sensitivity settings, as set by detector hardware switches.

Figure 4 — Example of a flame detector field of view

As each detector model has a different detection envelope, it is important to model the correct detector type. Optical flame detector mapping should include the following information to model the coverage characteristic:

- detector make and model;
- detector sensitivity setting; and
- fuel (for the fire hazard being modelled).

The mapping methodology should also take account of degraded performance in operational conditions, e.g. by factoring in a reduced detection distance to allow for dirty optics. Where the detector manufacturer does not publish sensitivity distances for the precise fuel type, the nearest one should be selected (see [7.2.4](#) for guidance on spectral characteristics).

Detector sensitivity is not a standardized parameter with flame detection. Flame detector manufacturers publish performance claims to a set of tested fires. Performance of the device in the field is determined against these, typically using the inverse square law. This means a fixed detection cone is rarely an acceptable way of modelling coverage as the device might be able to see a fire at a much greater distance if the radiant heat output (RHO) of the flame is sufficiently high.

For example, a square foot n-Heptane pan fire is widely accepted to generate an RHO of approximately 40 kW. If the data sheet claims detection of this fire at 30 m, it can be modelled that the flame detector detects a 160 kW fire at 60 m (see [7.3.7.1](#) for guidance on the inverse square law).

6.2.3 Fire detection

The main fire detection technologies in general industrial use are:

- a) optical flame detection;
- b) heat detection; and
- c) smoke detection.

Mapping assessment methodologies are typically applied for optical flame detectors in order to optimize numbers and ensure coverage targets are met. Heat detection is currently not regarded as suitable for mapping optimization but its contribution to fire detection coverage can be taken into account during mapping, assuming adequate risk assessment has been carried out.

NOTE Smoke detection is outside the scope of this British Standard.

6.2.4 Optical flame detection

Optical flame detection consists of the following types of sensor:

- a) infrared (IR) sensors;
- b) ultraviolet (UV) sensors; and
- c) imaging/visual (camera-type) sensors.

IR and UV are further categorized as:

- 1) Single IR (see [6.2.5](#));
- 2) Multi-spectrum IR (see [6.2.5](#));
- 3) UV/IR combined (see [6.2.7](#)); and
- 4) UV (see [6.2.6](#)).

6.2.5 Infrared (IR) and multi-spectrum (MSIR)

IR flame detectors can be “single IR” or “multi-spectrum IR”. Single IR devices use a single IR channel filter to “look” for the IR signal emitted when a hydrocarbon fire burns. Signal processing is used to reduce the response from other IR sources (such as hot compressor or engine exhausts and sunlight). Multi-spectrum IR devices “split” the IR signal into channels to monitor different frequency ranges. In so doing, they attempt to look at the IR signal due to hot CO₂ (or in some cases water) and therefore attempt to further reduce false alarms from hot surfaces and background radiation.

6.2.6 Ultraviolet (UV) flame detectors

UV flame detectors respond to the UV radiation emitted from a fire. UV flame detectors are typically fast responding devices. Their location should be carefully planned as they are susceptible to false alarms from electrical arcing and lightning and also lose sensitivity if their lenses become coated in a film of hydrocarbons. It is more common for UV detectors to be used indoors where the surrounding environment is controlled.

6.2.7 Combination ultraviolet/infrared (UV/IR)

These devices combine the detection principles of a single IR and UV detector with the aim of reducing false alarms. They do this by applying an AND gate; both sensors need to detect a presence of their respected radiation in order to output the fire event. These combined devices can be an advantage in some specific applications to reduce false alarms; however, the UV channel can lose sensitivity if the lens is covered by a thin film of hydrocarbon, and IR detectors can be desensitized by multiple environmental factors, so their use is often restricted.

6.2.8 Visual flame/CCTV

Visual flame detectors are designed to respond to the visual footprint of a flame within the visual/near IR region of the electromagnetic spectrum. This region allows the devices to distinguish between real flames and other, non-visual, background radiative sources (i.e. flare radiation, turbine exhausts). This also means the detectors are unable to see fires which burn invisible to the naked eye, i.e. methanol, hydrogen. Some units can provide live CCTV images, and video recording of fire events. These devices are typically applied on facilities with/near a flare, or where the environment can be challenging (i.e. offshore oil and gas, onshore external exposed environments with hot processing equipment/background radiation, etc.).

6.3 Heat detection

COMMENTARY ON 6.3

Within this British Standard, the term heat detectors include fusible plug, linear wire, frangible bulb, pneumatic systems, bimetallic strip, IR heat sensor and thermistors.

Heat detection devices are generally considered as point detectors and respond when exposed to an elevated temperature at a predetermined set point.

6.4 Flammable gas detection

6.4.1 General

For all detector types the manufacturer's and third-party performance testing of detector performance for different gases and vapours should be taken into account and guidance on installation followed.

Point flammable gas detectors can be pellistor or IR. It should be noted that, unlike IR point detectors, pellistor detectors rely on both gas and air to be present to detect the gas component and might not therefore respond in situations where large gas releases rapidly displace the air. IR detectors operate in 100% v/v gas environments. Relative sensitivities for pellistors and IR detectors change significantly across the broad range of hydrocarbons, e.g. pellistors are more sensitive towards lighter hydrocarbons than heavier hydrocarbons which is the inverse for IR detectors. IR detectors cannot detect hydrogen.

6.4.2 Pellistor

The pellistor sensor works on a Wheatstone bridge principle where one of the four resistors react to the target gas. This chemical reaction results in a rising temperature that causes an electrical

change across the bridge. Certain airborne contaminants can affect the chemical reaction on a temporary or permanent basis, so regular maintenance is a supporting requirement when using pellistor technology.

NOTE Pellistor is also referred to as catalytic.

6.4.3 Thermal conductivity sensor

The operation of a thermal conductivity sensor depends upon the change of heat loss by conduction of an electrically-heated element referenced against another electrically-heated element which is not exposed to the measurement gas or vapour.

6.4.4 Infrared (IR) point

The principle operation of an infrared sensor is IR absorption at specific frequencies referenced to non-absorbent frequencies. A point detector detects gas at the point of its sensing element. IR devices need to be selected to detect all hazardous gases within the target zone and the alarm set point selected to ensure adequate response to the least sensitive gas likely to be encountered in a hazardous situation.

6.4.5 Line of sight (open-path)

6.4.5.1 IR based

IR LOS detectors transmit an IR beam between a transmitter and a receiver. LOS detectors can have a path length between 0 m and 250 m and the maximum allowable path length should be taken into account.

IR devices might show a differing response to lower explosive limit (LEL) levels of different gases and devices need to be selected to detect all hazardous gases within the target zone and the alarm set point selected to ensure adequate response to the least sensitive gas likely to be encountered in a hazardous situation.

LOS detectors measure the integrated gas concentration within the beam path (beam path absorption), e.g. if a well-mixed gas/air accumulation of 10% LEL concentration was enclosed and fully distributed throughout a module in which the following LOS detectors were installed, the following signals would be recorded from each device:

- a) LOS 1: 10 m long would record $10 \text{ m} \times 10\% \text{ LEL} = 100\% \text{ LEL m}$ or 1 LEL m; and
- b) LOS 2: 30 m long would record $30 \text{ m} \times 10\% \text{ LEL} = 300\% \text{ LEL m}$ or 3 LEL m.

6.4.5.2 Laser based

Laser LOS detectors are very similar to IR LOS detectors in respect to mounting arrangement and path lengths. However a key difference between laser and IR technology is that laser technology is gas-specific, therefore it is essential that the target gas is matched to the laser device. Alongside the gas-specific characteristics of laser-based technologies lies greater sensitivity in the ppm/metre range, however overall path lengths might need to be reduced to gain this extra sensitivity.

6.5 Toxic gas detection

6.5.1 General

For all detector types, the manufacturer's and third-party performance testing of detector performance for different toxic gases, including cross sensitivities, should be taken into account and guidance on installation followed.

6.5.2 Electrochemical sensors

Electrochemical sensors consist of two or more electrodes, sensing electrodes, reference electrodes and measuring electrodes. This type of sensor works on the diffusion of the target gas into the sensor resulting in a change of electrical current proportional to gas concentration.

The operation of an electrochemical sensor depends upon the electrical parameters of electrodes placed in an electrolyte due to redox reactions of the gas on the surface of the electrodes. As with all types of sensors, the cross sensitivity towards other gases needs to be known before use and this information can be found in the manufacturer's supporting data.

6.5.3 Semiconductor sensors

The operation of a semiconductor sensor depends upon changes in electrical conductance of a semiconductor due to chemisorption of the gas being detected at its surface. Many semiconductor sensors require higher gas concentrations to initiate a response due to the "sleep effect" of semiconductor materials.

6.6 Ultrasonic (acoustic) gas leak detectors

Ultrasonic gas leak detectors (UGLD) "listen" for the characteristic sound of gas under pressure being released through an orifice.

UGLD can be considered as a "first line of defence" device as they do not measure the concentration of a potential hazard but do respond quickly to pressurized gas leaks, depending upon the built-in time delays to guard against spurious trips. While UGLDs can be used to detect the release of gas, mapping of UGLD is outside the scope of this British Standard.

7 F&G mapping techniques

7.1 General

Three different mapping methods are available:

- a) prescriptive method;
- b) volumetric-based method; and
- c) scenario-based method.

Prescriptive methods are the most simplistic, with the level of complexity and effort increasing progressively towards the employment of scenario-based methods. The mapping methods can be considered analogous to qualitative (prescriptive), semi-quantitative (volumetric-based) and fully-quantitative (scenario-based method) methods applied to risk assessment. The concept of proportionality should be applied with the methodology chosen being appropriate to the level of risk. Further detail on these approaches is provided in [7.3](#), [7.5](#) and [7.6](#).

7.2 Flame properties

7.2.1 General

Where the potential for a flaming fire exists and traditional fire detection technologies cannot be relied upon for detection (e.g. in an external area/an area where [BS 5839-1](#) is not applicable), the use

of optical-based flame detection should be determined. In determining the risk posed by such flaming hazards, a number of factors should be taken into account, including, but not limited to, the:

- a) nature of the anticipated fire (gas jet, liquid spray, pool fire, etc.) (see [7.2.2](#) and [7.2.3](#));
- b) burning characteristics of the material in the area (whether the processed material contains materials which burn with a unique characteristic making detection through optical means difficult, e.g. hydrogen) (see [7.2.4](#));
- c) escalation potential of a secondary fire, such as escalation potential associated with a particular piece of equipment or area, e.g. if a 100 kW RHO flame impinges a particular vessel, is there potential for escalation which would be unacceptable against the organization's risk philosophy, e.g. boiling liquid expanding vapour explosion (BLEVE);
- d) dimensions of the graded volume for a piece of equipment [e.g. at what distance from a vessel would the target fire (e.g. 100 kW) not lead to escalation]; and
- e) fire load/inventory size, e.g. fire duration.

Accounting for the factors in a) to e) allows the designer to apply escalation grades/performance targets (see [7.3.2](#)) when prescriptive measures are not applied.

7.2.2 Pool fire

Where an ignition occurs after the leak has accumulated as a liquid pool, a pool fire can occur. Pool fires either spread under gravity and reach an equilibrium pool size (where the burn rate of the fuel equals the spill rate) or are limited by geometric features local to the release (e.g. contained within a bund or the walls of a room).

Detectors should be placed to take account of the location of liquid inventories and the likely areas where leaks would accumulate, e.g. bunds, sumps, ditches, below grated decks.

7.2.3 Momentum driven jet/spray fire

Pressurized releases that ignite result in a momentum driven jet/spray fire. The jet can be either gas or liquid or a mixed phase resulting in a jet/spray fire. Secondary pool fires should also be taken into account.

Jet/spray fires can be in any direction as they typically occur due to the failure of flanges or instrumentation. Their momentum also means they are more susceptible to impingement on equipment and walls than pool fires. If conducting volumetric mapping, this should be taken into account when specifying the volume over which the detector coverage is assessed.

7.2.4 Spectral characteristics

Dependent upon the characteristics of the material burning, flame characteristics vary. Pool fires and jet/spray fires (gas, liquid, mixed) generate radiation to varying extents. The carbon content of the flame also has an impact on the capability of differing technologies to respond, and also impacts the response time of detection. Certain fuels that are free from impurities (e.g. methanol and hydrogen) burn "cleanly" with a flame colour with limited luminosity. The flames from these fuels might appear "invisible" to some technologies, so the detection characteristics for the expected fuel (including purity) should be determined and the technology selection (UV, IR or visual) should be suitable for the anticipated burning characteristics of the fuels of concern.

7.2.5 Environment

The environment in which optical flame detectors are to be applied should be taken into account, including seasonal weather conditions, e.g. will the detector operate in warm/cold temperatures, when there is fog/mist, when the sun is shining/modulating?

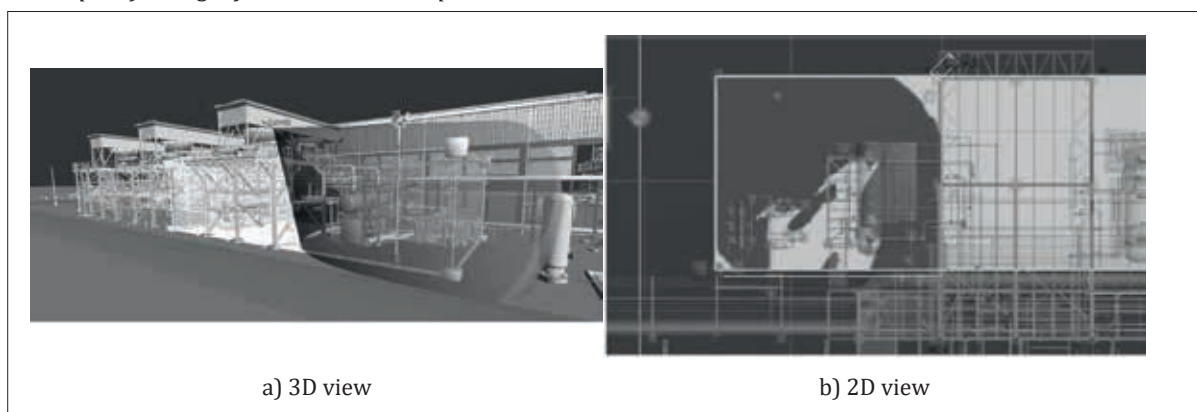
7.3 Flame detection

COMMENTARY ON 7.3

This subclause primarily addresses optical flame detection, however this does not repudiate taking credit for heat detection within the process area. Heat detection can be mapped in process areas, typically giving credit for coverage of the area in close proximity to the location of the sensors (see 6.2.3). This method of design needs to be reviewed against the risk assessment process and facility approach to risk (see Clause 4 and Clause 5), taking into account the reduced sensitivity and detection capability of heat detection in external areas.

An example detection envelope of a flame detector is shown in [Figure 5](#).

Figure 5 — Example of a single flame detector map shown in 3D and 2D views



7.3.1 Set points and voting

Flame detectors do not have set points and are binary in their response, i.e. they either generate an alarm or they do not. They do not respond to different RHO levels of a flame (in the way a gas detector provides different responses to %LEL). The differing level of RHO, however, allows detectors to be positioned closer to/further from the source depending on the target fire size. The multiples of RHO commonly seen (10 kW – 500 kW) should be calculated from a baseline sensitivity, e.g. the 1 ft² n-Heptane pan fire. This detection distance should then be extrapolated to larger or smaller fires using the inverse square law (see [7.3.7.1](#)) to provide an accurate measure of detection capability to the fire size of concern.

To minimize the potential for spurious trips within a mapping zone, detector voting can be employed. As an example, an alarm would be generated if gas was detected by one detector (1ooN, where N is the number of devices in the voting group) followed by an executive/control action on two detectors detecting the gas (2ooN).

NOTE Voting can also be applied when using multiple detectors at a specific point, e.g. 3 × smoke detectors located within a vent duct.

7.3.2 Graded volumes and performance standards

When conducting flame detection mapping it is typically necessary to define a target fire size. This can be based on protecting personnel, avoiding escalation, ensuring business continuity or protecting the environment. This fire should be justified based on the flammable materials handled in the area, the conditions of the process and the corporate approach to risk.

When selecting a target fire size, it should be noted that the geographic volume of a flame is not the only determinant of the escalation potential posed by a fire. A jet fire with the same base area as a pool fire provides significantly different consequences. The surface emissive power is, typically, higher for jet fires. For this reason, where the hazard is a flaming fire, a fire hazard can be defined by its RHO, specified in kW, which is verified by third-party certification testing (i.e. BS EN 54-10

and FM 3260 [4]). RHO provides a good indication of the potential damage and likelihood of escalation and loss.

NOTE 1 When applying RHO as a metric, the designer is required to implement an RHO value for the type of fire specified. There is no industry consensus or standard which specifies how to do this, but methods include extracting values from test burns, fire models, traditional fire dynamics calculations, etc.

Grades and performance standards should avoid using target fires which flame detectors have not been certified to detect (i.e. large fires which result in sensor saturation/blinding). The target fire sizes used in the placement of flame detectors are typically smaller than those associated with escalation, i.e. the intention is to detect the fire before escalation occurs, not to detect the fire which could credibly occur in a worst-case scenario. Care should be taken if determining the placement of flame detectors based on worst-case fires calculated as part of the predictive aspects of a *Control of Major Accident Hazards Regulations 1999 (COMAH)* [5] assessment or a fire risk assessment/quantitative risk assessment (QRA) conducted to support an offshore safety case. If detectors are located on this basis, smaller fires that could potentially cause escalation might not be detected, and the design could be based on fires the detectors are not certified or capable of detecting.

Multiple target fire sizes are often taken into account, including:

- a) target fire size for alarm (where the fire is not significant enough to result in automated actions and can be dealt with manually); and
- b) target fire size for executive action (when an operator can no longer be relied upon to initiate the mitigating action manually to reduce/prevent escalation, i.e. the fire has grown to a level where automated action is required).

These fire sizes have different values (based on RHO), e.g. an alarm is generated at 50 kW RHO fire and executive action at 100 kW RHO.

Where the escalation potential and methodology warrant it, graded volumes can be defined within the mapping zone. The size of the fire to be detected within the graded volume can be based on the escalation potential, if that fire occurs within that volume, allowing optimization of the layout, i.e. smaller fires can be targeted for high escalation areas.

Where the escalation potential and method warrant the application of graded volumes, representation should allow for presentation of coverage beyond a binary “coverage” or “no coverage” (e.g. a fixed XX m detection cone is rarely a suitable metric of coverage). To design a performance-based flame detection system based on a review of escalation potential there are multiple factors to take into account, including but not limited to percentage coverage of:

- 1) alarm only;
- 2) control action;
- 3) late alarm; and
- 4) minimal coverage to all target fire sizes.

NOTE 2 “Late alarm” represents a location where the target fire size for control action will be detected by two devices, however any smaller target fire size for alarm will not be detected to provide a fast response alarm without executive actions.

NOTE 3 “Minimal” coverage represents a location where the target fire size is not within range, but if it grows beyond the specified target, it will be detected by a single device (100N) as a clear line of sight exists to that point.

7.3.3 Prescriptive

Prescriptive flame detector mapping refers to the use of qualitative evaluation and judgement-based placement in the locating of flame detectors. Examples include:

- a) proven design that is reused for similar applications; in many instances this would not require a full modelling activity;
- b) prescriptive ruleset provided by the operator; compliance can be verified using a checkbox type review to ensure the rules are met; and
- c) for open facilities with minimal obstruction to the field of view, it might be sufficient to place detectors at the extremities of the area where fires could exist, taking account of the detector range.

Software flame and gas modelling tools and metrics (e.g. fire size/percentage coverage) are not required when using the prescriptive approach and compliance can often be checked using a checkbox-type approach against the rule set that has been employed.

7.3.4 Volumetric-based method

Volumetric-based mapping assesses the capability of a system to detect a fire of a certain size and presents the results in terms of the coverage of a given volume. The size of fire can be based on different methods, including effective viewing distance to the flame of specified size, or RHO of a fire.

Volumetric-based flame mapping is performed using software packages which verify the coverage achieved by the system.

The advantages of volumetric-based flame mapping include:

- a) 3D model can be applied for visualization purposes;
- b) information on process incorporated, e.g. heat and mass balance data, can be used to show stream pressure and composition which influences the potential for escalation and therefore the selected target fire size;
- c) no modelling of specific scenarios is required;
- d) analysis is relatively quick;
- e) consistent designs;
- f) easily audited;
- g) coverage can be validated against certified detection capability; and
- h) it is less subjective than scenario-based type approaches.

A disadvantage of volumetric-based mapping is that results can be influenced by the size of volume assessed, i.e. the designer can expand the graded volume into open space to increase the resulting percentage coverage achieved. [7.3.6](#) gives guidance on assessing adequacy and the issues surrounding the using percentage coverage as the sole metric of adequacy.

NOTE An example of volumetric-based flame mapping is given in [Annex B](#).

7.3.5 Scenario-based method

Scenario-based methods in the analysis of flame detection coverage are not commonly used due to a perceived lack of benefit for the additional complexity. Guidance on their application is therefore omitted.

7.3.6 Assessing adequacy of coverage for volumetric mapping

Where modelling is carried out, the resultant coverage should be reviewed for adequacy. The designer should assess the percentage coverage (if applicable) of each volume and determine if the coverage is acceptable.

Any predetermined target for coverage should not be used as a definitive pass/fail criterion. Percentage coverage alone might not be a suitable measurement against which to determine acceptable coverage in isolation.

It is important that each modelled volume is reviewed both in isolation and in the greater context of the facility to determine if the coverage is acceptable. This review process should have input from all relevant disciplines.

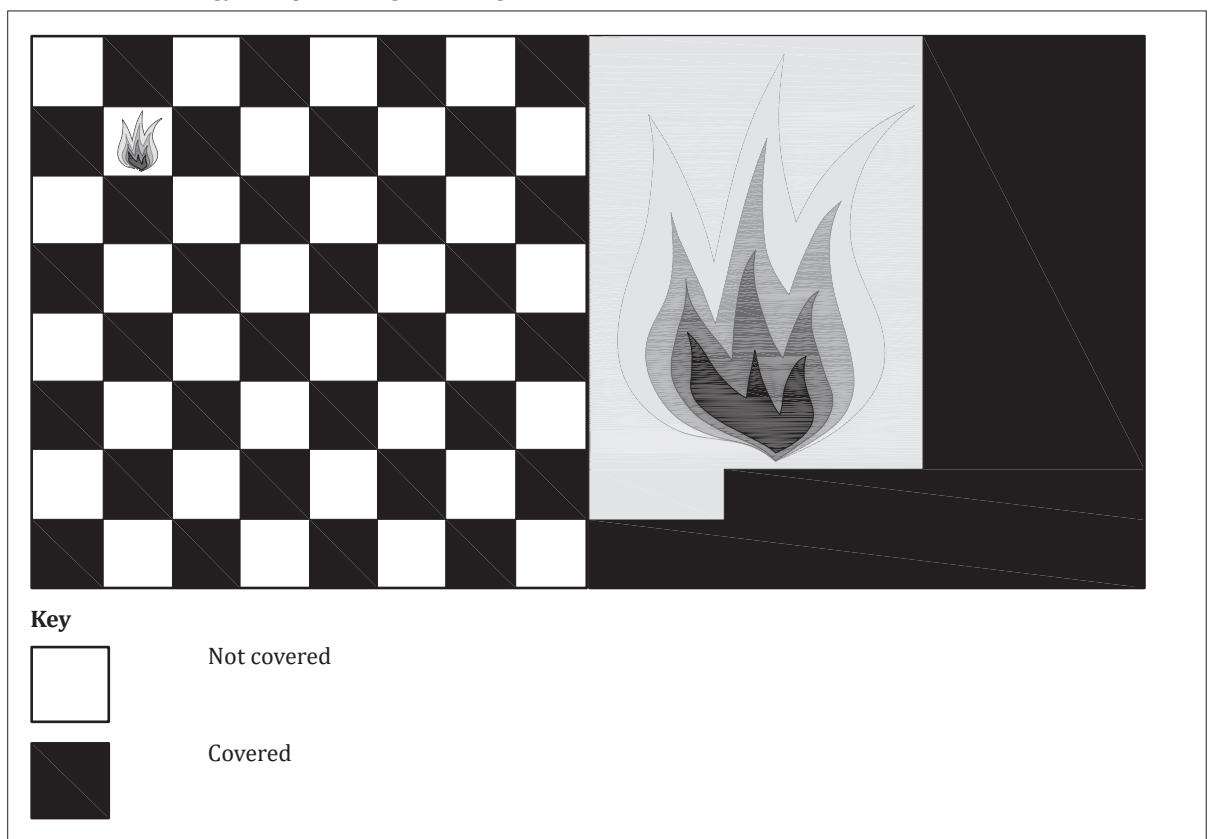
This approach ensures that:

- a) the project team are fully aware of any potential gaps in coverage, or areas where the system could be over-engineered; and
- b) the resultant design is appropriately optimized by reviewing areas of suitable coverage, blind spots where fires might remain undetected, and how each detector contributes to coverage.

Reviewing detector contributions allows the designer to identify when the point of diminishing returns is reached with respect to the addition of more detectors.

Alongside the concept of reducing the risk to ALARP, the addition of detectors should take into account the benefit in coverage provided for each additional device proposed. Percentage coverage alone is not a suitable metric for this (however it can be used as a guide), as the designer needs to review the nature of the area covered to determine if the cost of additional detectors is suitably offset by the reduced risk of a hidden fire at that specific location. This assessment of acceptability can change throughout the same volume, for varying degrees of percentage coverage achieved. For example, a device which provides only a 5% increase in coverage for an area where numerous fires can originate with significant escalation potential would be preferable over a detector which would provide an additional 10% coverage of an open area with limited escalation potential.

The deficiencies of analysing solely on percentage coverage are illustrated in two dimensions using the analogy of a chess board in [Figure 6](#). [Figure 6](#) shows that in both instances there is a 50% coverage. The example on the right, however, has much larger “gaps” and therefore a significant fire could go undetected. Percentage coverage can give an indicative metric for the effectiveness of fire detection systems but should not be used in isolation. The designer should review gaps in the detector layout, as these are the weak points, and assess if a fire could remain undetected at those locations.

Figure 6 — Chessboard analogy and percentage coverage

7.3.7 General considerations

7.3.7.1 Inverse square law

The inverse square law provides an approach that can be used to represent detection capability for a range of fire sizes. This allows multiple fire sizes to be included in a single assessment, if required, in accordance with 7.3.2. If multiple fire sizes are to be modelled, it is necessary to select tools/methods capable of calculating this, for example where a piece of equipment is assigned a graded volume to detect 10 kW for alarm (100N) and 100 kW for control action (200N). For a performance-based flame detection assessment, coverage should be calculated based on the inverse square law, not an assessment of a binary closed cone of vision.

7.3.7.2 Flame detection response time

Desired response time from the flame detector should be based upon the hazard intended to be detected and the escalation potential. Designers should ensure the response time is acceptable in allowing effective mitigation measures to be applied.

Designers should verify that the response time across the full claimed field of view is suitable. Any reduction in detection distance to meet the desired response time should be accounted for in the mapping where software modelling is carried out.

7.3.7.3 Point/volume source modelling

When mapping is carried out, flame detection can be modelled by mapping points in space, verifying if a clear line of sight exists to that point (point source modelling), or by representing the fire as a 3-dimensional shape (volume source modelling) and verifying that there is a line of sight between the detector and that volume or a portion of that volume.

Coverage represented using the point source approach requires a clear line of sight to the location of the targeted fire to be represented as successful detection. Engineering judgement is then applied to analyse adequacy of the layout and subsequent coverage (see 7.3.6).

Care should be taken when using volume source modelling as the result could differ depending upon flame detection technologies and/or models (e.g. the region of the electromagnetic spectrum the device operates, specific filters applied, signal processing firmware, burning fuel).

EXAMPLE Two “triple IR” flame detectors, despite being the same technology, might use different filters along the mid IR range (impacting solar blindness, sensitivity, false alarm immunity, etc.), analysing the signals received in a unique way. Where the flame volume is partially obscured by geometry, the features of the specific flame detector being modelled (including sensitivity to the portion of the flame visible, the environmental conditions of the facility and any desensitization the flame detector might be subjected to) should be taken into account.

NOTE These data are often protected by the flame detector manufacturer.

7.3.7.4 Detector sensitivity

The viewing characteristics of a flame detector typically presented by its manufacturer are those under ideal test conditions.

Each flame detection technology and model are desensitized in a unique way and, to a varying degree of severity, dependent upon the technology, model, facility and environment.

Calculation of the “in-field” viewing distance to a target fire (i.e. the 1ft² n-Heptane pan fire) should be taken into account. This is not the maximum viewing distance at which the detector detects a fire. The in-field detection distance should be a facility-tailored baseline sensitivity. If using a flame detector modelling tool, this value should be extrapolated to suit the grade on any given piece of process equipment/volume and its subsequent target fire size applying the inverse square law.

The following factors should be determined, as a minimum.

- a) Reduction in sensitivity to genuine flame in the presence of unwanted/false alarm stimuli. Values for this can be taken from the manufacturer or detector manual, if disclosed. Unwanted stimuli can include sunlight (direct, modulated, unmodulated, reflected), welding activities, blackbody radiation (modulated/unmodulated), florescent/incandescent lamps, shielded and unshielded quartz halogen lamps.
- b) Dirty optics. Optical fault occurs at a manufacturer-set degradation in field of view. The impact of dirty optics should be taken into account in the calculation of flame detection effectiveness. The maintenance periods and detection technology’s susceptibility to dirty optics can be taken into account in the value applied.
- c) The filter edge effect represents the reduction in sensitivity across the claimed field of view from the maximum at the centre line. This value should be justified from an analysis of the flame detector’s specific detection cone of vision as this differs between devices. If modelling is applied, this should account for the shape of the detector field of view. The filter edge effect can also have an impact on detection response time and affects devices in a unique way, dependent upon the filters and sensors used. The device manual and third-party certification data (i.e. FM approvals test report) should be consulted to review the impact of the filter edge effect on devices.

7.4 Fluid states and dispersion

7.4.1 General

Fluids vary in their nature dependent on their state, pressure and temperature. The nature of the fluid being released influences how the gas/vapour disperses and has a direct effect on where

gas/vapour accumulates and hence where detectors are placed. This is equally applicable to both flammables and toxic gasses. The different physical states are:

- a) a gas, which might be at an elevated temperature or pressure;
- b) a gas liquefied by the application of pressure, e.g. LPG, CO₂;
- c) a gas which can only be liquefied by refrigeration, e.g. methane; and
- d) a liquid with an associated release of flammable or toxic vapour.

In addition to fluid state, ventilation also affects where gas accumulates and how gas/vapour disperses to a safe limit (i.e. below the flammable or toxic limits). The different types of ventilation are:

- 1) local extract ventilation;

NOTE 1 This is generally installed local to a release point to remove gas/vapour and exhaust it to a safe location.

- 2) bulk mechanical ventilation;

NOTE 2 This is typically installed to ventilate rooms or enclosures and, if designed correctly, prevents dead spots (areas of low ventilation or recirculation) where clouds could accumulate, dilutes gas to a safe limit and/or exhausts it to a safe place. Bulk mechanical ventilation can influence detector placement as gas/vapours are likely to be carried by the flow, hence detectors need to be placed taking this into account.

- 3) natural ventilation;

NOTE 3 For enclosed plants, natural ventilation is provided by either flow-through louvres or advantageous openings around doors, windows or penetrations.

For open plants, the degree of natural ventilation depends on external buildings (if significant), large items of equipment, cladding and prevailing wind conditions.

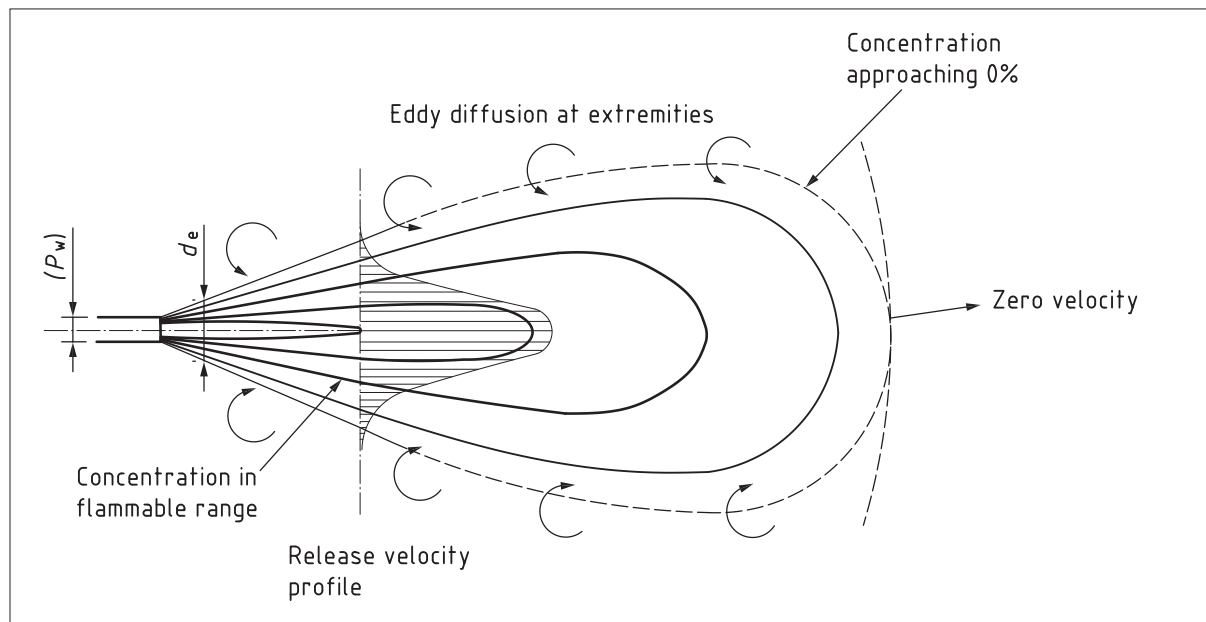
With respect to natural ventilation and the placement of detectors, the greater the natural ventilation, the more dilute the cloud and the “harder” it is to detect. Conversely, for a given release, the lower the consequence as the flammable cloud is smaller. Therefore, for a plant that has significant ventilation, detection might be impractical as it is unlikely to detect releases unless they are catastrophic.

Prior to detector mapping being carried out the nature of the release, ventilation and hazard range should be determined. The hazard range can be used to inform the maximum distance from any release point that detectors should be considered.

Accounting for the factors in 7.4.2 to 7.4.6 allows the designer to apply escalation grades/performance targets where prescriptive measures are not applied.

7.4.2 Gaseous releases

Gaseous releases result in a jet or plume, dependent on the storage pressure. Above the critical pressure the release is choked and a sonic jet forms; below the critical pressure the release forms a plume with the velocity dependent on the storage pressure. For unobstructed, external jet releases in the near field (i.e. close to the release) the dispersion is driven by the entrainment of air into the jet caused by the differential velocity between the jet and the surrounding air (see [Figure 7](#)).

Figure 7 — *Jet entrainment*

For plumes, velocities are lower and therefore entrainment due to differential velocity is lower and dispersion is governed by the bulk air flow (provided either naturally or via forced ventilation).

If releases impinge on geometry this reduces the momentum, and hence the velocity, leading to less entrainment of air, larger cloud sizes and a break-up of the jet/plume type structure. If releases are inside buildings, and the gas is not removed, there is the potential for gas to build up, leading to a background concentration of gas; this leads to larger cloud sizes as air that is entrained into the plume or jet already contains gas. Detectors should be placed in areas where gas is likely to accumulate based on the geometry and/or room. The buoyancy of the gas also affects the size, shape and location of where gas accumulates. Buoyancy tends to have more of an effect for plumes, obstructed or internal releases. For unobstructed jets, the momentum and direction of the release tends to dominate over any buoyancy affects.

7.4.3 Location of detectors near pressurized leak sources

Point detectors which are placed with the purpose of monitoring specific leak sources of pressurized gas should not be located too close to those leak sources.

A pressurized gas plume is narrow near the initial release point and expands with distance due to the air entrainment (see [Figure 7](#)).

Therefore, gas detectors (particularly if point detectors are used) are best placed some distance from the leak source – typically, detectors should be placed away from a collection of sources such as a skid with multiple flanges and instrument fittings (the distance to be agreed by the project team and justification documented). This is less applicable when a volumetric-based approach is applied in an area where there are leak sources across the mapping zone.

If pellistor point detection is used, the gas plume is oxygen-deficient prior to air entrainment. Pellistor detectors require the presence of oxygen to function and have poor performance in the gas-rich zone close to the leak point.

7.4.4 Gases liquefied by the application of pressure

For gases liquefied by pressure (e.g. propane and butane), an aerosol is formed on release. The aerosol is made up of droplets of varying sizes. Small droplets might vaporize, forming a flammable cloud, whereas larger droplets might coalesce and rain out onto the ground, forming a pool. The pool

would then absorb heat from the surroundings, leading to evaporation and the formation of vapour that can add to the cloud formed from the aerosol.

7.4.5 Gases liquefied by refrigeration

Gases that are liquefied by refrigeration evaporate on release. Small releases might absorb sufficient energy from the environment to form a vapour without pooling. Larger releases typically pool before forming a vapour cloud. Once released, the pool boils as heat is absorbed from the surroundings, leading to a vapour cloud. Initially the vapour is cold and tends to be heavier than air but heats up as air is entrained, leading to a cloud that transitions to being neutrally or positively buoyant depending on the fluid released. The vapours from releases of this type have low momentum and are non-directional so are not likely to be affected by obstructions in the near field, and the area where vapour is likely to accumulate is driven by the bulk air flow (provided either naturally or via forced ventilation).

7.4.6 Liquid release with an associated release of flammable or toxic vapour

For these types of release the vapour evaporates from the liquid pool and the extent of the cloud is dependent on the vapour pressure and its flammable range and/or toxicity. The likely release points and where they are likely to accumulate should be taken into account given that liquids are likely to follow the terrain and topography of a facility. The vapours from releases of this type have low momentum and are non-directional and are not likely to be affected by obstructions in the near field, and the area where vapour is likely to accumulate is driven by the bulk air flow (provided either naturally or via forced ventilation) and the buoyancy of vapour. For external releases, in the near field, vapours which are negatively buoyant tend to accumulate close to the ground whereas buoyant vapours tend to rise; within buildings or enclosures the location of the cloud is driven by a combination of the buoyancy and the bulk air flow (provided either naturally or via forced ventilation). In the far field, the initial buoyancy of the vapour is less dominant as the plume entrains air and becomes more neutrally buoyant.

7.5 Flammable gas detection

7.5.1 Set points and voting

Set points for flammable gas detectors are set below the LEL of the targeted gas (0%-100% LEL). In the event of a gas release, the lower the set point the earlier the detection system responds.

NOTE 1 Guidance on set points for point flammable gas detectors is taken from HSE publication, "The Selection and Use of Flammable Gas Detectors" [6]. This states:

"The gas detector should be set to alarm at a level low enough to ensure the health and safety of people but high enough to prevent false alarms. False alarms are most likely to be caused by fluctuations in sensor output due to environmental changes (e.g. ambient temperature, pressure or humidity), sensitivity to other gases or vapours, or sensor drift. If false alarms are a problem then one option is to use two detectors - the alarm level must be registered by both detectors before the alarm activates.

"In determining the required alarm levels for fixed gas detection systems, the following should be taken into account:

- a) any industry standards and recommendations;
- b) the LEL of the gas or vapour;
- c) the size of the potential leak and the time to reach a hazardous situation;
- d) whether the area is occupied;
- e) the time required to respond to the alarm;

- f) the actions to be taken following the alarm; and
- g) the toxicity of the gas or vapour.

A suitable safety margin also needs to be incorporated to account for ventilation dead spots, where vapours could accumulate, and the variability of natural ventilation. One option is to set two alarm levels. The lower alarm could act as a warning of a potential problem requiring investigation. The higher alarm could trigger an emergency response, such as evacuation or shutdown or both. For leak detection purposes (i.e. not process monitoring), the first alarm level is to be set as low as practicable, preferably not higher than 10% of the LEL. The second alarm level is to be no more than 25% LEL.”

Similar guidance is given in BS EN 60079-29-2 with respect to having set points as low as possible but avoiding false alarm. It gives specific values not to go below in order to avoid potential false alarm:

- 1) 5% LEL for methane;
- 2) 10% LEL for propane; and
- 3) 20% LEL for gasoline vapours.

The guidance in BS 60079-29-2 and from the HSE [6] is consistent, except for gasoline vapours. BS 60079-29-2 gives a higher set point than the minimum recommended value from the HSE.

Open-path detector set-points are in LEL/m, i.e. concentration (LEL) is integrated across the path length (m). The same principles apply as point detectors in terms of setting set points as low as practical without encountering false alarms. Guidance given for low- and high-level set points for open-path detectors is:

- i) low-level – 1 LEL m; and
- ii) high-level – 2 LEL m.

NOTE 2 See HSE research report RR1123 [7].

The set points to be used should be decided before the mapping is completed and should form part of the gas mapping philosophy. This should be used as input if quantitative gas mapping (volumetric or scenario-based method) is being carried out.

If quantitative gas mapping is carried out, sensitivity analysis could be performed to understand the improvement in detection of using lower set points to understand the trade-off with the increased potential for false alarms.

To minimize the potential for spurious trips within a mapping zone, detector voting can be employed. As an example, an alarm would be generated if gas was detected by one detector (1ooN, where N is the number of devices in the voting group) followed by an executive/control action on two detectors detecting the gas (2ooN).

NOTE 3 Voting can also be applied when using multiple detectors at a specific point, e.g. 3 × smoke detectors located within a vent duct.

7.5.2 Graded volumes and performance standards

If volumetric or scenario-based mapping (see 7.5.7.1 for information on the use of target gas cloud size in scenario-based mapping) is being carried out, it is necessary to define a target gas cloud size to detect, based on the philosophy for the gas detection system and what it is trying to protect against. For example, it should be determined if the aim of the gas detection system is to protect against personnel fatalities or escalation, asset protection, business continuity or a combination of each. If prescriptive mapping is being carried out, this target gas cloud is often used as a guide for a set detector spacing.

Historically, detectors are perceived to be applied on (5 × 5) m spacing criteria based on *HSE research report, Report OTO 93 002* [8]. This research showed that the flame speed from a 6 m cloud would not exceed 100 m/s and not produce overpressure of more than 150 mbarg for a methane or propane cloud for a given geometrical configuration. Overpressures of this magnitude are lower than those expected to lead to personnel fatalities.

NOTE See also HSE Technical Report SPC/Tech/OSD/30 [9].

The HSE research allowed for a moving scale of congestion/confinement changes which would subsequently alter the targeted gas cloud size.

HSE research [9] can be used as a basis for detector spacing and risk-based target gas cloud size if handling similar materials and looking to protect against personnel fatalities. If handling different materials (e.g. hydrogen) or looking to protect against escalation (e.g. cloud size that would not lead to failure of a blast wall or temporary refuge), a range of different approaches could be used, including:

a) Qualitative:

The target cloud size to be detected is a function of:

- 1) the overpressure being protected against;
- 2) the material being handled;
- 3) congestion;

NOTE 1 Congestion increases the rate of combustion and hence overpressure as turbulence is generated when the flame front interacts with geometry, increasing the rate at which combustion products are mixed with unburnt fuel.

- 4) the confinement;

NOTE 2 Confinement is the degree to which the expansion of combustion products is inhibited by solid boundaries, e.g. walls, decks or ceilings.

NOTE 3 A qualitative assessment would consider either the work by the HSE, other experimental work or previous project experience to set a target cloud size based on a consideration of the parameters listed.

b) Semi-quantitative:

Semi-quantitative methods are defined as the use of empirical or phenomenological tools to define a target gas cloud for a given overpressure, material, congestion and confinement. Methods that could be employed are the multi-energy method and Game theory. Although these methods are quantitative, they do not offer a rigorous determination of congestion and confinement and the choice of the parameters used in these models can be subjective.

Use of the blockage ratio as a value, correlated to an anticipated explosion overpressure, based on this congestion and confinement can also be applied to determine a target gas cloud performance target.

c) Fully-quantitative:

Fully quantitative methods utilize computational fluid dynamics (CFD) explosion analysis to quantify the target gas cloud size for a given overpressure in a specific area.

NOTE 4 CFD modelling in this context (explosion analysis) is not to be confused with CFD dispersion modelling (traditional scenario-based probability analysis of dispersion characteristics).

CFD methods take 3D models of a facility (hence congestion and confinement are explicitly included) and can capture the influence of the material being handled within the calculations. If CFD modelling is to be employed, the influence of the following should be investigated, including:

1) Cloud position:

The location of the cloud with respect to congestion should be investigated (i.e. multiple simulations should be run) to understand the influence on results.

2) Ignition location:

The influence of ignition location within the cloud should be investigated. In general, larger overpressures are produced for ignition at the edge of a cloud as the flame has a greater run-up distance through which to accelerate.

If CFD analysis is employed, multiple scenarios should be run and an appropriate target gas cloud size selected based on the results. For example, it might not be appropriate to set the target gas cloud size on the worst-case result if this far exceeds other scenarios modelled, and a more pragmatic approach could be to choose an average of the modelled scenarios. If one scenario far exceeds the others modelled it shows that this is based on a unique set of parameters (e.g. gas cloud in a particular location and ignition location) which might be unlikely to occur. Care should be taken, however, in applying such assumptions.

7.5.3 Prescriptive

Prescriptive flammable gas detector mapping refers to the use of qualitative evaluation and judgement-based placement in the locating of gas detectors. Examples include:

- a) proven design that is reused for similar applications;
- b) ruleset-based approach, e.g. detectors at a spacing of X m; and
- c) evaluation of the factors in [7.4](#) and detectors placed at locations where gas is likely to accumulate; this is most appropriate for applications that do not contain many potential leak sources and releases likely to behave in a predictable manner, e.g. no areas of recirculation.

Software modelling tools and metrics (e.g. target gas cloud size/percentage coverage) are not required when applying the prescriptive approach; compliance can often be checked using a checkbox-type approach against the ruleset that has been employed.

7.5.4 Volumetric-based method

Volumetric-based mapping assesses the capability of a system to detect a target gas cloud of a certain size and presents the results in terms of the percentage coverage of a given volume. Volumetric-based mapping is typically performed using software packages that verify the coverage achieved.

The size of the target gas cloud size can be based on different methods, as detailed in [7.5.2](#). The shape of the cloud is typically assumed to be of uniform concentration.

In its simplest form, volumetric-based mapping assumes a gas cloud size can be present at any point within the three-dimensional space being analysed. The total volume to be assessed should encompass all potential release locations, but subdivision might be appropriate where release locations are not evenly distributed. More complex applications can include a dilution factor to account for a central high concentration point, diluting to low concentration the further it moves from the centre.

When conducting volumetric-based mapping, detector set points (see [7.5.1](#)) can be accounted for by scaling the target gas cloud by the dilution factor (e.g. low-level alarm based on cloud size of 15 m and high-level alarm based on cloud size of 5 m); if consequence analysis has not been conducted this factor has to be set based on designer judgement.

The following advantages and disadvantages are generated based on common applications where the method is selected. The critical factor is use of the most appropriate methodology to the specific application, e.g. the exclusion of subjective factors such as wind direction, leak orientation, gas release rate, is a strength in the volumetric-based approach as it aims to detect dangerous accumulations which could result in explosion, rather than calculate how this occurs. If the application is a controlled environment, however, where variables are reduced (i.e. an internal volume with few sources of release), the exclusion of those factors could be considered a disadvantage.

The advantage of volumetric-based mapping include:

- a) 3D model can be applied for visualization purposes;
- b) detailed information on process is not always required, e.g. heat and mass balance data;
- c) subjective features of analysis are reduced by comparison with scenario-based mapping, i.e. where leaks originate, likelihood of pipe rupture, etc.;
- d) no complex modelling of specific scenarios is required;
- e) analysis is relatively quick;
- f) consistent designs;
- g) easily audited;
- h) coverage detects the “dangerous” accumulation of gas, with respect to explosion potential, with a higher degree of certainty than alternative methods as it assumes accumulation has taken place and aims to detect the critical cloud volume; and
- i) change management is easily incorporated (i.e. detailed analysis does not need to be rerun as a result of minor alterations in the process area).

The disadvantages of volumetric-based mapping include:

- 1) results can be influenced by the size of volume assessed;
- 2) there is no clear way of determining where the graded volume is to extend to;
- 3) subjective features of analysis are increased by comparison to prescriptive methods, i.e. various cloud sizes, coverage adequacy interpretation, etc.; and
- 4) behaviour of releases might not be taken into account.

NOTE An example of volumetric-based gas mapping is given in [Annex B](#).

7.5.5 Scenario-based method

Scenario-based mapping requires information specific to the process and is analogous to the approach taken in a QRA, i.e. it considers both frequency and consequence to quantify risk. For this reason, scenario-based mapping is often carried out where QRA has been conducted previously. The QRA can be used to provide the frequency of individual scenarios. Dispersion modelling might have been completed as part of a QRA, but this might not be at the required resolution for scenario-based mapping so that additional dispersion modelling is often required.

Scenario-based method mapping models individual scenarios using a range of different parameters such as:

- a) release location;
- b) release direction;
- c) wind speed and direction; and
- d) hole size,

and quantifies the potential for releases to be detected in terms of either percentage of scenarios detected or the frequency of releases that remain undetected. If conducting scenario-based mapping, event trees can be used to calculate the frequency of each discrete scenario. Event trees consider the initiating frequency of a release and the probability of each individual branch (e.g. release direction, wind speed and direction). The probability of ignition should be accounted for when calculating the frequency of each scenario. Guidance on such calculations is outside the scope of this British Standard.

It should be noted that fire safety engineering applications often assume a probability of the central initiating event (i.e. the fire) being 1.0 to avoid dilution of the holistic risk (across the prevention and mitigation spectrum) of the facility, thereby potentially reducing the emphasis on an effective and critical mitigation system.

The scenarios are modelled using dispersion analysis which calculates the gas concentration in 3D space. Dispersion analysis can be conducted using either phenomenological models or CFD. Phenomenological models are faster to run but in general do not account for obstructions which can have an influence on the size and shape of a cloud following a release. CFD offers a more rigorous modelling approach with the geometry being imported into the modelling software and both impingement and the influence of geometry on upstream wind conditions being captured.

The dispersion analysis also gives more detailed information on each scenario modelled, such as:

- 1) time of detection;
- 2) cloud size versus time, i.e. this can highlight the importance of ensuring a fast system response; and
- 3) number of gas detectors that alarm for a given scenario highlighting redundancy in a system, e.g. this information could be used to justify continued operation if a given detector is in a fault condition.

The following advantages and disadvantages are generated based on common applications where the method is selected. The critical factor is use of the most appropriate methodology to the specific application, e.g. the more thorough consideration of release scenarios is a strength in the scenario-based approach, but if the application is a standard application (i.e. a tank storage facility), expending resources on a more thorough assessment of risks could be classed as a disadvantage.

The advantages of scenario-based mapping include:

- I) 3D model can be applied for visualization purposes;
- II) if dispersion analysis has been conducted, this can be used directly to account for different set points of detectors as the concentration in 3D space is calculated, i.e. judgement is not required in terms of the difference in the size of a cloud at low and high set points; and
- III) more thorough consideration of different release scenarios through consideration of specific release conditions, including wind direction, release orientation, gas release rate etc. [see, however disadvantage 3)].

The disadvantages of scenario-based mapping include:

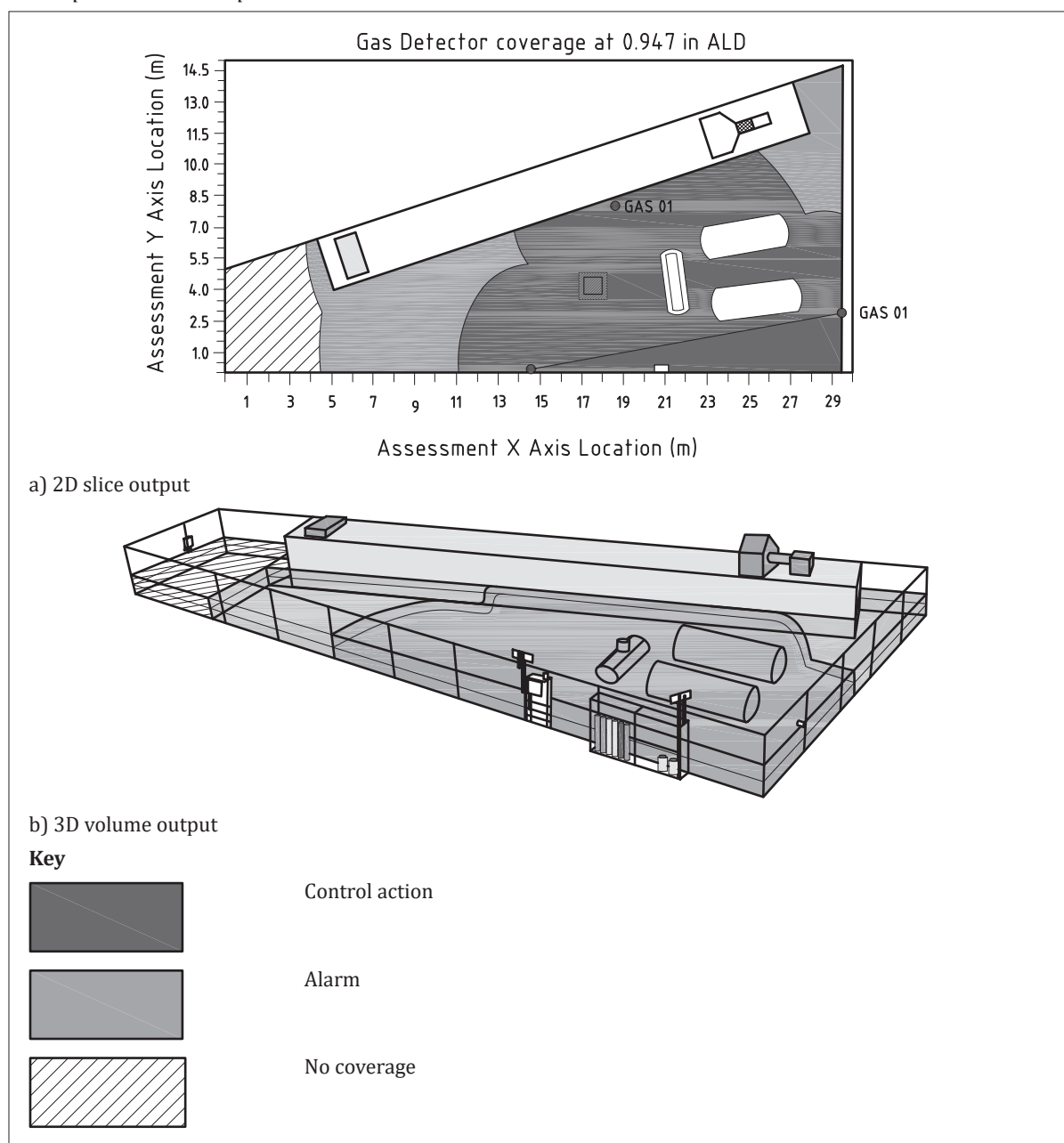
- dependent on specifics of design; if the design changes, analysis might have to be rerun, so it is of more value for mature designs, and less appropriate at the concept stage of a project;
- it can be dependent on the number of scenarios considered; the designer should ensure that results are independent of the number of scenarios (see [7.5.6.2](#)). The result is based on the number of identified conditions and a range of assumptions (inputs). The number of inputs is open and variable and changing the input causes inconsistency of results due to the higher degree of inputs, creating greater uncertainty compared to alternative methods. The impact of inaccurate assumptions, for example, should be determined (e.g. frosting of the release source making release trajectory unreliable, wind gusts rather than a modelled steady prevailing wind direction etc.);
- more time-consuming than volumetric-based mapping;
- minor changes in the area (i.e. addition of scaffolding/ temporary habitat) can impact the gas detection design. The engineer should assess if the design is adequate or if a revalidation process is required. Each change determined to impact on the detection layout requires the engineer to undergo the assessment and revalidation process (including additional detection or relocating existing detection if required). Designs based solely on dispersion can be more susceptible to additional/ altered obstructions in the area. Update of assessments, revalidation and modification can be onerous dependent upon the risk philosophy; and
- detectors can be recommended in areas where no explosion hazard exists when based solely on gas migration behaviour (see [7.5.7.6](#)).

7.5.6 Assessing adequacy

7.5.6.1 Volumetric-based mapping

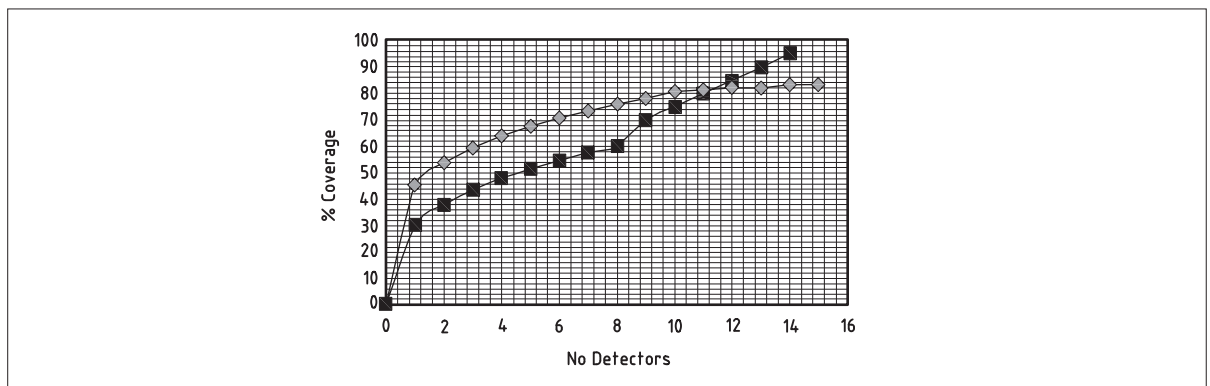
In volumetric-based mapping, coverage is quantified in terms of the percentage coverage of the volume being assessed. It is important to note that percentage coverage is not a probability of detection as, typically, volumetric-based mapping does not consider frequency and assumes that gas can be located at any point within the volume being analysed. Volumetric-based mapping provides an increased probability of detection specifically for “clouds of concern” regarding explosion overpressure. The probability of detection of incipient-type releases is not addressed. Inherent within the philosophy of volumetric-based detection is the notion that if the idealized cloud which is deemed to create an explosion overpressure exists, it needs to contact a device in that volume.

The percentage coverage for each volume should be reviewed to determine if the coverage is acceptable. Any predetermined target for coverage should be a baseline target (a rough preliminary target) as, in isolation, it is not a suitable guide against which to measure adequacy of coverage as it does not provide a robust demonstration that risks have been reduced to ALARP, e.g. if the target is 90% and a coverage of 89.9% is achieved it is not clear if this is sufficient. To provide a more thorough demonstration that risks are ALARP, the benefit of additional detectors should be reviewed for each volume being studied. This can be achieved by plotting the results as shown in [Figure 8](#), visualizing the results of the volumetric-based mapping as 2D or 3D outputs.

Figure 8 — Example detection output

[Figure 9](#) shows two example configurations against % coverage; the initially steeper gradient of the darker shaded area (with diminishing returns around eight devices), maintains dominance over the lighter shaded area until 11 detectors are added, for example.

When assessing the contribution of individual detectors and coverage of the volume, metrics should not be reviewed in isolation as the nature of the area being covered should also be reviewed, i.e. a given detector might be in an area with only limited congestion/confinement. Therefore, the benefit of additional detectors should be reviewed in parallel with 2D or 3D results (see [Figure 8](#)) to understand potential gaps in coverage, or areas where the system might be over-engineered. This approach ensures the resultant design is suitably optimized by reviewing areas of suitable coverage, blind spots where gas clouds might remain undetected, and how each detector contributes to coverage for both 100N and 200N for both Hi and Lo alarm, where necessary. Reviewing detector contributions allows the designer to analyse when the coverage is subject to diminishing returns for the addition of more detectors.

Figure 9 — *Detector contributions analysis*

The chessboard analogy in [7.3.6](#) is also applicable in gas detection mapping.

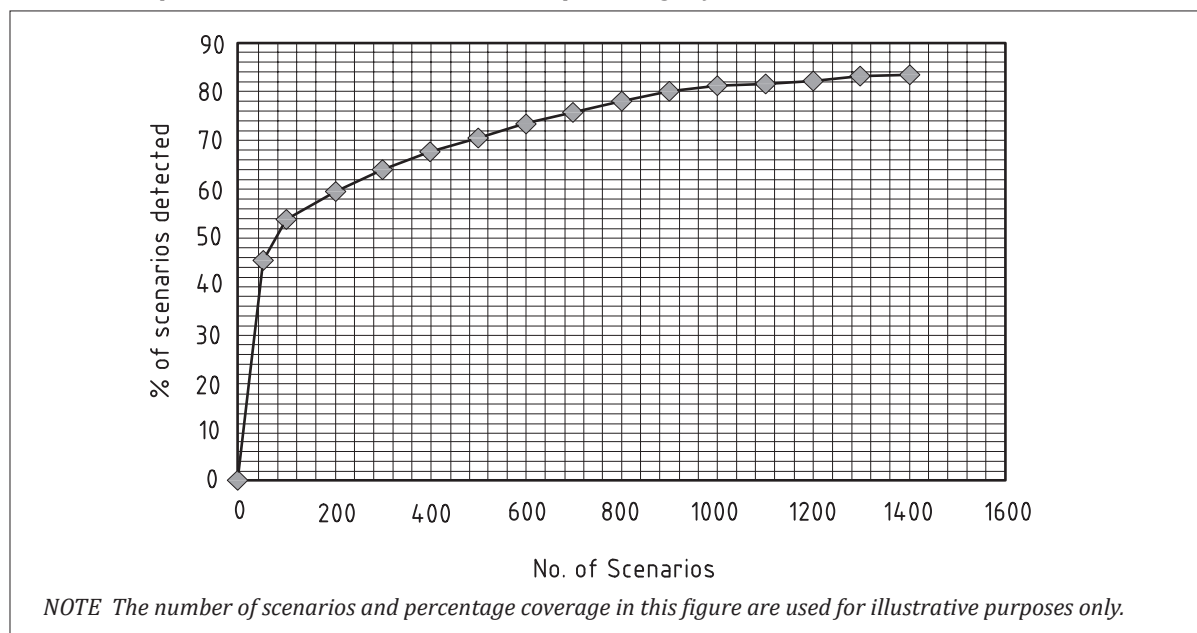
7.5.6.2 Scenario-based mapping

If conducting scenario-based mapping, the adequacy is assessed against the percentage of releases that are detected or the frequency (per year) of releases that go undetected. When presenting these metrics, scenarios that give cloud sizes smaller than the target gas cloud size (i.e. have zero or low escalation potential) should not be included.

The frequency of releases that go undetected should be used in preference to percentage of scenarios detected. This is a more appropriate metric as assuming that all scenarios have equal probability of occurrence is an oversimplification. Furthermore, presenting the results in terms of the frequency of releases that go undetected allows acceptability to be linked to individual or societal risk criteria. Similar to volumetric-based mapping, it is important to not just consider a single value for the metric being used to assess adequacy as it does not present a robust demonstration that risks are ALARP. The contribution of additional detectors should be understood (see [Figure 9](#)).

When assessing adequacy using the percentage of scenarios undetected or the frequency of undetected releases, the releases that go undetected could potentially be low-frequency, high-consequence scenarios, e.g. platform collapse leading to multiple fatalities. As part of the philosophy for the gas detection system, it should be documented if this is acceptable.

The advantage of scenario-based mapping, over volumetric-based mapping, is that representative releases are simulated and the ability of the system to detect these releases is assessed directly. Despite this, it is imperative that a demonstration can be made to show that the final detector layout is independent of the scenarios modelled, i.e. if additional scenarios were modelled the layout would not change. An infinite number of scenarios can occur, so it is necessary to discretize each parameter (e.g. choose wind speeds at X m/s intervals) ensuring a suitable range is modelled for each, e.g. if only two directions were considered, this could greatly affect the proposed detector layout and would not be appropriate. Justification should be provided for the scenarios selected and it should be demonstrated that results will not change if more scenarios are considered. [Figure 10](#) shows way of demonstrating scenario independence, i.e. as more scenarios are modelled the percentage coverage reaches a plateau.

Figure 10 — Relationship between scenarios considered and percentage of scenarios detected

The number of scenarios required vary depending on the facility/installation in question. Factors that would lead to more scenarios requiring analysis include:

- a) number of release sources;
- b) variance in process conditions, i.e. a greater range pressure, temperatures, fluid compositions requiring more scenarios to be analysed;
- c) variation of meteorological conditions, i.e. facilities with greater seasonal variance in temperature or wind speeds/directions, require more scenarios to be analysed;
- d) potential for wind “gusts”;
- e) potential for the release to create freezing conditions at the orifice, resulting in changes to the release trajectory; and
- f) equipment size, i.e. if pipework is small bore the range of hole sizes to consider is lower.

Although scenario-based mapping and dispersion modelling considers representative releases, simplifications are still made, and models are not validated for all conditions. The following should be taken into account for the facility/installation being designed:

- 1) metrological conditions, e.g. wind speed, direction and temperature, are assumed to be constant for a given scenario; and
- 2) many models are not validated for low wind speeds.

However, analysis such as that presented in [Figure 10](#) cannot be compared to analysis of a volumetric layout as in [Figure 9](#), i.e. analysing the number of scenarios detected when detection is designed on a volumetric basis. To compare the percentage of scenarios detected with the percentage of the volume covered is misleading as the approaches are fundamentally opposed. A design which performs strongly against one might perform poorly against the other.

7.5.7 General considerations

7.5.7.1 Dispersion-based scenario mapping

Detectors should not be placed solely on probabilistic dispersion scenarios. The consequence of a release igniting and subsequent overpressure should also be determined, i.e. the target gas cloud size

should be defined (see [7.5.2](#)). If detectors are placed solely on the results of probabilistic dispersion analysis there is the potential to skew the detector coverage towards “small” releases as these are of high likelihood but do not lead to adverse consequences.

7.5.7.2 Change management

Change management during design and in facility operation can be more challenging and costly when applying scenario-based methods, compared to volumetric-based modelling. Scenario-based mapping is more specific to the installation in question as detectors are placed at locations where gas is likely to accumulate, e.g. a more rigorous review of the gas detector locations might be required if a temporary habitat is to be installed in an offshore module as this could change the basis upon which detectors have been placed, potentially requiring a re-analysis. If the detectors had been placed based on volumetric mapping, using a target gas cloud (TGC) approach, the layout is likely to be less sensitive to changes and re-analysis is not always required.

7.5.7.3 Modelling of open-path detectors

The concentration across the beam length should be taken into account with regard to open-path detectors in volumetric or scenario-based detector mapping (beam path absorption). Open-path detectors do not alarm when gas simply touches the beam. The concentration should be integrated across the beam length and presented in LEL/m. If a cloud is simplified to spheres of uniform concentration, the portion of the cloud that intersects the beam should be calculated and converted to an LEL/m value. If a cloud is calculated based on dispersion modelling results, the concentration varies and this varying concentration should be integrated across the beam length to calculate the LEL/m value.

7.5.7.4 Practical considerations

Gas detectors should be placed as appropriate for their environment. The following factors should be taken into account when defining the initial layout, rather than once mapping has been completed, to avoid analysis having to be run multiple times:

- a) low-level devices might be subjected to splash/dirt/mud ingress;
- b) open-path devices should not be placed where obstructions can cause a beam block;
- c) ease of maintenance ([Clause 11](#)); and
- d) the need to avoid exposure to environmental stimuli where the device can be prone to false alarm or fault.

7.5.7.5 Detector response time

Response time of a gas detector can refer to the response of the device to respond to gas when exposed, or the response time of the gas detection system from release:

time taken for gas to reach the detector + time taken for detector to respond + time taken for control action to be implemented.

Response time considerations vary dependent on several factors, including but not limited to:

- a) the method applied (prescriptive, volumetric, scenario);
- b) the approach to risk;
- c) the safety actions implemented on the facility;
- d) the principle of gas detection, i.e. optical, pellistor (catalytic), electrochemical sensor, etc.; and
- e) the accessories connected to the gas detector.

Volumetric approaches typically focus on the response time of the gas detection devices (i.e. ensure the target cloud is detected and the devices installed are of a sufficiently fast response), whereas scenario-based method approaches can calculate the response time of the system from release of the gas to detection for the finite number of scenarios analysed.

7.5.7.6 Perimeter/boundary monitoring

Where the risk to or from a facility is the migration of gas to an adjacent area (whether part of the same facility or otherwise), the migration of gas into the facility, or where migration of gas would require an alarm to be raised, perimeter gas detection can be applied. Perimeter detection can be applied either through prescriptive, volumetric-based or scenario-based methods.

Perimeter monitoring is typically achieved using open-path gas detectors to monitor large (open) spaces for the presence of gas migrating from or to a facility or area. If the perimeter is contained or short, e.g. openings between buildings, the use of point gas detectors is possible.

When placing perimeter detection, factors to take into account include the:

- a) nature of the areas to which the gas is migrating:
 - 1) non-hazardous buildings;
 - 2) adjacent process areas;
 - 3) whether the threatened buildings are pressurized or otherwise; and
 - 4) public spaces, etc.;
- b) prevailing wind direction;
- c) nature of the fluid being released (see [7.4](#)); and
- d) practical considerations of placing open-path gas detectors (mounting elevation, avoiding obstructions, etc.).

7.5.8 2D versus 3D mapping representation

Where modelling is implemented, the resulting assessments can be presented in either a 2D contour or a 3D volume. A section can also be taken through the results of a 3D assessment to present results at a given elevation in 2D.

Where a 3D model is available and computational mapping is to be performed, a 3D coverage analysis is preferable.

Where a facility does not have a 3D model, the following two approaches can be taken:

- a) a 2D assessment (which should still account for blockages in the three-dimensional space); and
- b) a hand-built 3D model.

NOTE Simplified 3D modelling (hand-built) is a recognized and valid approach. However, care needs to be taken to factor in the loss of detail, such as cable trays and small-bore pipework. Misleading coverage factors can be produced requiring site-specific details (survey/photographs) to be examined.

7.5.9 Other methods

7.5.9.1 Smoke testing

For smaller and enclosed spaces where air movement is under some form of motive force, the use of practical testing is possible. Practical tests for normal and abnormal operations should be carried out.

The use of artificial smoke is a frequent occurrence, with the movement of smoke indicating air currents, dead spaces and areas of accumulation. This results in an understanding of air flow pattern

which can be used to inform device location. However, this technique is flawed if the following parameters are not taken into account:

- a) artificial smoke might not simulate the density of the target gas;
- b) artificial smoke might have a different temperature at the source of generation than that of the temperature of gas at a release point; and
- c) artificial smoke might have a different motive force.

NOTE Some forms of artificial smoke are very acidic and could cause damage to plant and equipment.

7.5.9.2 Gas testing

When gas release scenarios are known, the simulation of such releases using either the target gas or a less dangerous surrogate gas is possible. However, such tests need to ensure additional safety precautions are undertaken.

7.6 Toxic gas detection

7.6.1 Set points and voting

7.6.1.1 Point detectors

Lower set points generally result in a faster response and greater protection to personnel.

Personal detectors (those worn in the breathing zone) tend to measure time weighted average (TWA) concentration and guard against health effects. Fixed detectors should measure instantaneous concentrations and typically have higher set points than personal detectors as they initiate executive action.

To define the set points, the occupational exposure levels for the material in question should be reviewed. Many different metrics are used to measure the impact of toxic materials on personnel.

NOTE Information on workplace exposure limits (WEL) can be found in the HSE document EH40 [10]. This defines a long-term exposure limit [LTEL (8 h)] and a short-term exposure limit [STEL (15 min)]. These are based on TWA values rather than instantaneous values but are recommended as a starting point for setting low- and high-level set points for toxic gas detectors. Where LTEL and STEL limits are not available, information from other countries could be used. Where only STEL data are provided the low set point can be set based on 33% of the STEL (typical relationship between STEL and LTEL).

Defining the set points at the LTEL and STEL can be impractical due to the following reasons:

- a) the technology might not be capable of reliably alarming at low concentrations;
- b) spurious trips;
- c) the detector can be triggered during normal operation, e.g. filling/unloading; and
- d) background concentration of the target gas.

Therefore, the practicalities of setting the low and high levels at the STEL and LTEL should be reviewed and it might be more appropriate to use immediately dangerous to life or health (IDLH) values. To guard against spurious trips due to short-lived peak concentrations, a time delay could be incorporated into the alarm (e.g. if the set point has to be maintained for X seconds, the time would be dependent on the particular operation) or voting could be employed, e.g. 2ooN detectors required to alarm.

The set points for toxic gas detectors should also be reviewed against the criteria used to justify the number of detectors for a particular area or operation (see 7.6.2), e.g. areas where egress is difficult and/or takes an especially long time might require lower set points than those where egress is unimpeded.

The set points should be monitored over time to ensure they are appropriate to the operation. If detectors are continuously alarming there is the potential that they will be inhibited by operators and hence not provide the intended function.

Unlike set points for flammable detectors, toxic set points could change as more research and data are collected on the physiological effects of different materials. Therefore, toxic set points should be reviewed periodically to meet current guidance.

7.6.1.2 Open-path detectors

The set points for open-path detectors are measured in ppm/metre and hence do not measure the concentration directly. The concentration for detectors should be based on practical experience, or modelling can be used to calculate the potential size and concentration of clouds to inform the set point to be used.

7.6.2 Graded volumes and performance standard

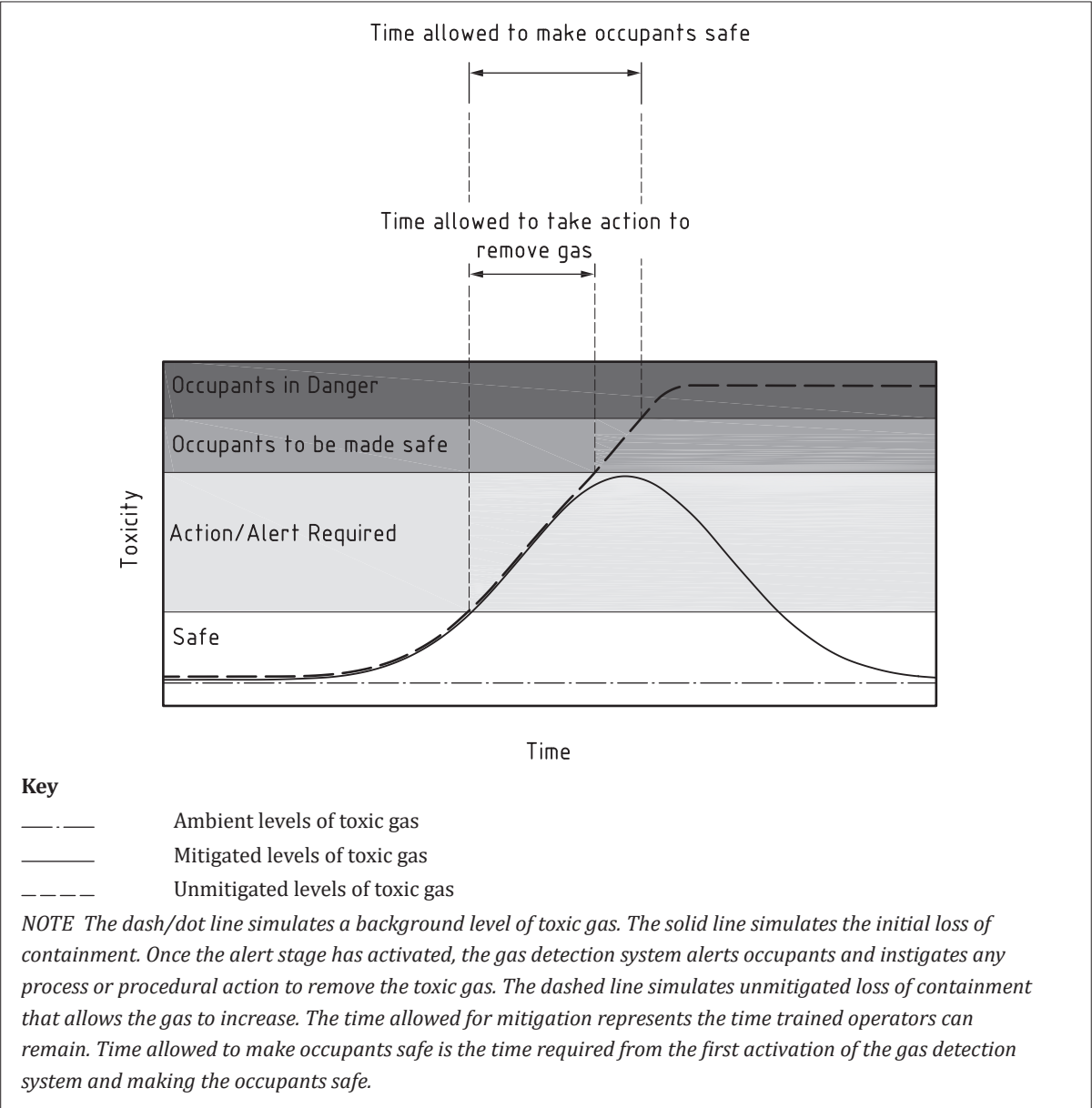
Volumetric and scenario-based design of fixed gas detector location takes into account multiple factors, defines the area into graded volumes and sets detection/performance targets. The following factors should be taken into account in setting a grade for the volume that requires detection.

Once containment is lost, the fixed gas detection system should provide enough time for operational procedures to kick in and the occupants to safely escape from the area.

[Figure 11](#) provides an example of a graphical representation of how different factors affect the time required to escape from the area with the loss of containment. The horizontal bands represent the concentrations of gas (e.g. ppm), where:

- a) safe: the levels of gas are permissible for occupants to operate;
- b) action/alert: all occupants to be alerted to the danger, the gas detection system should activate any operational/emergency shut-down procedures to mitigate or reduce the toxic gas; and occupants who are not trained with respect to the hazards of the toxic gas in question should begin taking steps to make themselves safe;
- c) occupants made safe: all occupants should be taking steps (e.g. evacuation) to make themselves safe; and
- d) occupants in danger: any occupants in the area should be evacuated or made safe by other means (e.g. breathing apparatus).

Figure 11 — Response time with a steady release of toxic gas



[Table 1](#) provides guidance on determining how much time is required to make occupants safe, e.g. a congested area might take more time to evacuate. The time allowed to evacuate can be increased by lowering the low threshold of the action/alert levels.

Table 1 — *Factors in risk and mitigation*

Occupancy	The type of occupant and level of knowledge impact the level of risk. Typically, industrial applications have levels of control of occupants and training.	
	Active occupants	Occupants who are aware of the risk and are trained how to respond in the event of loss of containment.
	Passive occupants	Occupants who are not aware of the risk or trained on how to respond in the event of loss of containment.
	Number of occupants	The number of occupants (passive or active) determines the amount of time required to evacuate to a place of relative safety. Areas not regularly occupied should be taken into account.
Means of escape	Calculating the travel distance from an area of risk requires an understanding of the physical restrictions of the area and the toxicity/availability of the gas. For grading, the graded volumes should be identified (for example “Open is within X Metres of the Exit”).	
	Open	Near to a place of relative safety or fresh air.
	Access	Two routes to a place of relative safety that can be reached without any obstruction or trip hazards. Or: A single shorter escape route to a place of relative safety that can be reached without any obstruction or trip hazards.
	Remote	Beyond the limits of the access-graded volume, or on ladders or obstruction on the floor.
Purpose of area or zone	How the area is used impacts on the level of the risk and the grading. Where escape routes do not exist, their use should be determined.	
	Operational	An operational area that has X number of occupants as the typical occupation.
	Escape route	Designated escape routes from the area of risk.
	Access routes	An area that is used to pass through the area of risk (typically pedestrian).
	Neighbouring risk	An area outside the area of risk that is near to the risk and could have untrained (passive) occupants.
Nature/toxicity of the gas	The risk gas should be identified with any relevant documentation (safety data sheet), including a document clarifying the effects of toxicity that can be anticipated in the event of LOC. This should include: • the PPM or percentages that cause harm (or toxicity) to the occupants; • the effects of inhalation; and • whether the gas can be seen or smelled (colourless and odourless gas does not naturally notify the occupants of the hazard). Density – while some gasses can be heavier or lighter than air, variances in temperature, humidity, ventilation, carrier gas, etc., could affect the density of the gas or the gas might “settle”. Typically, the gas detectors should be placed at “normal occupant breathing level”. <i>NOTE: The PPM or percentages regarding toxicity are set against average capacity and do not account for occupants who might suffer from respiratory difficulties.</i> The equipment or process should also be clearly labelled in the reports and accompanying drawings.	

Table 1 (continued)

Supply pipe Integrity	The risk assessment needs to identify whether the supply is part of the risk. This can be mitigated by welded joints and regular testing. If the supply pipework is not tested a strategy should be identified to manage the risk.	
Inventory size	It is important to consider the supply of toxic gases and the volume of gas that even a small container of gas might produce, e.g. a relatively small cloud of gas at 100% of concentration can dissipate to an exponentially larger cloud with a harmful level (ppm) of gas.	
	Set points	The set points of the gas detectors should reflect the risk and alert the occupants to the presence of toxic gas with a significant margin before harm.
	Response time	The response time for the gas detectors should be incorporated into the overall response time of the occupants, e.g. detectors with a slow response time need to be closer together to increase the time to evacuate. Gas detector response time is measured as T90 (the time taken to reach 90% of the applied concentration is used). Different manufacturers use different test methods to determine the T90 time. Accordingly, the selection of devices selection should be confirmed in the report. An audit of an existing system should incorporate the response time.
	Executive action	Once gas is discovered, if the control panel takes any executive action (shut off the source of the gas) to prevent more gas entering the area it can increase the time allowed to evacuate.
Facility geometry	The physical layout of the site impacts how the gas behaves in the event of LOC.	
	Access	By restricting access to the area of risk the site can prevent uncontrolled access and give rise to procedures to better protect the occupants.
	Containment	In the event of loss of containment (LOC) the gas might migrate or settle in a specific area. In the event of a risk of the build-up of gas, a volumetric-based approach to the area provides a demonstrable solution to the performance targets.
	Internal/external	Internally the site can affect more control over the environment. If the building area is secure the perimeter of the risk can be protected. The external environment can require the use of bunds to contain any toxic liquids or gas. Climate (wind speed, direction, temperature, hours of sunlight, etc.) cannot be predicted and, consequently, the use of CFD for external applications cannot provide sufficient information to make a verifiable safety case.
	HVAC	See forced/natural ventilation.
Mitigation	Procedures and systems can be in place to provide additional safeguards. Procedures (occupant interaction) need to be monitored and verified as part of the health and safety of the site. Systems need to be monitored and failure of a system should activate executive action to protect the occupants.	

Table 1 (continued)

Training	It is important to identify what training the occupants undertake on action to take in the event of the LOC and the gas detection system (GDS) activating. Specifically: • basic level training should be undertaken for every occupant of the site on action to take in the event of the GDS activating (on induction and refreshed annually); • operative level training should be undertaken on action to take (i.e. interrogate panel, investigate, etc.) in the event of the GDS activating (on induction and refreshed annually); and • site first responders training – some sites have a team of people who are required to take action or rescue occupants.
Forced/natural ventilation	HVAC is often be sited as mitigation to remove or disperse gas. In such cases, the HVAC is required to be monitored or to be part of a fail-safe system that prevents the availability of the risk gas. Information should be provided that indicates that the flow is being monitored as sufficient (as installed in a safety system).
Pressurized vestibules	The risk of gas can be contained in a room by pressurizing a lobby or vestibule to create an airlock. This can prevent gas from spreading beyond the room of risk; however, protection or detection is still required prior to occupants entering the area of risk.
Vapour sealed doors	Some areas have doors that prevent gasses from migrating from one area to another (typically fitted as fire doors). These need to be fitted correctly with no space between the cold smoke seal and the frame or opposite door.
Fire stopping	When pipes and cables penetrate a wall or barrier, fire stopping can be used to prevent the migration of gas where a sufficient seal is provided. Fire compartmentation repairs are to be undertaken by a competent person who understands the importance of correct installation of the product in the context of fire safety.
Water curtains	Water curtains can be used to prevent the spread of gas but should only be used if there is no reaction to the gas. The control for the water curtain should be automatically activated via the gas control system.
Operational safety	Sites could mitigate the risk of toxic gas by procedures and permits to control occupancy and process.
Emergency response	The site needs to develop and implement action to take in the event of LOC and the gas detection system activating.
Personal gas monitoring	Dependent upon the risk, facility and factors listed in this table, portable gas detectors can be applied as the only safeguard for detection of toxic gases, as well as supplementary to the fixed toxic gas detection system.

There are multiple harmful gasses which can harm occupants. Moreover, the risk gas could be altered by a carrier or atmospheric gas which changes its nature and risk.

Comprehensive empirical data are not available for all toxic gases and the environments of application are not repeatable for reasonable study. Consequently, toxic gas detection location design is likely to be targeted to achieve specific goals for the occupants to take action and/or get to a place of safety. Such goals can be presented using either prescriptive, volumetric-based methods or scenario-based method targets. [Table 1](#) provides a guide for the gas detection location designer to determine the performance targets for the gas detection system.

Toxic gas detection can only detect loss of containment; it cannot protect an occupant who is geographically closer to the loss of containment than the fixed gas detector. Consequently, the fixed gas detector should have a quick response time and detection setting that give early warning.

The performance targets reflect the risk and mitigation as outlined in [Table 1](#), e.g. a single application of a toxic gas next to a walkway could have various separate performance targets:

- a) scenario – to detect gas near the pumps and tanks operating at pressure;
- b) volumetric – a stricter performance target to reflect that a trained operative is in the area during normal working operations;
- c) escape/communications route – a volumetric performance target to protect occupants passing through or near to the risk; and
- d) operational procedure – to protect occupants who might be working in an area that is not normally occupied and therefore does not require permanent detection.

The performance targets are to be drafted and agreed prior to the design of the gas detector locations.

NOTE [Annex D](#) provides further guidance on risk and mitigation identification.

7.6.3 Prescriptive

Prescriptive toxic gas detector mapping refers to the use of qualitative evaluation and judgement-based placement in the locating of gas detectors. Prescriptive toxic gas detection should have supporting documentation that explains the design practice in accordance with [Table 1](#). Examples of prescriptive toxic methods include:

- a) proven design that is reused for similar applications; in many instances this would not require a full modelling activity;
- b) ruleset-based approach, e.g. all escape routes will have a gas detector every X metre; and
- c) evaluation of the factors in [7.4](#) and detectors placed at locations where gas is likely to accumulate; this is most appropriate for applications that do not contain many potential leak sources and releases are likely to behave in a predictable manner, e.g. no areas of recirculation.

Software modelling tools and metrics (e.g. percentage coverage) are not required when using the prescriptive approach and compliance can often be checked using a checkbox-type approach against the ruleset that has been employed.

7.6.4 Volumetric-based method

Volumetric-based mapping assesses the capability of a system to detect a toxic cloud of a certain size and presents the results in terms of the coverage of a given volume. The size of toxic cloud size can be based on the cloud size and gas detection risk assessment (see [Clause 4](#)).

Volumetric-based toxic mapping takes account of where personnel requiring protection are normally present, without falling under a specific permit to work system and associated safe working practices (e.g. restricted access, breathing apparatus, portable detection).

Focusing on where personnel are present, volumetric-based mapping therefore typically concentrates on regularly-used walkways and escape routes through the process area, and entrances/exits. Control

measures to restrict occupancy of areas not normally occupied (outside the assessed volumes) can include permit to work, removing the risk of gas, portable protection, etc.

Volumetric-based mapping is typically performed using software packages that verify the coverage achieved.

The following advantages and disadvantages are generated based on common applications where the method is selected. The critical factor is use of the most appropriate methodology to the specific application, e.g. the exclusion of subjective factors such as wind direction or leak orientation, is a strength in the volumetric-based approach, but if the application is a controlled environment where variables are reduced (i.e. an internal volume with few sources of release), the exclusion of those factors could be considered a disadvantage.

The advantages of volumetric-based mapping include:

- a) 3D model can be applied for visualization purposes;
- b) incorporates information on process, e.g. heat and mass balance data to determine constituents of a potential cloud (e.g. pressure, molecular weight and temperature) to set elevation/location of the graded volume;
- c) no detailed knowledge of where leaks originate is required;
- d) subjective features of analysis are reduced by comparison to scenario-based mapping, i.e. where leaks originate, likelihood of pipe rupture, etc.;
- e) relatively quick to design;
- f) consistent designs; and
- g) easily audited.

The disadvantages of volumetric-based mapping include:

- 1) results can be influenced by the size of volume assessed;
- 2) no clear way of determining where the graded volume should extend to; and
- 3) subjective features of analysis are increased by comparison to prescriptive, i.e. various cloud sizes, coverage adequacy interpretation, etc.

7.6.5 Scenario-based method

Scenario-based mapping requires information specific to the process and is analogous to the approach taken in a QRA, i.e. it takes account of both frequency and consequence to quantify risk. For this reason, scenario-based mapping is often carried out where QRA has been conducted previously. The QRA can be used to provide the frequency of individual scenarios. Dispersion modelling might have been completed as part of a QRA, but this might not be at the required resolution for scenario-based mapping so that additional dispersion modelling is required.

The following advantages and disadvantages are generated based on common applications where the method is selected. The critical factor is use of the most appropriate methodology to the specific application, e.g. the more thorough assessment of risks is a strength in the scenario-based approach, but if the application is a standard application (i.e. a tank storage facility), expending resources on a more thorough assessment of risks could be classed as a disadvantage.

The advantages of scenario-based mapping include:

- a) 3D model can be applied for visualization purposes;
- b) if dispersion analysis has been conducted this can be used directly to account for different set points of detectors as concentration in 3D space is calculated; and
- c) more thorough review of different release scenarios through consideration of specific release conditions, including wind direction, release orientation, gas release rate, etc. [see also disadvantage 3)].

The disadvantages of scenario-based mapping include:

- 1) it is dependent on specifics of design, so if the design changes significantly, the analysis has to be rerun; therefore it is of more value for more mature designs, and less appropriate at the concept stage of a project;
- 2) it can be dependent on the number of scenarios considered; the designer should ensure that results are independent of the number of scenarios (see [7.5.6.2](#));
- 3) the result is based on the number of identified conditions and a range of assumptions (inputs). The number of inputs is open and variable and changing the input causes inconsistency of results due to the higher degree of inputs creating greater uncertainty compared to alternative methods. The impact of inaccurate assumptions, for example, should be determined (e.g. frosting of the release source making release trajectory unreliable, wind gusts rather than steady prevailing wind direction etc.);
- 4) more time-consuming than volumetric-based mapping;
- 5) minor changes in the area (i.e. addition of scaffolding/ temporary habitat) can impact the toxic detection design. The engineer is required to assess if the design is adequate or if a revalidation process is required. Each change determined to impact on the detection layout requires the engineer to undergo the assessment and revalidation process (including additional detection or relocating existing detection, if required). Designs based solely on dispersion can be more susceptible to additional/ altered obstructions in the area. Update of assessments, revalidation and modification can be onerous dependent upon the risk philosophy; and
- 6) detectors can be recommended in areas where no personnel exist when based solely on gas migration behaviour.

NOTE See [7.5.5](#) for additional considerations when applying a scenario-based method approach, whether for flammable gas detection or toxic gas detection design.

7.6.6 Assessing adequacy

The design of the gas detection system should be assessed as part of the original design and reviewed regularly or whenever a significant change occurs in the process or occupancy.

Adequacy is heavily based on the performance requirements specified as a result of the considerations given in [Table 1](#).

7.6.7 General considerations

7.6.7.1 Response time

All gas detectors are supported by the manufacturer's technical data sheets which state response times. Response times are typically in seconds and refer to a T90 of a fixed measuring range or referenced to a response time to specific alarm levels when the gas detector has a non-linear characteristic.

A gas detector should not be selected during a mapping study based on response time only as there is no consistency in current test methods to determine response times of a gas detector, i.e. different test methods give different results.

If accessories are used in normal operation these should be included in the mapping study, e.g. effects on speed of response or gas detector availability.

7.6.7.2 Use of accessories

Irrespective of whether an accessory is supplied by the gas detection manufacturer or is custom-made for a specific application, an accessory should not compromise the mapping study results. Accessories can be categorized as follows:

- a) weather protection: sun/rain/snow shades guard against the general weather conditions and might or might not affect speed of response or deflect the amount of gas reaching the gas detector;
 - b) general protection: dust guards prevent or slow down the clogging of sintered disks or gas membranes which would otherwise greatly affect the gas detector's speed of response, but in many cases dust guards are not included in a gas detector's diagnostics routine;
 - c) insect protection: insect guards prevent small insects from obscuring the small optical paths within a point IR gas detector, potentially increasing the frequency of optical path fault conditions and decreasing the availability of the gas detector;
 - d) calibration adaptors: calibration accessories which are permanently installed, e.g. for ease of calibration for gas detectors in an inaccessible location, might affect the speed of response;
 - e) collection cones: for ambient areas where the gas to be detected is lighter than air, e.g. ammonia, a collecting cone can help to motivate the gas towards the gas detector; and
- NOTE If collecting cones are used in air currents, checks need to be employed so that the gas does not deflect across the gas detector which results in low measurement levels.*
- f) sampling points: for aspirated systems, the use of integral dust filters, ball floats, calibration valves, etc., might or might not affect the performance.

7.6.7.3 Recovery

The recovery of a gas detector after a deluge/high exposure to gas is important for the overall mapping study with respect to gas detector availability.

Certain gas detector technologies have a long recovery duration if over-exposed to gas and are therefore unable to detect reoccurring gas leaks until a full recovery has been realized.

Gas detectors which are over-exposed might not recover and need to be replaced.

7.6.7.4 Concentration

No gas release has a uniform concentration once dispersed into the ambient environment. At points closest to a release the concentration is much higher than monitoring points at a distance from a release.

High gas concentrations do not typically improve response times of the gas detector but can improve the time to alarm. Background or ambient gas concentrations might cause spurious trips if measuring ranges are too small.

7.6.7.5 Calibration

All gas detectors should be calibrated for the target gas they are intended to detect. For a calibration to be correct, the composition of the calibration gas should be known and validated with a gas composition certificate – this includes the balance gas composition, e.g. the make-up of synthetic air.

Many gas detectors have known cross-sensitivities which give either a positive or negative effect, e.g. ammonia electrochemical sensors are cross-sensitive to many amine gases, so known cross-sensitivities should be accounted for during the mapping study.

NOTE If surrogate, less dangerous gases are used for calibration, the effects of a surrogate calibration with respect to accuracy and response times might need to be determined during the mapping study.

7.6.7.6 Inferring toxic gas

In certain circumstances, dedicated fixed toxic gas detection is not required. If the toxic gas can be more reliably detected by other means (i.e. through detection of another gas within the stream), this can be used to infer the presence of toxic gas, e.g. if hydrogen sulfide (H_2S) is present in a stream of natural gas and infrared gas detection is already provided in the area (which detects the presence of the natural gas in the event of a release), the presence of H_2S can be inferred through the detection of natural gas.

Factors to consider to safely infer toxic gas releases through other means include, but are not limited to:

- a) the concentration of toxic gas in the stream. If the concentration of toxic gas is above the facility's acceptable exposure level, without annunciating the flammable gas detectors, the use of dedicated toxic gas detection should be assessed;

EXAMPLE

500 ppm H_2S in a stream of methane. Flammable gas detectors set at 20% LEL alarm threshold. Assuming methane LEL is 5% by volume in air (noting there can be a discrepancy in the value), with a set point of 20% LEL, 1% gas in the air is being detected. This can be used to infer detection of 1% of the H_2S content, i.e. 5 ppm H_2S . If this is an acceptable concentration to alarm, this can be considered acceptable inferential gas detection, subject to the other factors in b) to d).

- b) the set point of the flammable gas detectors;
- c) the location of the flammable gas detectors (whether they are in the same location where both flammable and toxic gas presence is intended to be detected); and
- d) the facility philosophy with respect to toxic gas detection and protective measures.

7.6.7.7 Perimeter/boundary monitoring

Where the migration of a cloud of toxic gas is the hazard of concern, perimeter toxic gas detection can be used. If there is no risk to personnel within the area of toxic gas release, migration-based detection can be implemented to detect when this cloud migrates to an area where it presents a hazard.

Detection should be determined at the boundary of the area where the release might occur, at the location where the toxic gas might present a hazard, or both.

Typically, open-path toxic gas detection should be taken into account for these areas. These devices are more optimal at detecting clouds and covering a larger area while optimizing the number of detectors required.

NOTE An example of toxic gas mapping is given in [Annex C](#).

8 Detailed engineering

8.1 General

Typically, before detailed engineering begins, preliminary F&G layouts are issued. These layouts are required to be produced early in the project life cycle to enable purchase of long lead items such as F&G systems cabinets. F&G layouts are typically produced using engineering judgement and a more prescriptive approach. However, once the design has become fixed and stable, more detailed mapping work can be undertaken. Irrespective of the approach selected, the guidance provided in [8.2](#) to [8.4](#) is very important during detailed engineering.

8.2 Schedule constraints

8.2.1 General

Selection of the appropriate mapping technique is important when determining a project's execution schedule. Many projects are scheduled such that construction commences prior to full completion of the design. This is an issue particularly during early work scopes, such as civils and underground work. Furthermore, purchasing of F&G-related equipment, such as safety and marshalling cabinets, can occur early in the design process. This means that further changing of detector numbers or types can greatly impact the project. These factors are important when selecting an appropriate method for locating F&G detectors. Selecting a method which is time-consuming or requires a high level of design definition, such as scenario-based method mapping, might result in an overly negative impact on the project schedule. However, it is possible to use a two-stage approach whereby a conservative (high detector number) prescriptive-based method is used early in the design and then the design is optimized by using more detailed techniques (such as volumetric or scenario-based).

8.2.2 Completion of F&G detector layout design

Meeting the project schedule is the central challenge in completing a successful project. Instrumentation engineers typically require reasonable instrument input/output counts to size instrumentation rooms, procure cabinets and develop the F&G system architecture appropriately. This can lead to a design needing to determine how many detectors are required before any 3D model is even 50% complete. This can be achieved using a prescriptive approach in early design. Given the likely differing outcome between the prescriptive and modelling approaches, and further model development changes, the design should aim to err on the side of conservatism and add extra detectors to the total count. Even though change is never desirable during a project, deleting detectors is usually far easier to manage than adding new ones.

Schedule pressure might also require that the time taken to complete more complex mapping techniques has to be reduced. Examples of methods to reduce the time taken to complete mapping exercise include:

- a) preparing a bespoke 3D model file with all the hydrocarbon/toxic inventories highlighted and any virtual volumes deleted, which typically makes the geometric manipulation required by the consultant (or other third party) simpler;
- b) reducing the amount of CFD modelling to determine the hazardous gas cloud size by applying lessons from other mapping exercises in terms of the relative amounts of congestion and confinement; and
- c) producing a standard list of common assumptions.

8.2.3 Preservation of F&G devices

Typically for large projects (but also for other project types) the early procurement of F&G devices for scheduling purposes can result in the devices being put into storage for an extended duration. This can lead to a wide range of secondary issues, such as sensor element degradation or internal battery run-down leading to device functionality/set-point problems. During the detailed design phase, it is important to understand the procurement, shipping and handling process and develop suitable plans. These plans might include (but not be limited to) aspects such as staggered ordering of parts (e.g. the transmitter module ordered prior to the sensor element), careful preservation practices and careful unpacking and labelling. These plans should also apply to the ordering of spares.

8.3 Management of change

Once the F&G detector layout drawings are issued and approved for construction, ownership of the documentation needs to be determined. A competent F&G professional (See [Clause 12](#)) should be nominated for monitoring and controlling further changes to the documents. All proposed changes to the documents require prior approval from the nominated person, including any red line mark-ups. Sign-off of all future revisions of the documentation needs to include the nominated person's signature.

Some projects do not include on the 3D model every possible element of the final facility. Small-bore piping, insulation details, instrument tubing and minor support structures are typically left to the construction team to include. This might present additional risks to the F&G design, such as:

- a) optical type detectors might be presented with obstacles that can either restrict their effectiveness or cause them to not function at all;
- b) scenario-based method studies might not have included such obstacles with regard to additional leak sources or the migration of gas dispersion; and
- c) mounting location for other detectors can become unavailable, so other detectors might be repositioned without knowing the effect of such repositioning.

Accessibility might also be adversely affected for other types of devices and this should be taken into account during design.

All F&G detectors require a form of human intervention on occasion, e.g. access to carry out calibration and testing, cleaning of optical devices or realignment. Therefore, the physical location of any detector should be determined, including, where possible, mounting the detectors so that they can be easily accessed.

The first step in any management of change process should be to confirm that the responsible engineer for the detector layout is aware of any change that could affect the F&G detector layout drawing. Once aware, the engineer needs to assess whether the change is so significant that modification to layouts is required.

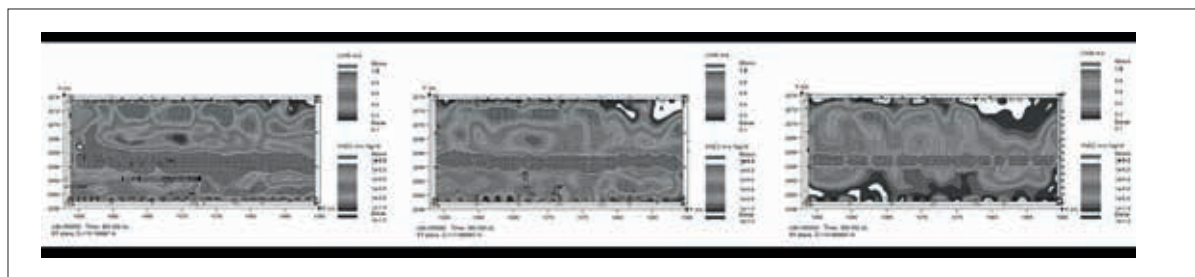
NOTE Changes can happen in the other direction, e.g. the F&G engineer might request changes to the facility to improve safety, requiring significant change and the mapping to potentially be preformed again.

If changes are required as a result of significant model changes it might not be practicable or possible to repeat mapping. However, by using gas concentration "heat maps" (commonly produced from dispersion mapping studies), an approximate assessment and change can be made using engineering judgement.

[Figure 12](#) shows an example of the level of ventilation within an enclosed facility. The warmer colours show high air velocities and the colder colours show low air velocity. A heat map (as shown in [Figure 12](#)) can be used to determine whether a change has occurred in a critical location and to

provide some understanding of the general physical processes that are dominant or occurring in a given location. In this way these maps can be an important tool in assessing and reacting to changes.

Figure 12 — *Example of ventilation analysis*



8.4 Human factors and ergonomics

Once F&G detectors are placed in their final positions they are to be located:

- a) where they can be easily accessed for inspection, testing and maintenance; and
- b) in a position which does not block escape routes, maintenance volumes and walkways.

Currently, no optimization algorithm can locate detectors in the most optimal position, while taking into account human factors and the issues surrounding adequacy (see [7.3.6](#), [7.5.6](#) and [7.6.6](#)).

Detailed guidelines on the human factor requirements for F&G detectors should be produced and issued, so that the consultants/F&G engineers understand how to fill the module with accessible detectors.

9 Installation and commissioning (verification)

9.1 Planning

Prior to construction, installation methods and commissioning procedures should be specified and include, as a minimum:

- a) installation activities, including the specific location, elevation for F&G detectors, orientation and field of view for flame detection;
- b) any special precautions required during installation (as recommended by the device suppliers);
- c) an approved schedule of persons, departments and organizations responsible for the installation activities;
- d) precautions to be taken when the installation is within a hazardous area;
- e) types of tests and checks to be performed on system start-up/commissioning, including checking all device specifications against the procurement list, and the final location, elevation, orientation and field of view for all F&G detectors;
- f) sensitivity settings for flame detectors, configuration and calibration of gas detectors and the time-delay settings of ultrasonic gas detectors;
- g) the PASS/FAIL criteria, including when to abort the tests, subject to a single or multiple test failure(s) or incorrect location, elevation, orientation or field of view of individual F&G detectors;
- h) checking for false alarm sources or sources of interference;
- i) ensuring that receiver /transmitters are not interfering with each other;
- j) procedures for the recording of test data/results and the hardware/software versions of the equipment installed;

- k) layout plans for the recording of the location, elevation, orientation and field of view of all F&G detectors;
- l) procedures for corrective actions, system modifications or changes to the location, elevation, orientation and field of view of any single or multiple F&G detectors;
- m) test gases to be used, supported by valid gas composition certificates;
- n) test torch or equivalent (manual walk/view exercise) to be used for checking the field of view for each individual flame detector; and
- o) any exclusions.

9.2 Execution

The installation should be conducted in accordance with the installation plan which gives details to support the location, elevation, orientation and field of view for all F&G detectors.

Commissioning should be conducted in accordance with the commissioning plan.

All installation and commissioning activities are to be documented and be fully transparent and relevant for future periodic inspection reviews.

Before installation and commissioning can be performed, all documentation is to be checked for:

- a) completeness; and
- b) correct revision, including cross references.

All documents are to be approved by a competent person.

If a device/system failure or location error is discovered during commissioning, the reason for the failure or the location error needs to be identified and documented. A decision needs to be taken as to whether to:

- 1) repair the failure or relocate a detector, including the adjustment of elevation, orientation or field of view and continue with the commissioning exercise; or
- 2) ignore, but document the failure or location error and complete the commissioning exercise; or
- 3) if the failure affects the total system or multiple detectors, abort the complete commissioning exercise, allowing the failure to be repaired or the location of detectors to be corrected, with a new commissioning date being arranged; or
- 4) partially complete the commissioning exercise and plan an additional final commissioning activity after corrective actions have been taken.

If during the commissioning exercise any modifications or changes are made to the system or the location of any gas or flame detector, these changes should be subject to a safety analysis to determine:

- i) the results of a new mapping exercise (if required);
- ii) the extent of impact on each safety function; and
- iii) the extent of re-test which should be defined and implemented.

All modifications are to be documented and approved by a competent person.

10 System validation

COMMENTARY ON CLAUSE 10

For the purposes of this British Standard, validation is defined as a process which confirms that the design intent of the fire and gas system has been met. This may include the mapping premises, design assumptions and checks that the implementation achieves desired coverage following commissioning.

This is different from “verification”, which in the context of F&G is a process to confirm that the design specification has been implemented correctly, and for the purposes of this document is met through commissioning and installation (see Clause 9).

10.1 Planning

Prior to commissioning, a system validation plan should be prepared and include, as a minimum:

- a) validation methods so that the installed F&G detection system delivers the coverage factor as specified in the mapping study;
- b) validation methods so that the installed F&G detection system operates correctly under:
 - 1) normal operation;
 - 2) abnormal operation (e.g. detector beam blocked by snow on transmitter/receiver optical window, obstructions being placed in the line of sight);
 - 3) special state condition;
 - 4) fault condition; and
 - 5) normal changes in use, occupancy and configuration of the monitored/process area;
- c) a method to include any system modification which might have been implemented during the factory acceptance test or system installation or commissioning;
- d) validation of the competencies of persons in attendance;
- e) procedures for the recording of test data/results and the hardware/software versions of the equipment under test; and
- f) layout plans for the recording of the location, elevation, orientation and field of view of all F&G detectors.

10.2 System validation test

The system validation test should be conducted in accordance with the system validation test plan.

Before the system validation test is performed, all documentation needs to be checked for:

- a) completeness; and
- b) correct revision, including cross references; and
- c) approval by a competent person.

If there is a failure during the system validation test, the reason for the failure needs to be identified and documented. A decision is to be taken as to whether to:

- 1) repair the failure and re-validate; or
- 2) ignore, but document the failure and complete the system validation; or
- 3) if the failure affects the total system, abort the complete validation program, allowing the failure to be repaired, with a new validation date being arranged; or
- 4) partially complete the validation and plan an additional partial system validation test.

If during the system validation test any modifications or changes to the system are made, these changes should be subject to a safety analysis to determine:

- i) the results of a new mapping exercise;
- ii) the extent of the impact on each safety function; and
- iii) the extent of re-test which should be defined and implemented.

Until all of the requirements detailed in the system validation test plan are validated, the F&G detection system is not operational.

11 Operation, maintenance and modification

11.1 Evergreening

It is important to check that the F&G mapping report remains relevant and that the F&G detectors continue to monitor the area/target identified during the F&G mapping review. Therefore, the F&G mapping report should be reviewed every five years or as and when changes to the plant/process layout are made.

11.2 Planning

The need for operation planning should be specified during the design phase of the F&G detection system and at regular intervals during the operational phase to ensure continued compliance.

Changes to any operational requirements or processes should take into account the guidance given in [Clause 10](#).

Planning should ensure, as a minimum, that:

- a) a procedure is in place so that only authorized personnel have direct access to F&G detector field devices;
- b) a system design criterion for normal operation is readily available;
- c) operational personnel have full knowledge of F&G detector objectives, operating principles and limitations and, if applicable, any relevant operational procedures per device;
- d) alarm shelving procedures include criteria and detection criticality factors of F&G detectors;
- e) how the number of demands placed on the F&G are recorded;
- f) if known, the number of missed detection scenarios are recorded;
- g) any misuse or abnormal operation of a flame or gas detector are recorded;
- h) applicable environmental data associated with each detector location are considered;
- i) the frequency of scheduled system maintenance activities is recorded;
- j) actions to be taken in an unplanned event are in place, e.g. if the F&G system fails during either operation, maintenance or modification, there is a back-up plan to deploy transportable gas detection zone monitors or fire watchers to observe the plant area for hazards; and
- k) regular reviews are carried out to determine whether any process changes have affected the F&G detector coverage factor.

11.3 F&G detection system – execution

The F&G detection system should be operated as detailed in the overall system operational and safety manual. As a minimum the manual should include:

- a) the gas detection risk assessment;
- b) the gas detection location/validation report;
- c) instructions on how to operate the panel; and
- d) any modifications to the system since the original design.

Individuals operating and responding to the F&G detection system should be competent and authorized to do so.

The F&G detection system should only be installed and operated within its design criteria.

11.4 Maintenance

11.4.1 Planning

The need for maintenance planning should be specified during the design phase of the F&G detection system and repeated during the operational phase, as necessary.

Changes to any maintenance activity should take into account the guidance given in [Clause 10](#).

The impact of peripheral and other maintenance activities on F&G detector coverage should be taken into account.

Planning should ensure, as a minimum, that:

- a) only authorized and competent personnel have direct access to F&G detector field devices;
- b) detailed maintenance activities and procedures are readily available and revision-controlled;
- c) the frequency of scheduled system maintenance activities is recorded;
- d) the frequency of scheduled system validation activities and reviews of coverage applicability is recorded;
- e) any abnormal maintenance schedules are authorized by competent personnel who have written authority to approve such activities;
- f) the maximum number of F&G detectors which are in override (inhibit) during any maintenance activities are known;
- g) additional operational measures to be taken during maintenance activities are known and used, e.g. temporary deployment of zone gas detectors;
- h) detailed maintenance records are kept, including device location and orientation, fitted accessories, device viewing angles, faults found, inconsistencies, corrective or repair actions taken, spare parts used, consumables used and any changes of system performance which might, in the future, affect the detector coverage factor;
- i) any identified environmental factors are identified which might affect the performance of a flame or gas detector; and
- j) detailed proof test results are available, including speed of response and recovery times.

11.4.2 Maintenance operation

Individuals maintaining the fixed gas detection system should be competent and authorized to do so.

The F&G detection system is to be maintained as detailed in the overall system maintenance plan. Only transmitters/detectors, spare parts and consumables listed in the safety manual or individual equipment maintenance manuals are to be used.

The F&G detection system is to be proof-tested as detailed in the system safety manual.

All activities are to be clearly documented.

11.5 Modification (management of change)

11.5.1 Planning

Modifications to any F&G detection system should be planned, reviewed and authorized prior to any modification being performed. The plan should demonstrate an acceptable level of safety during and after the modification.

Planning should ensure, as a minimum, that the following are in place:

- a) modification request documentation;
- b) impact analysis;
- c) validation methods so that any modification meets the existing or revised target coverage factors;
- d) detailed descriptions of the competencies of individuals who perform the modification;
- e) a documentation control process;
- f) continuation of the F&G detection safety function and device coverage during the modification process;
- g) associated hazardous area demands (explosion protection documentation);
- h) maintenance procedures and documentation;
- i) spare parts inventory;
- j) emergency plans if the modification is not performed on time or the modification cannot be completed, or an unavoidable event occurs, e.g. emergent work is revealed by the modification activities; and
- k) training of personnel after the modification is complete, including operational personnel, maintenance technicians and associated emergency response personnel.

11.5.2 Execution

A modification activity should not commence without proper authorization.

All modifications should be documented, verified and validated, and follow the modification plan, modification documents and modification instructions. The documentation should be revised and signed off by the person/department responsible for maintaining the document.

Any deviation from the modification plan should be authorized and, if necessary, a new impact analysis performed. If the impact analysis reveals unacceptable risks, the emergency plan should be followed.

Upon completion of the modification the entire part system or complete system should be validated to prove the safety function. An impact assessment is required to determine the extent of this validation. Modification is not complete until all training activities are complete.

12 Competence

Competence is:

- a) the ability to undertake responsibilities and to perform activities to a recognized standard on a regular basis; and
- b) a combination of practical and thinking skills, relevant experience and knowledge.

All persons, departments or organizations involved in F&G mapping should be able to demonstrate competence.

As a minimum, for the roles and responsibilities of individuals, the following is to be addressed when assessing competence:

- 1) knowledge of and education about the appropriate local, national and international standards applicable to F&G detection systems, including those standards set out in the bibliography;
- 2) appropriate knowledge of the legal and safety regulatory requirements pertinent to F&G detection;
- 3) engineering knowledge, education and relevant experience appropriate to:
 - i) the specific hazards and required risk analysis;
 - ii) the detection technologies of F&G detectors;
 - iii) the complexity and novelty of a F&G detection system design, which could be complex due to size and logic or relatively novel due to its design or application, requiring a higher degree of rigour with respect to competence; and
 - iv) the application or use of the F&G detection system;
- 4) understanding of the potential consequences of an event if a gas leak or open flame is not be detected; and
- 5) previous experience and its relevance to the specific duties performed to support and complete a F&G detection mapping study.

Annex A (informative)

Oxygen enrichment and deficiency

A.1 General

Typically, there are four main hazards or conditions when the concentration of oxygen needs to be measured:

- a) oxygen enrichment – hazard is increased risk of ignition or fire;
- b) oxygen deficiency – hazard is asphyxiation;
- c) condition monitoring – minimum oxygen content for explosion protection; and
- d) condition monitoring – guarding against oxidation.

The most frequent measurement of oxygen concentration is associated with confined space entry and working. For this application, the positioning of the gas detector, e.g. personal monitor, is within the breathing zone (carried on the person) or directly adjacent to the working space within the confined space environment. This application only requires a prescriptive safety procedure and is supported by the Safety, Health and Welfare at Work (Confined Spaces) Regulations 2001 [11]. Where forced ventilation is used for confined space working, any fixed oxygen gas detector is to be located in both the air intake and the air outlet as a minimum. A prescriptive safety procedure controls the location of an oxygen fixed gas detector.

Fixed gas detectors are not typically used to monitor confined spaces as the primary method of detection is by personal monitors. However, where fixed gas detectors are used to support personal monitors they are to be located to provide optimum coverage, e.g. in and around the work location.

Oxygen monitoring in the workplace typically occurs indoors where a prescriptive or volumetric-based method for detector placement is sufficient. Alternatively, basic air current testing using buoyant smoke tests can be useful.

A.2 Oxygen enrichment

Oxygen enrichment hazards are typically caused by stored oxygen leakages, e.g. medical oxygen, welding oxygen or chemical oxygen production. With oxygen levels greater than 20.9% v/v there is an increased potential of ignition or fire and, therefore, for concentrations above 23.5%, v/v special precautions need to be taken. All locations where there is a risk of oxygen enrichment are to be clearly identified and the safe handling of oxygen enforced. Oxygen leaks readily mix with ambient air, so the location of fixed oxygen gas detectors can be described in a prescriptive manner.

NOTE All % v/v figures are approximate and based upon standard temperature and pressure. Altitude greatly affects the partial pressure measurement of oxygen.

A.3 Oxygen deficiency

Oxygen deficiency hazards exist and can be displaced by other gases, consumed by oxidation (e.g. corrosion) or changes in ambient pressure. Oxygen levels below 19% v/v begin to alter the body's functions, with levels below 12% v/v causing an unconscious state with death at around 6% v/v. In the general working environment (confined spaces excluded), fixed gas detectors are to be installed within the breathing zone or at levels where it is known that a displacing gas causes an oxygen-deficient atmosphere at lower levels.

NOTE All % v/v figures are approximate and based upon standard temperature and pressure. Altitude greatly affects the partial pressure measurement of oxygen.

A.4 Explosion protection

The fire triangle includes an oxidizer, so removing the oxygen content of an environment, space or process decreases the chance of an ignition or fire. The “minimum oxygen concentration” (MOC) or “limiting oxygen concentration” (LOC) varies with different fuels. Therefore, an oxygen measuring range or alarm levels cannot be set until the fuel data are known. Environments where the removal of oxygen for explosion protection is practised support life as MOC/LOC levels are typically below 10% v/v. Explosion protection is typically triggered by the primary measurement of “fuel”, therefore oxygen monitoring is typically based on a deficiency basis as described in [A.3](#).

When the removal of oxygen for explosion protection is practised under the ATEX Directive [\[12\]](#) the oxygen measurement needs to be “performance approved” by a notified body in line with BS EN 50104.

A.5 Oxygen flushing

In many manufacturing processes there is the requirement to guard against product oxidization, so the process includes the flushing out of nominal air or oxygen content. Inert gases, e.g. nitrogen, are directly injected into the process or packaging cartons so that the oxygen concentration at this point trends towards 0% v/v. The monitoring of oxygen concentration at this location is not typically for personnel protection and is biased towards process monitoring and quality. The hazard associated with oxygen flushing is as described in [A.3](#).

A.6 Detection technology

Many different detection technologies for oxygen measurement are available and the correct detection technology for the application needs to be determined. When, for example, the hazard is oxygen deficiency due to the potential release of helium, the choice of detection technology needs to avoid capillary-based oxygen sensors due to helium fluxing of oxygen, e.g. for personnel protection it is imperative to choose partial pressure measurement of oxygen and not % vol.

A.7 Toxicity of inert gases

In oxygen-deficient applications which occur due to a displacement of oxygen by another gas, the toxicity of the displacing gas needs to be fully determined, e.g. the hazard of carbon dioxide displacing oxygen content needs to be based on carbon dioxide monitoring and not oxygen measurement due to the toxicity level of carbon dioxide being greater than the level of oxygen displacement.

Annex B (informative)

Flame detection – Example of volumetric-based mapping

COMMENTARY ON ANNEX B

This annex provides a worked example showing one possible representation of flame detection mapping.

B.1 General

It is assumed the area has been processed through a risk assessment (see [Clause 4](#) and [7.3.2](#)). Vessels V-001 and V-002 present a medium risk to the area (containing pressurized flammable liquid). Vessel V-003 presents a low risk as it is a storage vessel for flammable liquid at atmospheric pressure. [Table B.1](#) shows the performance targets applied to the area.

Table B.1 — Flame detection performance targets

Equipment tag	Grade	Graded volume extension from equipment	Fire size (RHO) alarm (100N)	Fire size (RHO) control action (200N)
		m	kW	kW
V-001	Medium risk (MR)	2	50	100
V-002	Medium risk (MR)	2	50	100
V-003	Low risk (LR)	2	250	500

[Figure B.1](#) represents the volume in digital form, where performance targets and detectors can be placed.

[Figure B.2](#) and [Figure B.3](#) represent the performance targets generated from the risk assessment.

[Figure B.4](#) represents the detector locations and [Figure B.5](#) and [Figure B.6](#) represent the assessment of coverage. A generic flame detector cone has been applied in the example. In reality, the specific flame detector being installed is to be modelled.

Figure B.1 — Volume to be mapped

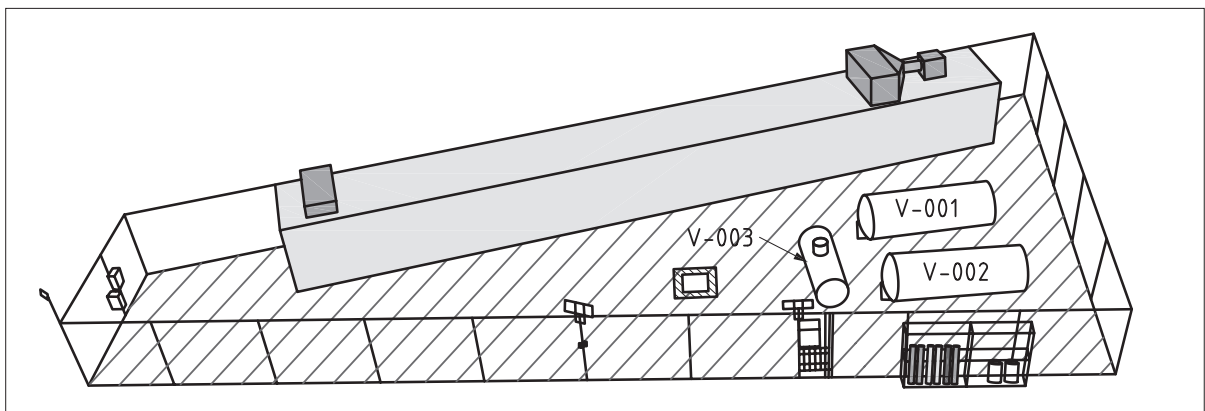


Figure B.2 — Grademap volume

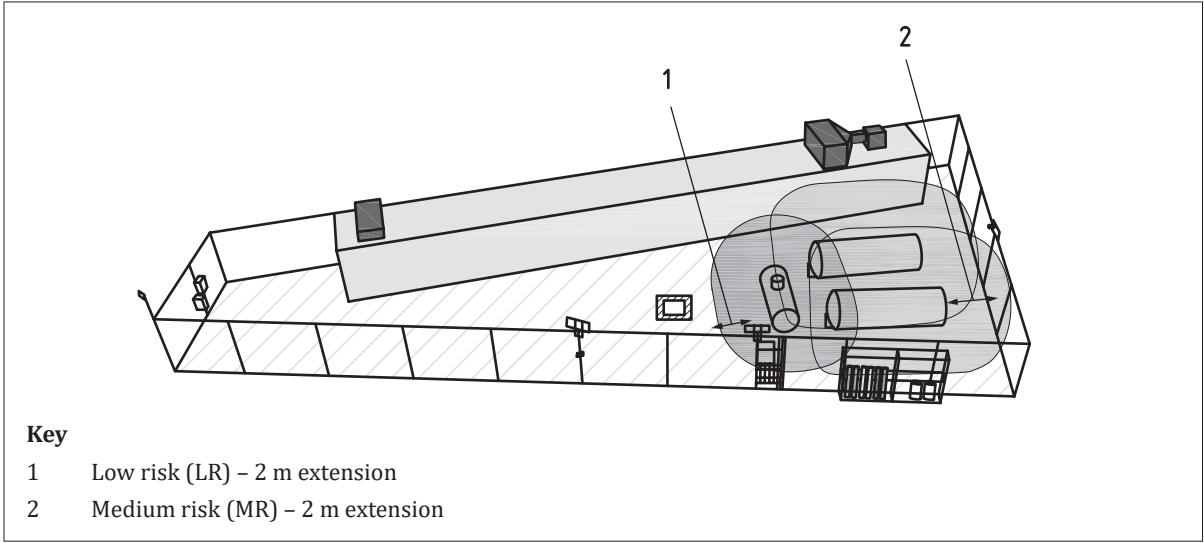


Figure B.3 — Risk grade map

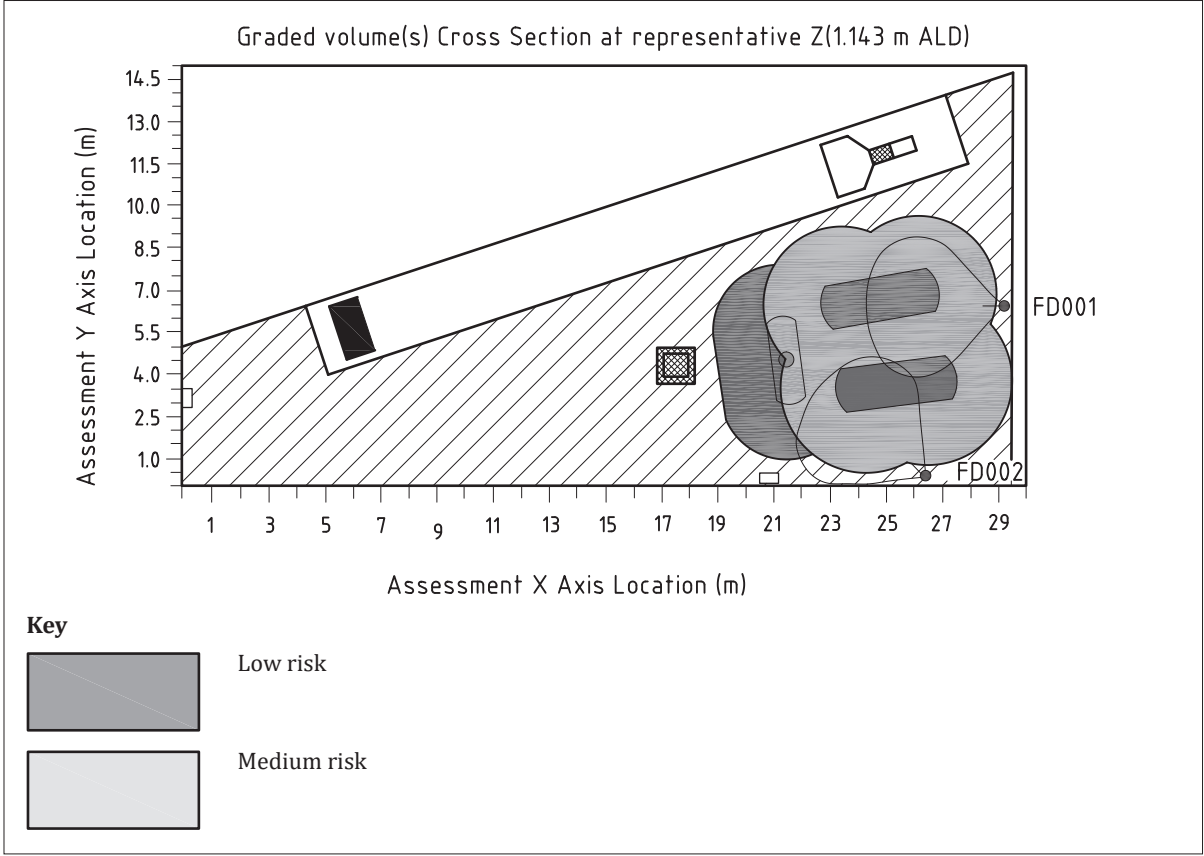


Figure B.4 — Existing devices coverage map

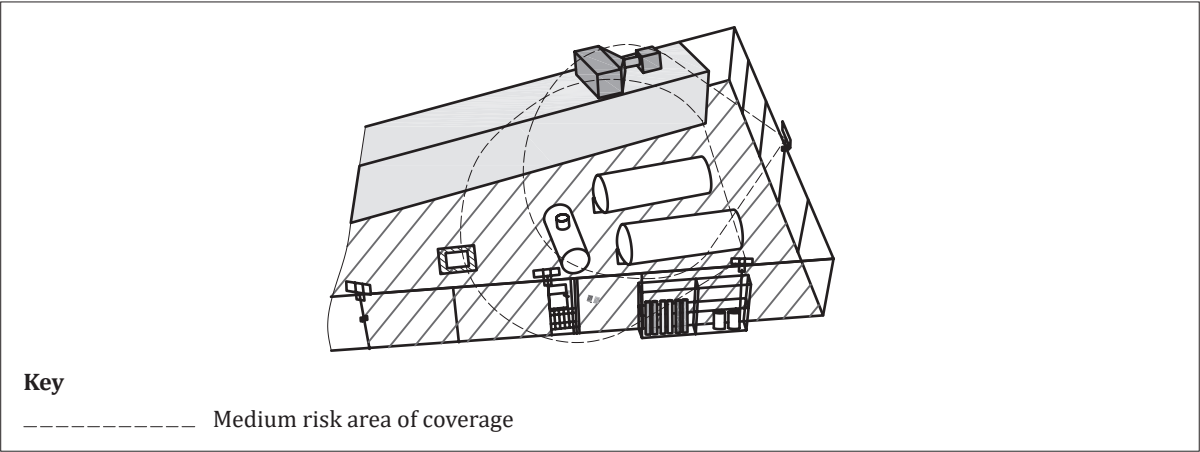


Figure B.5 — Existing flame detectors (coverage 1 of 2)

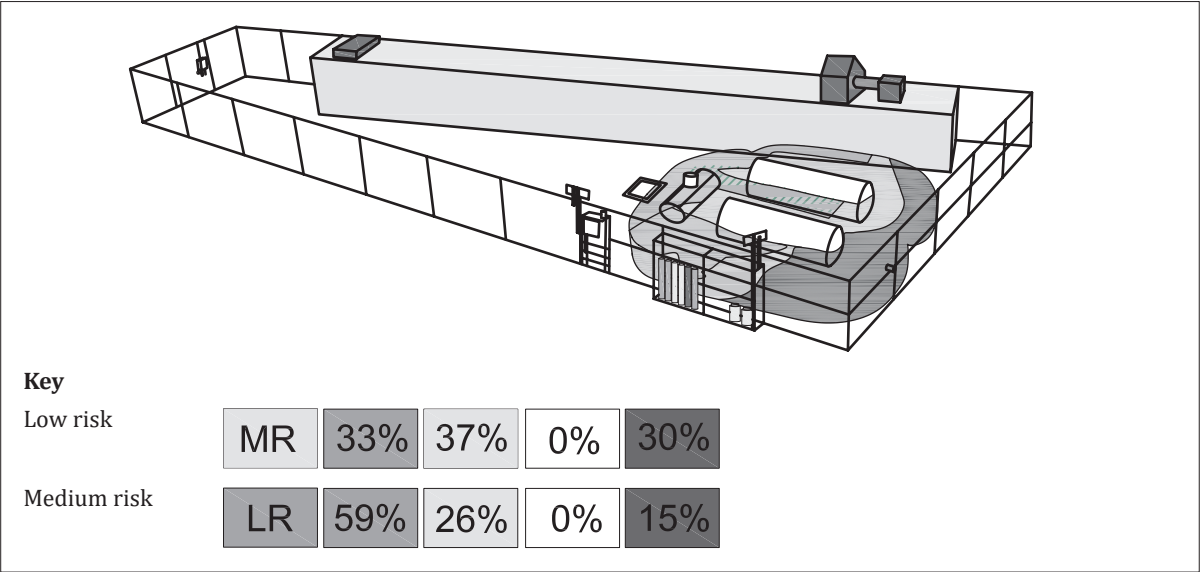
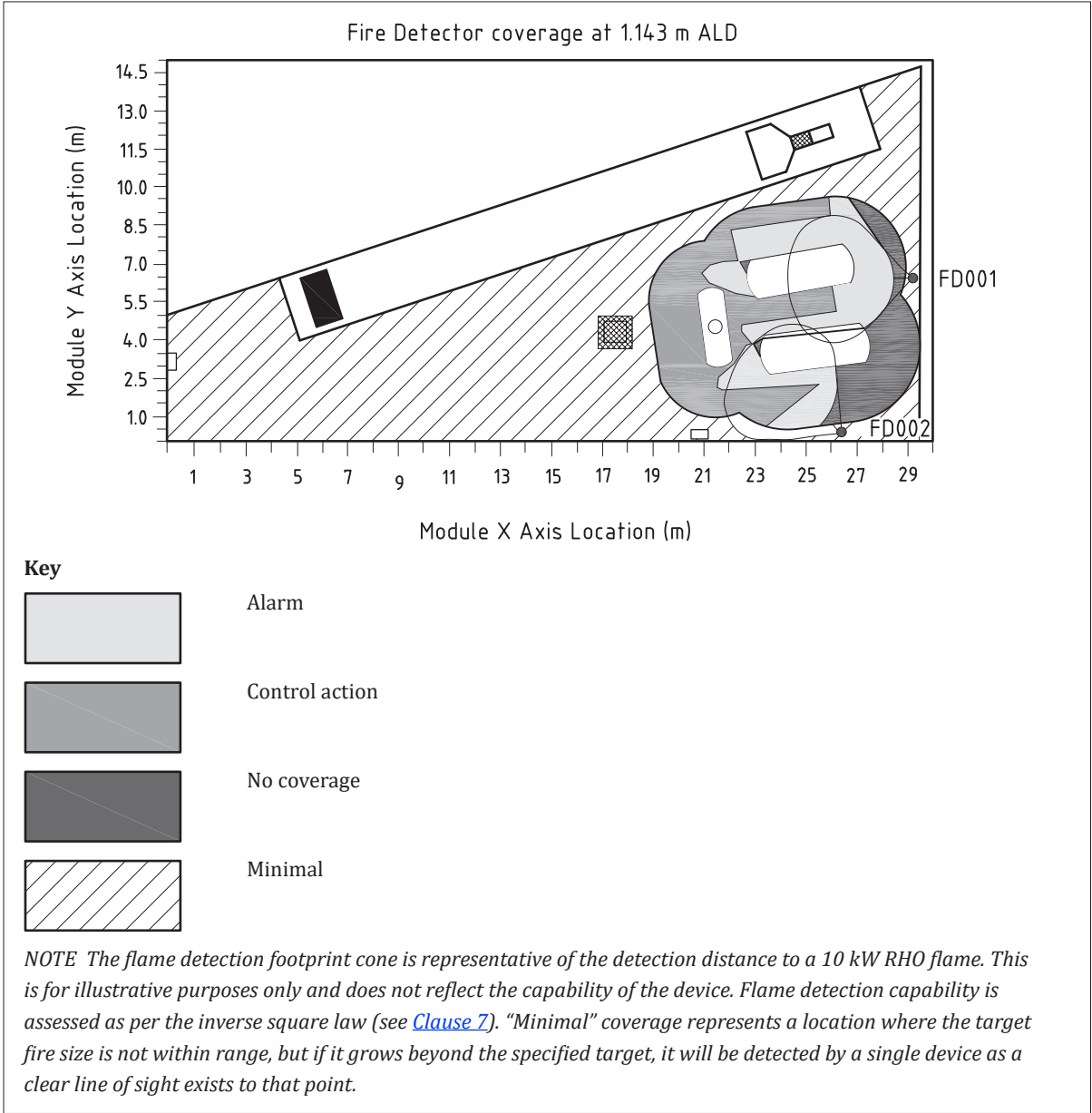


Figure B.6 — Existing flame detectors (coverage 2 of 2)



From this point, the engineer applies judgement to determine coverage adequacy. If significant gaps exist where fires could burn undetected, or redundancy is required, additional detection can be considered. An additional detector is added for clarity in [Figure B.7](#), with the resulting coverage impact represented in [Figure B.8](#) and [Figure B.9](#).

Figure B.7 — Proposed additional flame detector

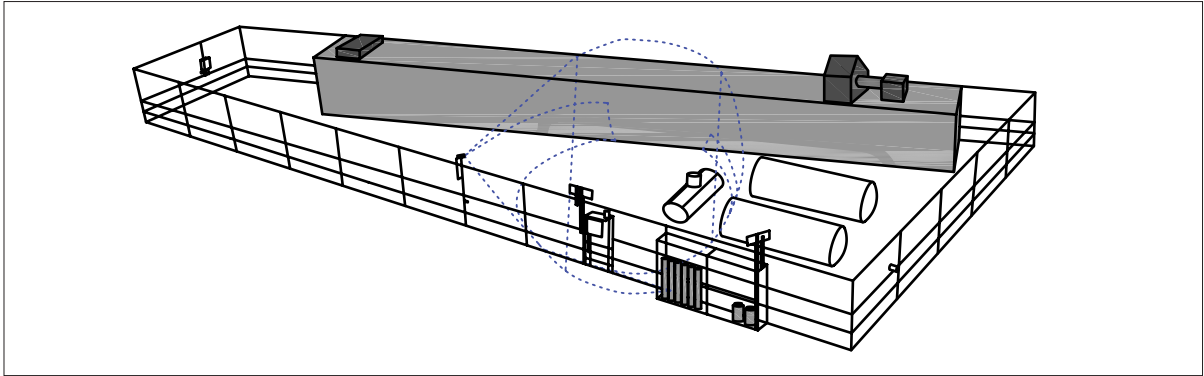


Figure B.8 — Updated assessment coverage map (1 of 2)

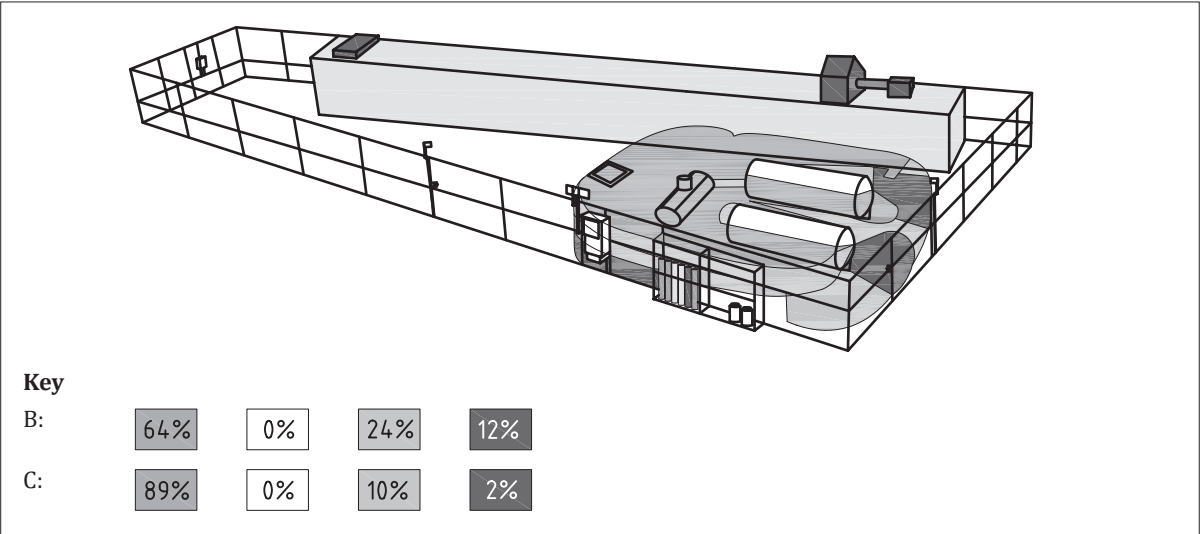
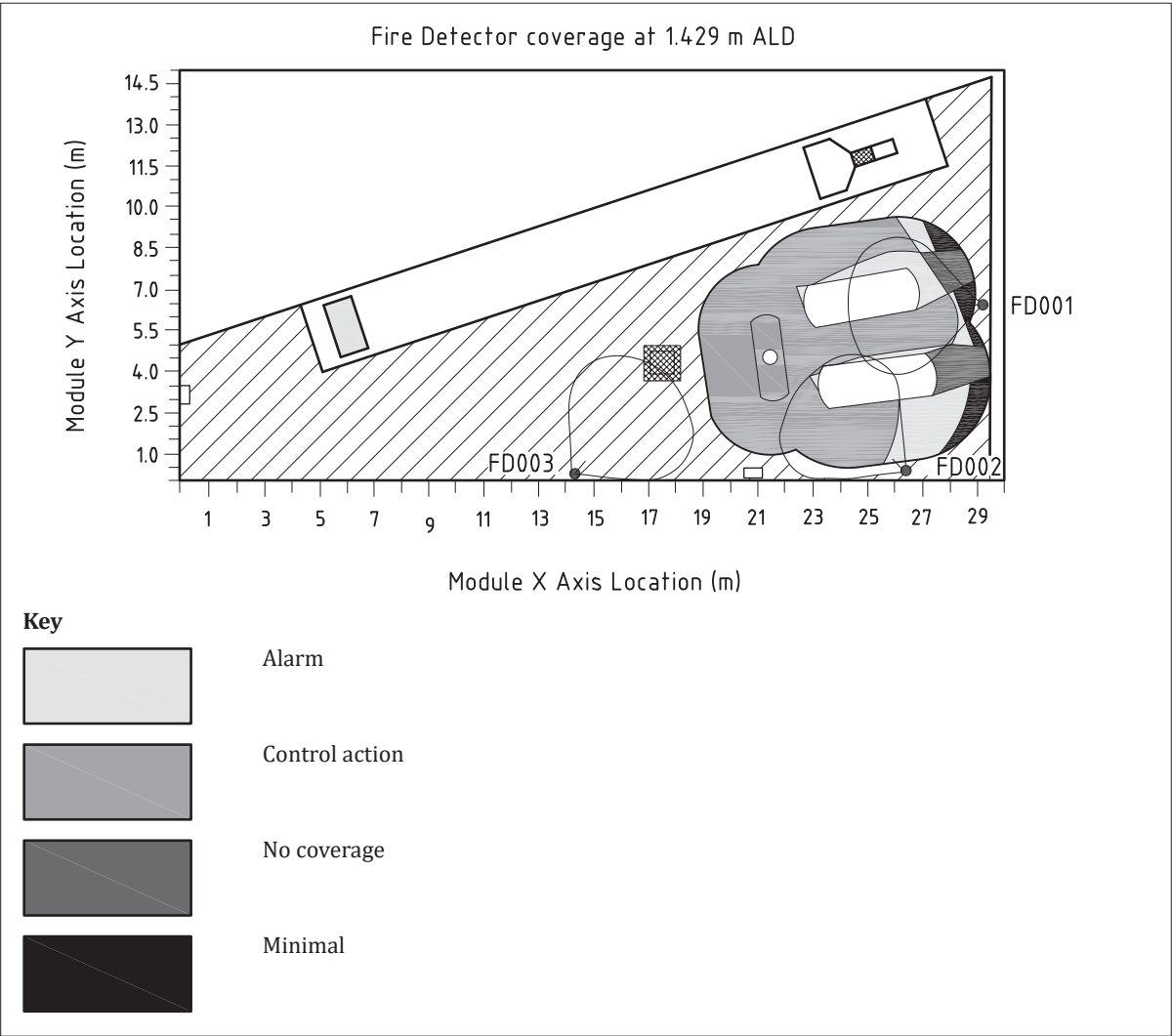


Figure B.9 — Updated assessment coverage map (2 of 2)



B.2 Example using alternative methodology which does not use grading approach

For methodologies which do not use graded volume, the 2D map can be represented simply by colour coding by the number of covering detectors (see [Figure B.10](#)).

An area is identified that requires some form of flame detection. [Figure B.10](#) to [Figure B.15](#) show an example of how 3D mapping tools can be used to demonstrate a target area is achieving a percentage coverage.

B.3 Flame detection mapping

Specific parameters are discussed, reviewed and selected, e.g. flame detector type, target fire size and a percentage coverage target, etc., and the mapping process can begin.

[Figure B.10](#) and [Figure B.11](#) show the coverage achieved using a single flame detector.

Figure B.10 — *Example of basic 2D representation of one detector*

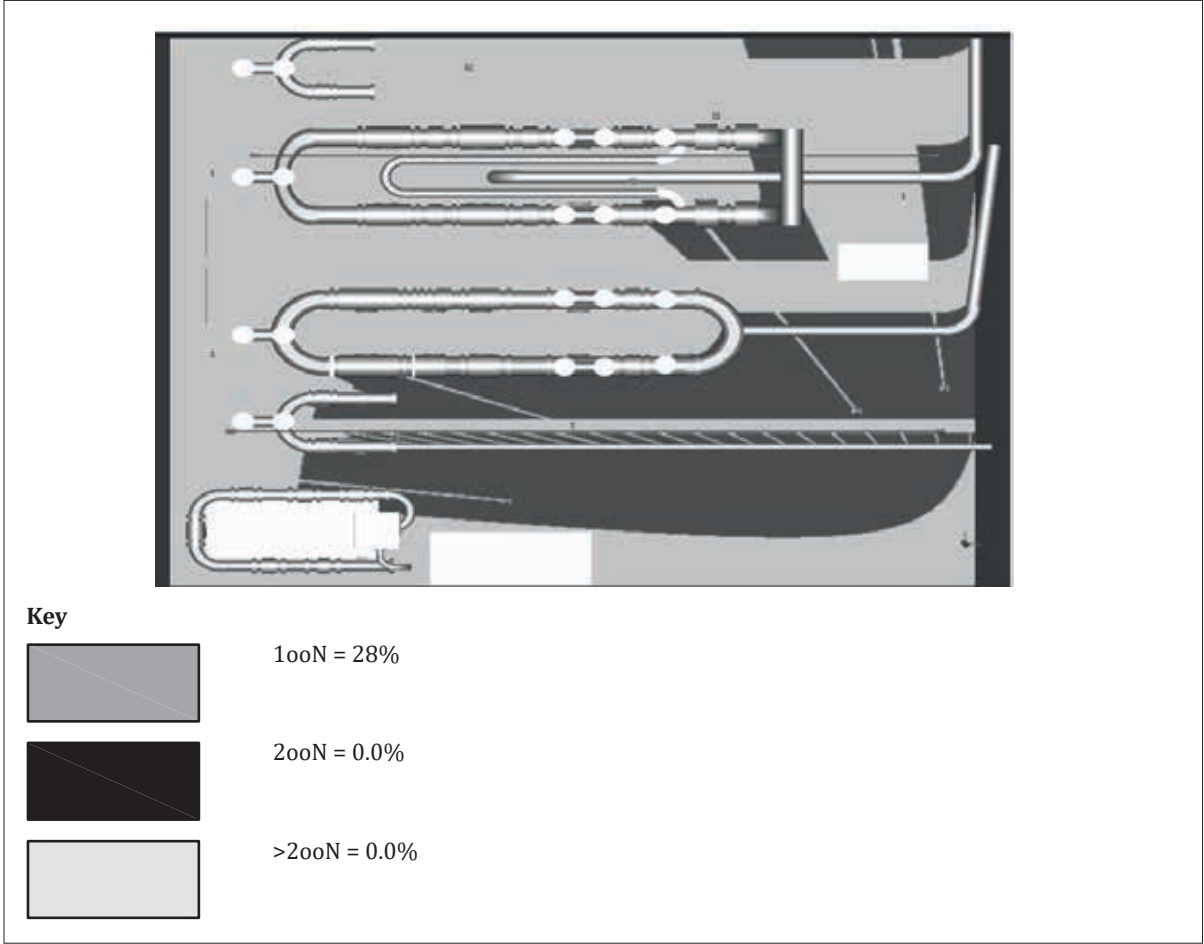
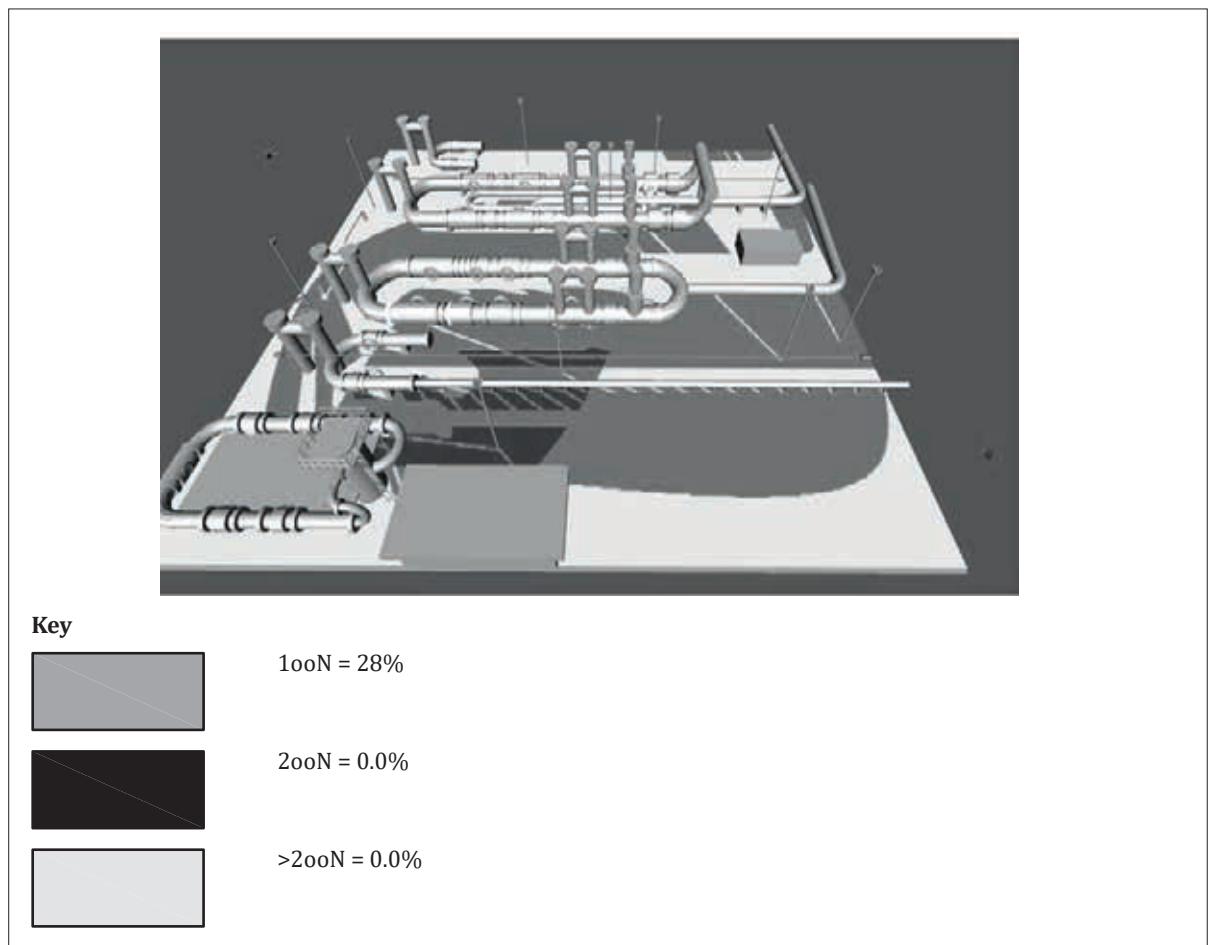


Figure B.11 — *Example of 2D plane of one detector through a 3D model*

[Figure B.12](#), [Figure B.13](#) and [Figure B.14](#) show the coverage achieved using two flame detectors.

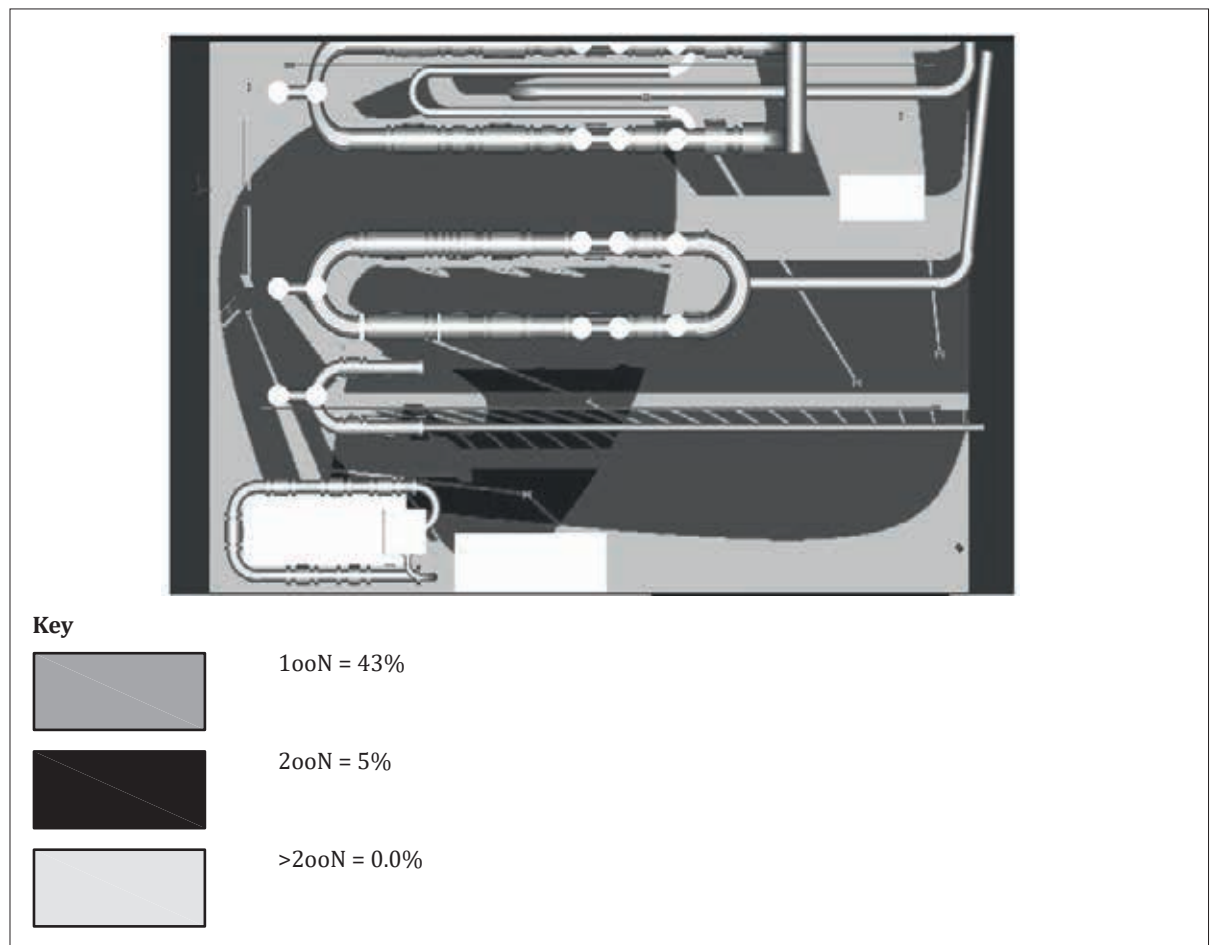
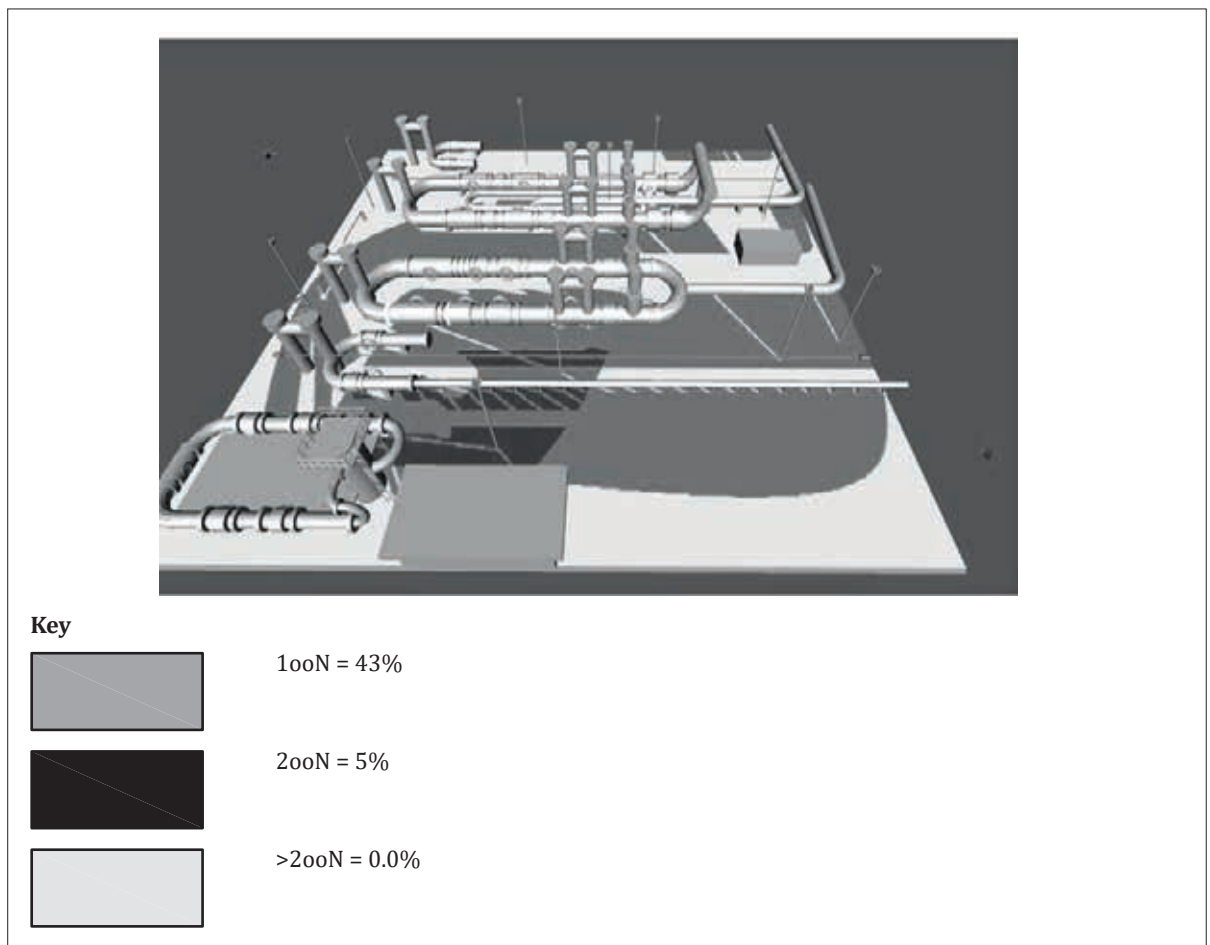
Figure B.12 — *Example of basic 2D representation of two detectors*

Figure B.13 — *Example of two detectors through a 3D model*

[Figure B.14](#) and [Figure B.15](#) show the coverage achieved using three flame detectors.

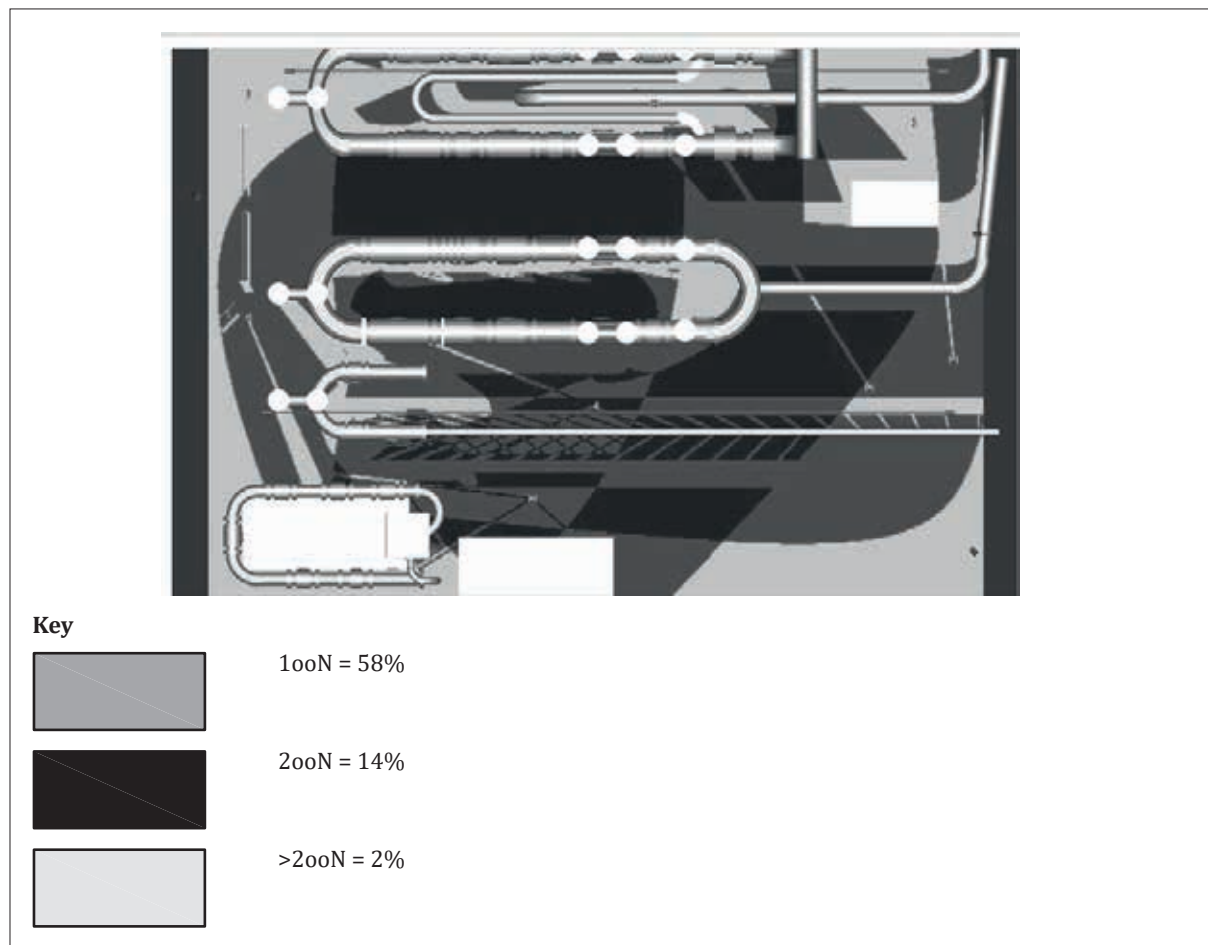
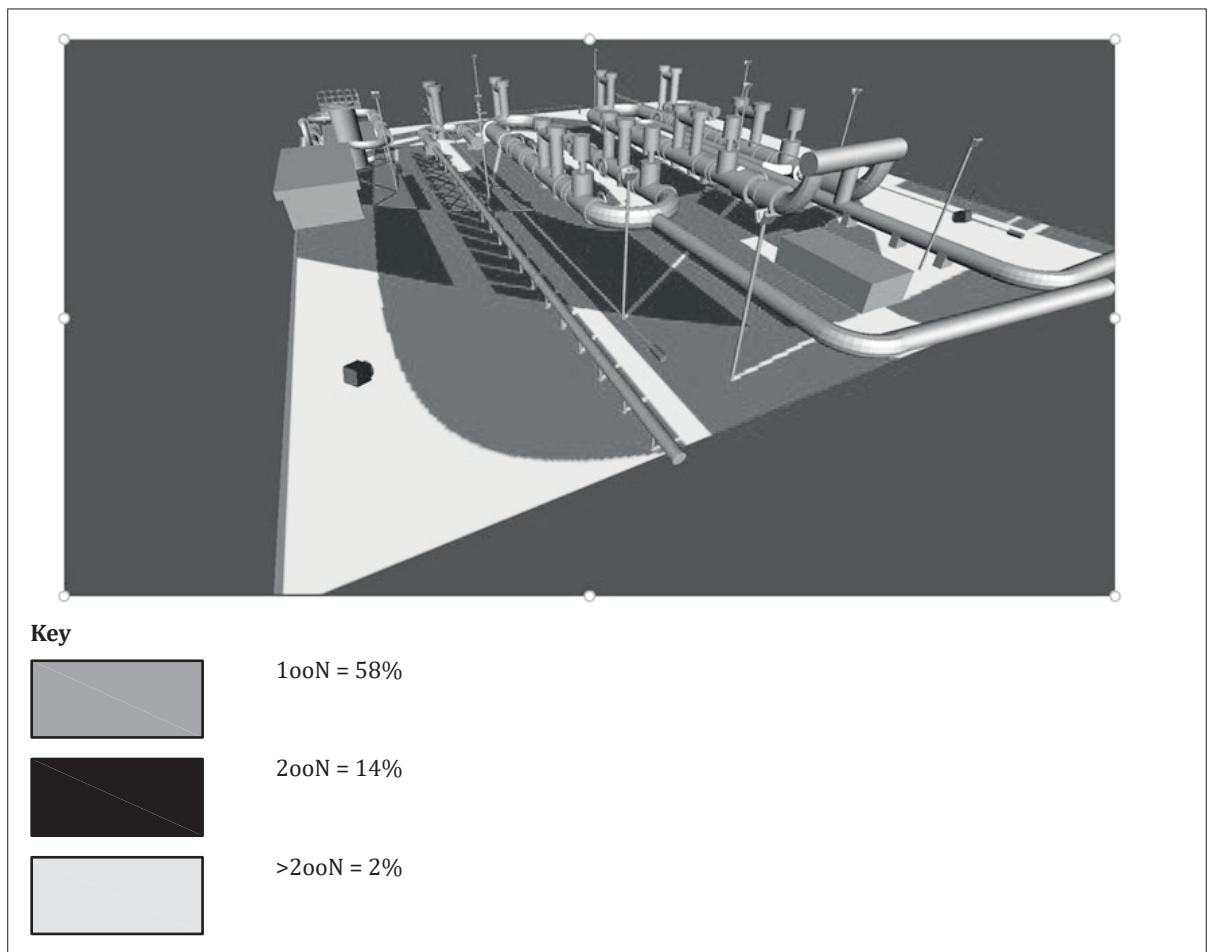
Figure B.14 — *Example of basic 2D representation of three detectors*

Figure B.15 — *Example of 2D plane of three detectors through a 3D model*

Annex C (informative)

Flammable gas detection – Volumetric-based mapping

COMMENTARY ON ANNEX C

This annex provides an example of volumetric-based mapping.

C.1 General

It is assumed that a risk assessment has been conducted and the requirement for gas detectors has been identified (see 7.5). The plant is outside and has low congestion. There is a concern that an accumulation of flammable gas could ignite, causing explosion overpressures exceeding 150 mbarg and leading to personnel fatalities and damage to the facility.

[Table C.1](#) shows the estimated blockage ratio within the area.

Table C.1 — *Estimated blockage ratio*

Orientation	Blockage factors	Blockage ratio	Combined blockage ratio	Flammable gas detection performance target
North	Container	0.07	0.27	Open
South	Open	0.00		
East	Vessels	0.03		
West	Open	0.00		
Elevation	Solid ground	0.17		
Ceiling/ roof	Open	0.00		

As the volume is relatively open with little congestion and confinement, the volume is classed as open. With the concern being the ignition of a volume of gas causing an explosion, judgement has been used to set the target gas cloud size as 10 m in diameter (twice the 5 m cloud size historically used).

The facility has a 2ooN voting configuration for both low- and high-level alarms. A dilution factor of three is applied to account for the difference in cloud size between low- and high-level alarms. The high alarm target gas cloud size is 10 m and the low alarm target cloud size is 30 m.

The target gas clouds sizes, detection technology and set points are summarized in [Table C.2](#).

Table C.2 — *Flammable gas detection performance targets*

Congestion category	Target gas cloud of concern	High alarm set point	High alarm target cloud	Low alarm set point	Low alarm target cloud
Open	10 m diameter cloud of 100% LEL	Point GD – 20%LEL	10 m	Point GD – 10%LEL	30 m
		OPGD – 3.0LELm	10 m	OPGD – 1LELm	30 m

[Figure C.1](#) shows the 3D representation of the plant.

Figure C.1 — Volume to be mapped

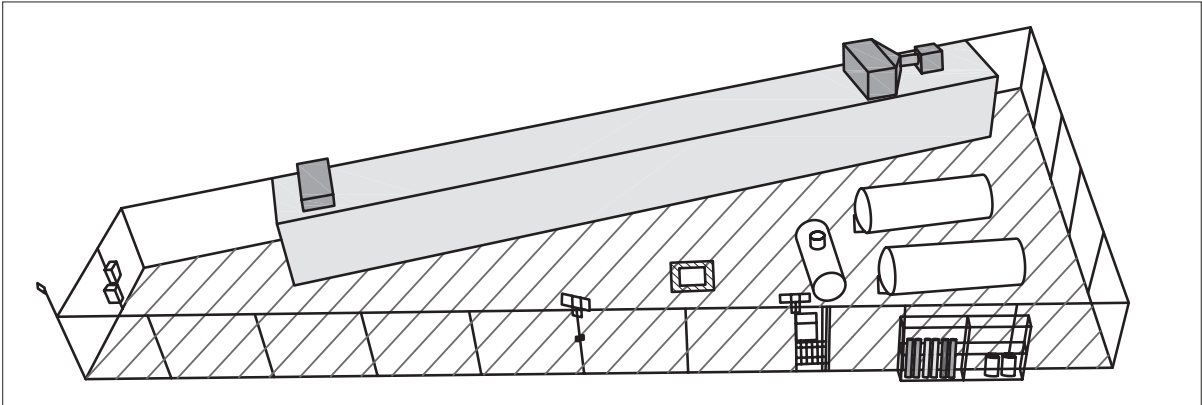


Figure C.2 represents the volume over which the volumetric-based mapping is performed, and adequacy is assessed.

Figure C.2 — Grademap volume

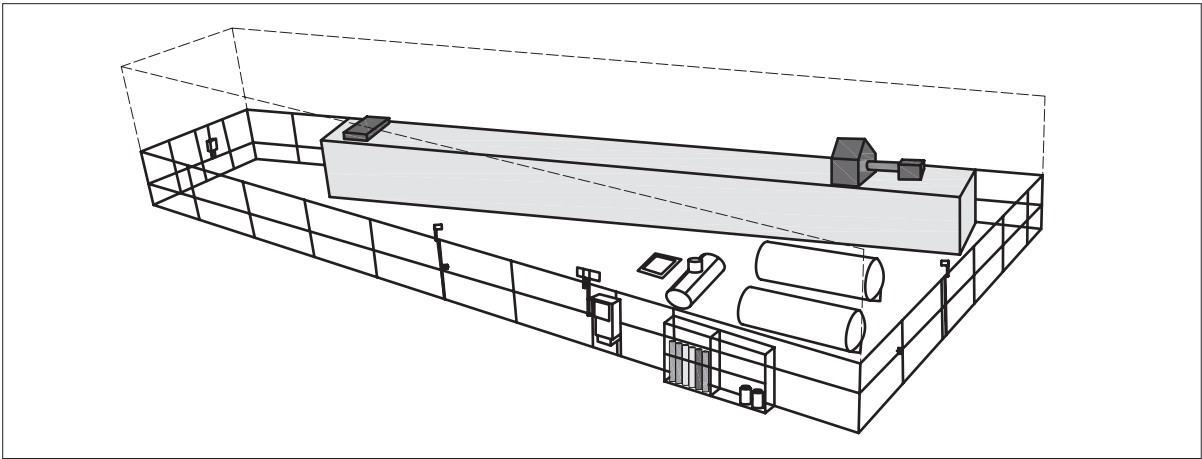
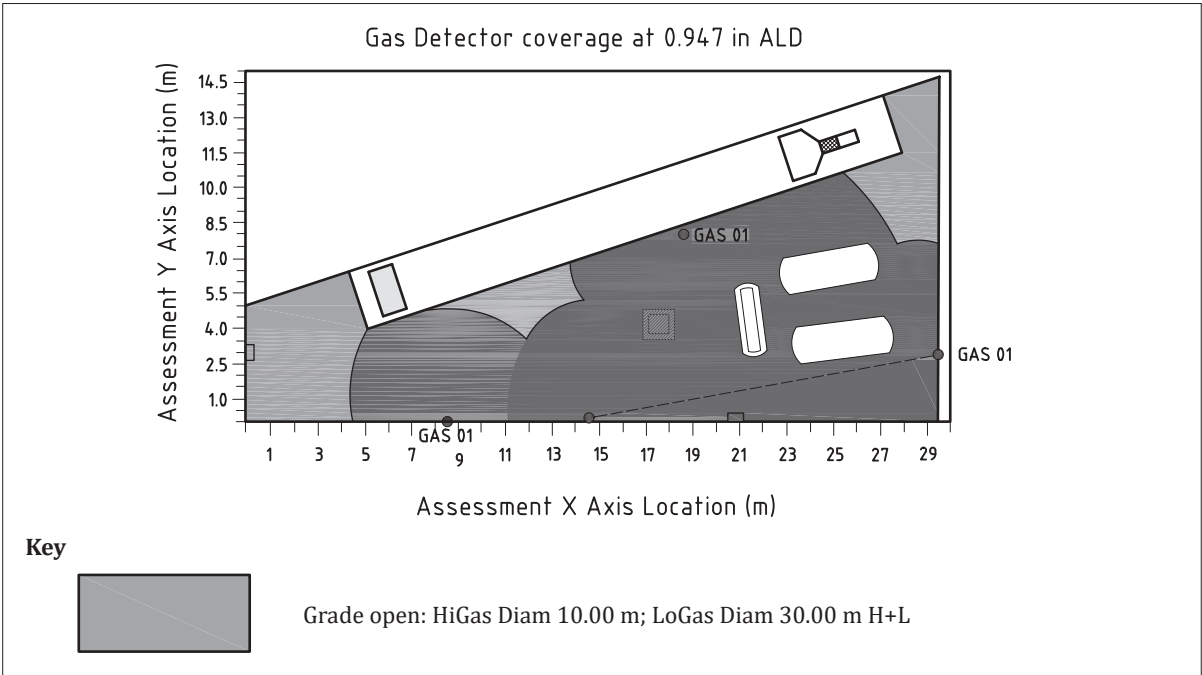


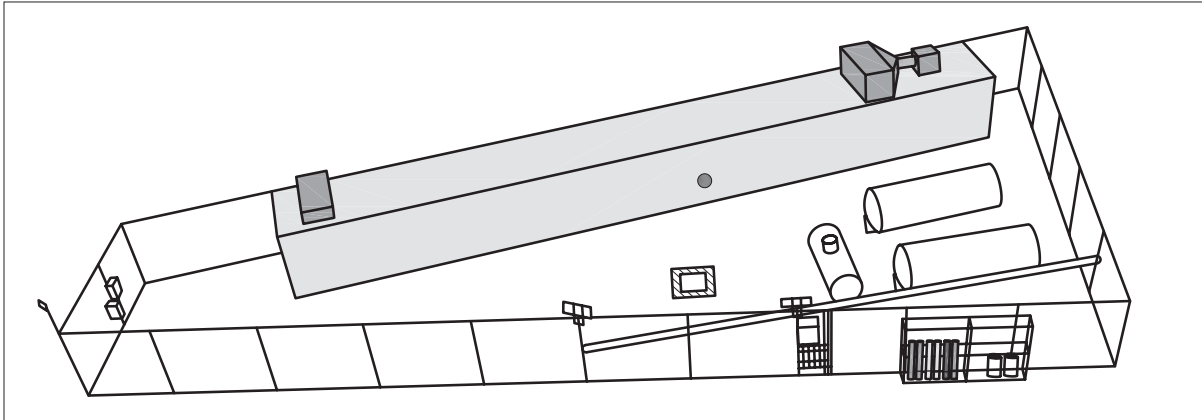
Figure C.3 represents the assessment of coverage.

Figure C.3 — Risk grade map



A combination of infrared open-path and infrared point gas detection technology is applied to the area, as shown in [Figure C.4](#).

Figure C.4 — Existing gas detectors



[Figure C.5](#) and [Figure C.6](#) represent the assessment of coverage.

Figure C.5 — Existing devices coverage map (1 of 2)

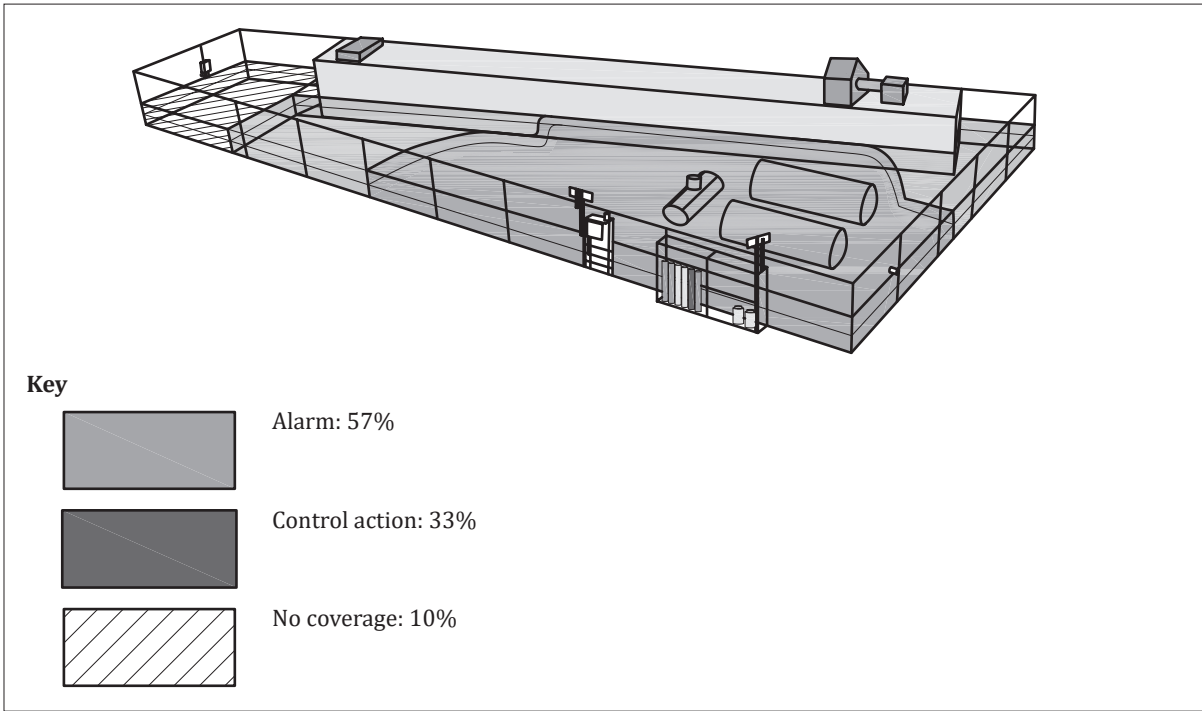
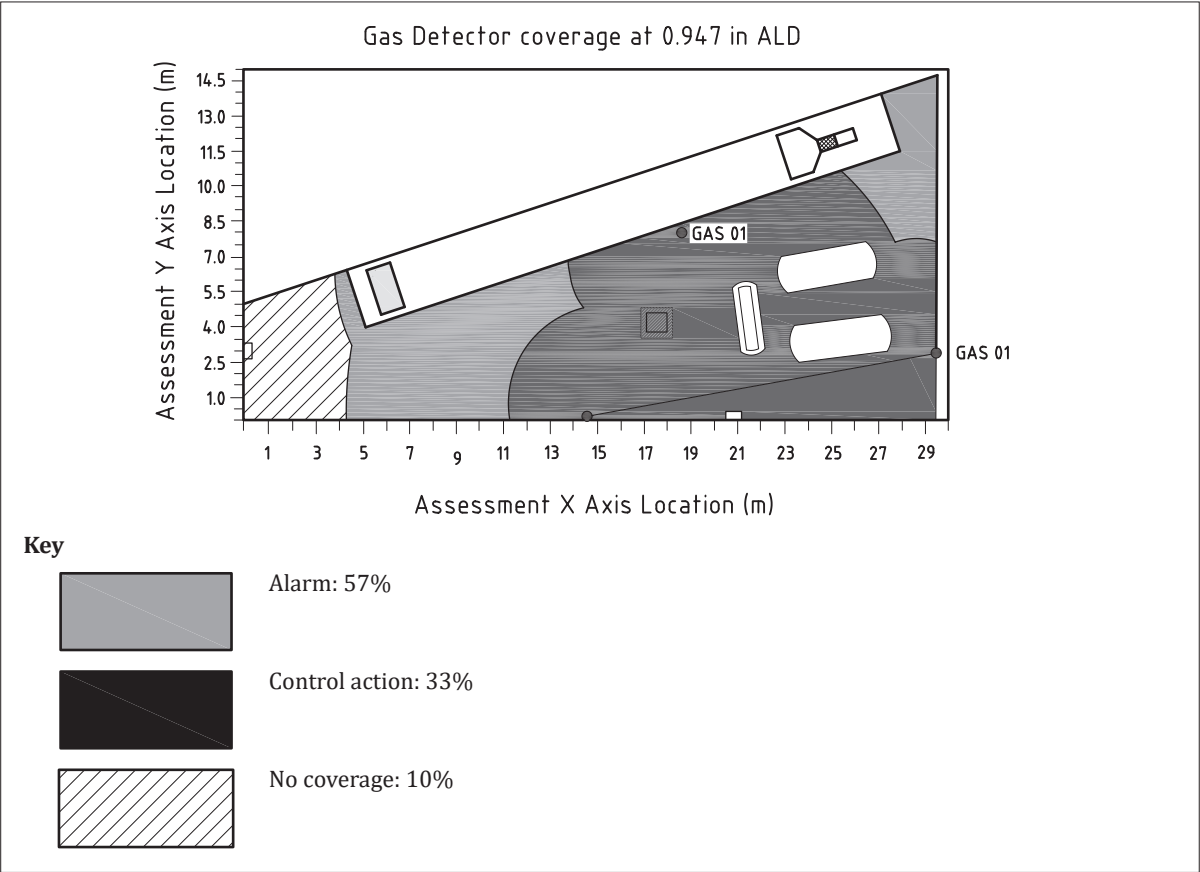


Figure C.6 — Existing devices coverage map (2 of 2)



As stated in 7.5.6, percentage coverage in isolation is not to be used to determine coverage adequacy. If significant gaps exist where the cloud volume of concern remains undetected, or redundancy is required, additional detection needs to be determined. An additional detector is added for clarity in Figure C.7. The updated coverage assessment is shown in Figure C.8 and Figure C.9.

Figure C.7 — Updated detection layout

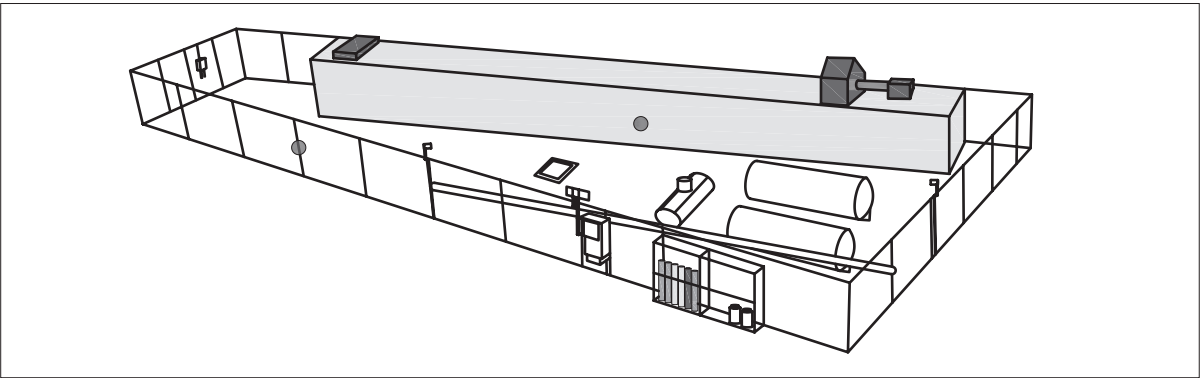


Figure C.8 — Updated assessment coverage map (1 of 2)

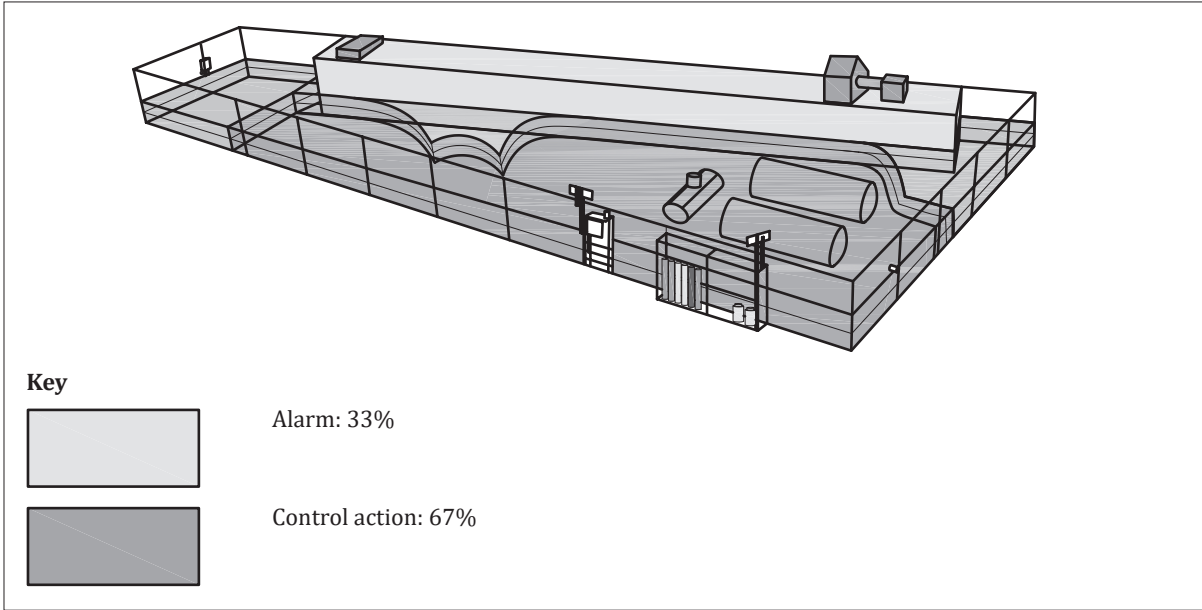
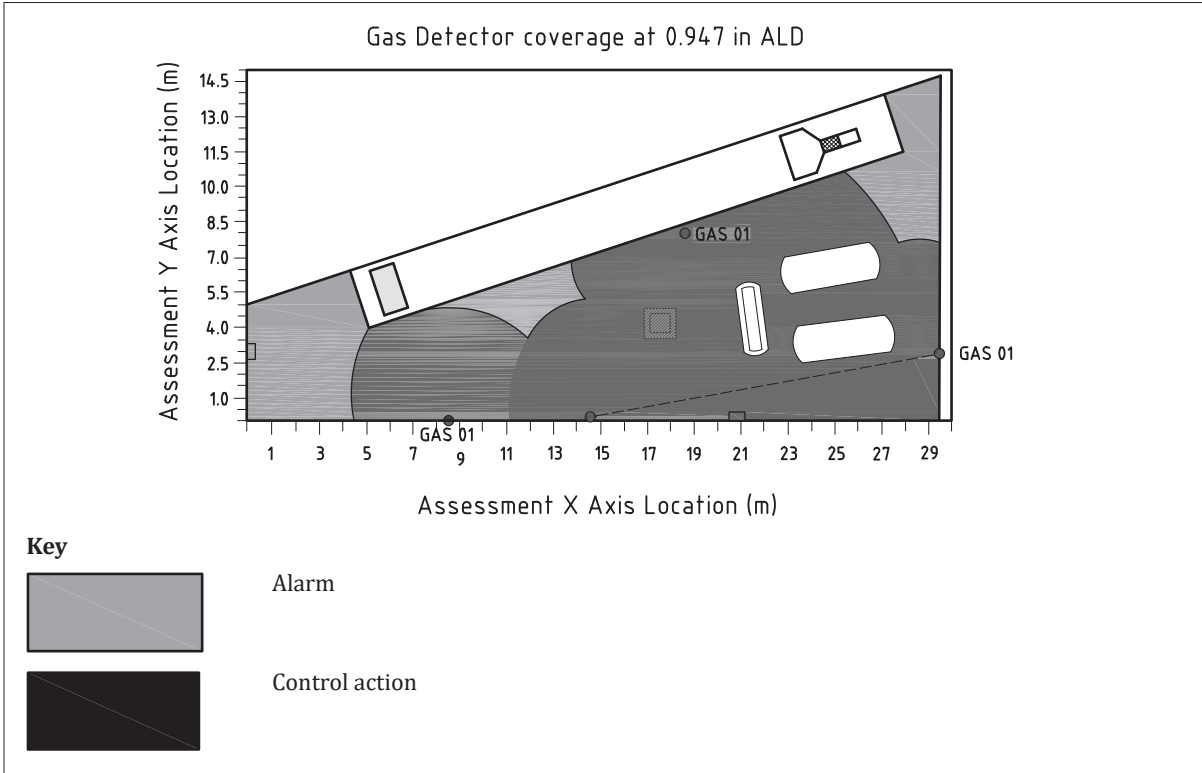


Figure C.9 — Updated assessment coverage map (2 of 2)



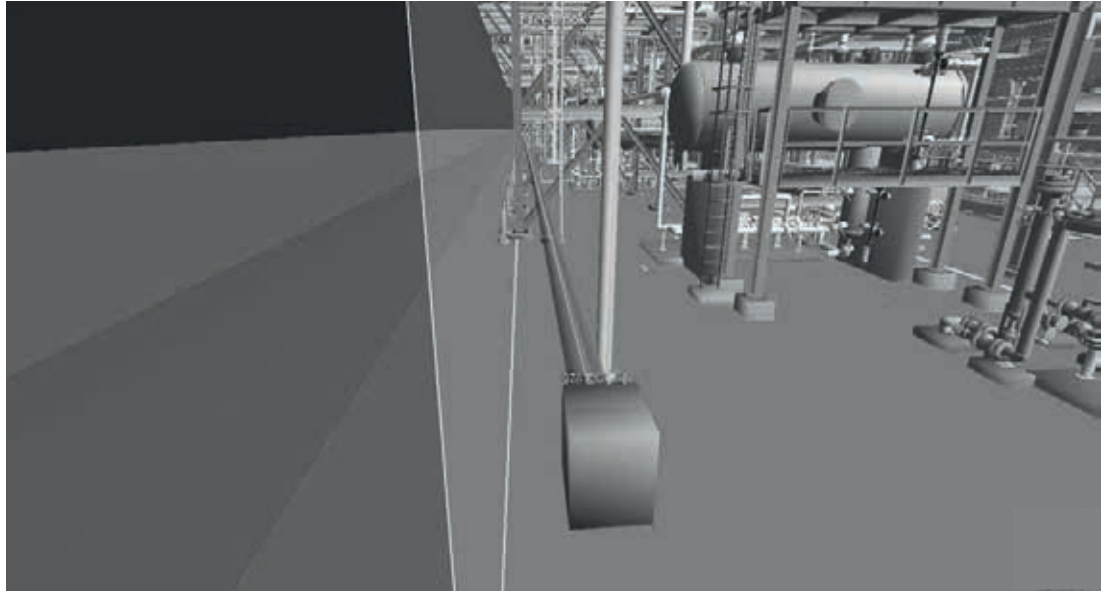
C.2 Example using alternative methodology which does not use grading approach

For methodologies which do not use graded volume, the 2D map can be represented simply by colour coding by the number of covering detectors. In the example in [Figure C.10](#), "100N" is dark grey, "200N" is black and ">200N" is black.

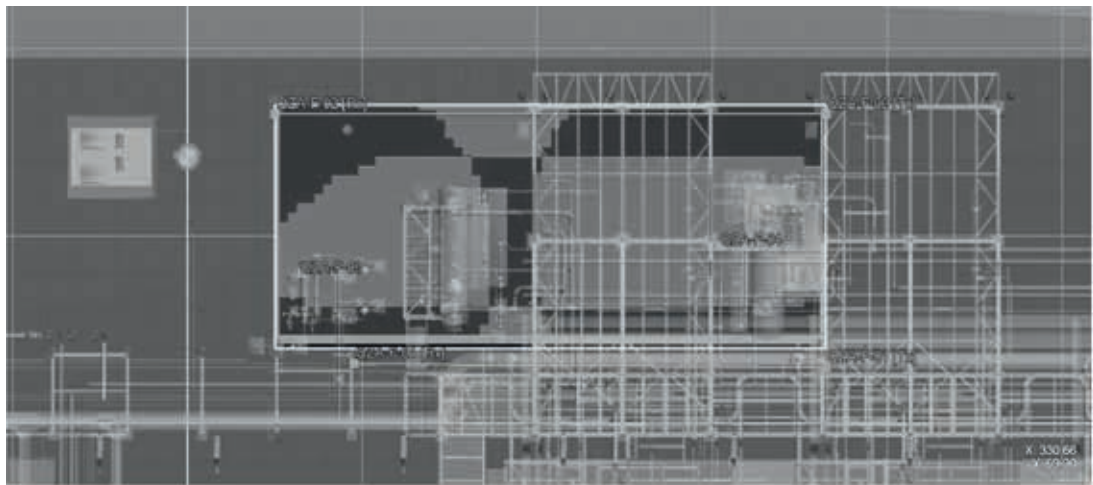
C.3 Further example of volumetric-based gas mapping

An area is identified that requires some form of gas detection. [Figure C.10](#) shows an example of how 3D mapping tools can be used to demonstrate a target area is achieving a percentage coverage.

Figure C.10 — *Example of OPGD placement in a 3D model*



a) Basic gas detection coverage through the volume



a) Basic gas detection coverage through the volume

Annex D (informative)

Risk and mitigation identification

D.1 General

The performance targets (PT) are to reflect the specific risks and mitigations for the site. The audit of the gas detection system and device locations is based on the risk mitigation and subsequent PT (see [Table D.1](#) for an example of facility information).

Table D.1 — *Example of facility information*

Toxic gas	Chlorine is present and being loaded into pressurized bottles. When liquid chlorine is released, it quickly turns into a gas that stays close to the ground and spreads rapidly. Chlorine gas can be recognized by its pungent, irritating odour. The site has large volumes of chlorine available in a pressurized and industrial environment. The strong smell is an inadequate warning to occupants that they are exposed. Chlorine gas appears to be yellow-green in colour. Chlorine can also react explosively or form explosive compounds with other chemicals, such as turpentine and ammonia. <i>NOTE 1 Chlorine is listed in HSE publication EH40/2005 [10], it addresses the risk of the loss of containment in the bottling area causing harm to the occupants.</i> The end-user is required to specify detection levels for the gas detection system. Guidelines to take into account include: <table> <tr> <td>1–3 ppm</td><td>Mild mucus membrane irritation that can usually be tolerated for approximately an hour.</td></tr> <tr> <td>5–15 ppm</td><td>Moderate mucus membrane irritation.</td></tr> <tr> <td>30 ppm and beyond</td><td>Immediate substernal chest pain, shortness of breath and coughing.</td></tr> <tr> <td>40–60 ppm</td><td>Toxic pneumonitis and/or acute pulmonary oedema can develop.</td></tr> <tr> <td>400 ppm and beyond</td><td>Generally fatal over 30 min.</td></tr> <tr> <td>1 000 ppm and above</td><td>Fatality ensues within only a few minutes.</td></tr> </table> <i>NOTE 2 These numbers are indicative for a “normal” occupant who is not suffering from any pneumatic disorders (asthma).</i> Gas detection can be set for two levels (concentrated gas and diluted gas) – early warning needs to be determined at the lower level ppm. The site might already have predetermined detection levels. The supply of chlorine from the site is to be considered unlimited.	1–3 ppm	Mild mucus membrane irritation that can usually be tolerated for approximately an hour.	5–15 ppm	Moderate mucus membrane irritation.	30 ppm and beyond	Immediate substernal chest pain, shortness of breath and coughing.	40–60 ppm	Toxic pneumonitis and/or acute pulmonary oedema can develop.	400 ppm and beyond	Generally fatal over 30 min.	1 000 ppm and above	Fatality ensues within only a few minutes.
1–3 ppm	Mild mucus membrane irritation that can usually be tolerated for approximately an hour.												
5–15 ppm	Moderate mucus membrane irritation.												
30 ppm and beyond	Immediate substernal chest pain, shortness of breath and coughing.												
40–60 ppm	Toxic pneumonitis and/or acute pulmonary oedema can develop.												
400 ppm and beyond	Generally fatal over 30 min.												
1 000 ppm and above	Fatality ensues within only a few minutes.												
Congestion/containment	The bottling takes place in a closed building; consequently gas can build up, causing a risk to occupants.												
Risk of false alarms	False alarms can foster occupant complacency and/or cost production time and product.												

Table D.1 (continued)

Occupants	All occupants in the area are trained to evacuate in the event of the alarms activating. Occupants are trained to use breathing apparatus in the event of the loss of containment for response and rescue. Each area has a sounder beacon to alert that an individual device has been activated. Outside contractors need to be briefed and inducted regarding the risk when working in the area. Normal toxic detection is fitted at 1 500 mm to 2 000 mm (normal breathing height). However, chlorine is heavier than air and detection at a lower height can be considered. If an occupant or outside contractor is required to work below 1 500 mm a permit to work system needs to be undertaken which includes personal gas detection.
Escape routes	The site has several escape doors; but the building has conveyors blocking escape routes, requiring the occupants to step over or crawl under which increases the time required to evacuate the building. Contractors' on site who are unfamiliar with the site require more time to evacuate. Additionally, emergency light/signage issues are highlighted in the fire risk assessment. The lack of lighting affects the time required to evacuate the building.
Executive action	No executive actions – emergency shutdown (ESD) have been defined.
HVAC	HVAC system is in place and specified to provide ten air changes per hour. The HVAC is not monitored, or action is defined in the event of failure.

Table D.1 (continued)

Gas detection system	Gas detection system is installed. There is no available documentation regarding the design of the installed system or detection targets. <i>NOTE 3 Attention is drawn to The Management of Health and Safety at Work Regulations 1999 [13].</i> There is no executive action (ESD) associated with this. Each area has a sounder beacon to alert that an individual device has been activated. The installed gas detectors are gas alarm systems model XXX. No information was provided by the site or the manufacturer regarding the type of gas detector installed. The part number listed is XXXX82-052; however a data sheet was not available publicly. It is reasonable to assume that the gas detectors fitted are electrochemical detectors. The data sheet does not provide response times or reliability verification/certification Electrochemical GDs use a porous membrane which is a physical/mechanical barrier. These detectors can trend to be more stable and reliable over the sensor's operational life. <i>NOTE 4 The sensors are subject to corrosive elements or chemical contaminations and might only last one to two years depending on the installed environment. A routine for servicing and testing the gas detector and control system is to be undertaken and recorded.</i> The gas detectors are reported to have only a single level of alarm.
Supply pipe integrity	The system is relatively new – this gas detection risk assessment does not consider the pipework feeding the process. However, the site needs to consider undertaking and recording a regular integrity test.

D.2 Recommended performance targets

The PTs are divided into areas with grades (and colour) with a description risk, an explanation of the reasoning, the target gas clouds, and the percentage of recommended coverage. The hazard and operability study (HAZOP) did not mention any specific risk of asphyxiation. The piping and instrument diagrams do have notes instructing gas detection be installed. No targets or philosophies were discussed.

The HAZOP does not directly address the possibility of the LOC for each area. The system is designed for 6 BARG. In the event of LOC, the gas could be propelled at least 3 m from the vessels and pipework.

The target size of the gas detection areas is based on the risk to the occupants and mitigations in accordance with [Table D.2](#).

Table D.2 — *Example performance targets*

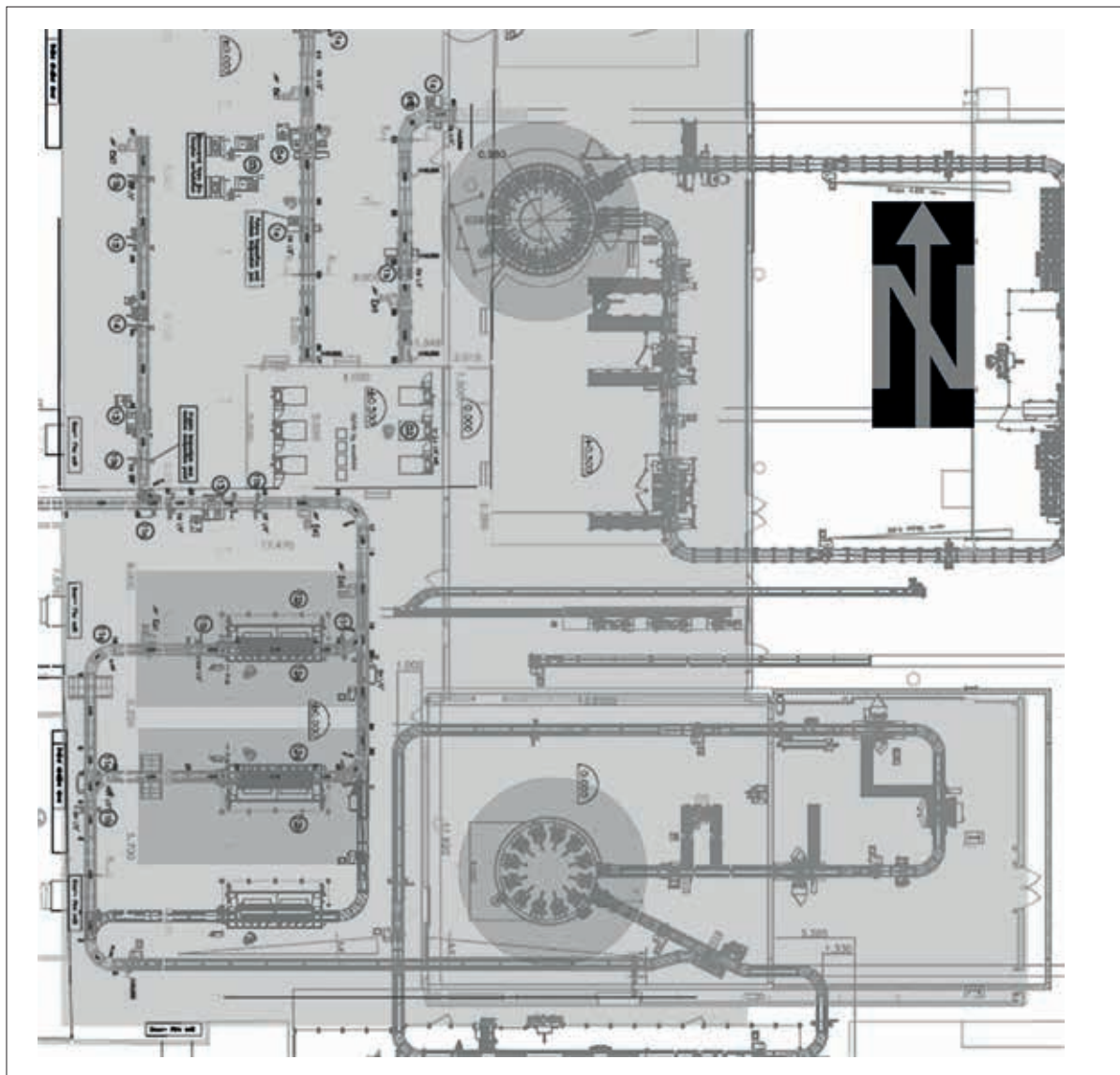
Area	Grade	Colour	Description
Bottling carousels	Pressurized LOC	Red	Bottling area where chlorine is bottled on an automated conveyor system. Occupants are working nearby. To reduce the risk of false alarms, voting of the detectors can be used to verify the loss of containment. Given: • the occupants working nearby; • the unknown response times of the GDs; • the operation of the HVAC is not monitored; • the chlorine is at least 130 psi pressure; and • the problematical escape routes, Fictional gas design company recommends: • detection target for the areas to be clouds or plumes of gas of 3 m or larger; diluted cloud of 9 m; and • 85% coverage of a 3 m gas cloud.
Bottle emptying/ inspection	Ambient LOC	Blue	Service area where bottles are emptied into a recovery unit – bottles inspected prior to reuse. Occupants are working nearby. Given: • the occupants working nearby; • the unknown response times of the GDs; • the operation of the HVAC is not monitored; and • the problematical escape routes. Fictional gas design company recommends: • detection target for the areas to be clouds or plumes of gas of 6 m or larger; diluted cloud of 9 m; and • 85% coverage of a 6 m gas cloud.
Building operational area	Confined area	Yellow	Building is enclosed. The risk of chlorine building up is present. Occupants are passing through with regularity but are not stationary in any single position. Given: • the occupants are not stationary; • the unknown response times of the GDs; • the operation of the HVAC is not monitored; and • the problematical escape routes. Fictional gas design company recommends: • detection target for the areas to be clouds or plumes of gas of 7.5 m or larger; diluted cloud of 15 m; and • 85% coverage of a 7.5 m gas cloud.

D.3 Main filling hall

The main filling hall is effectively enclosed and has (see the areas shaded darkest on [Figure D.1](#)):

- a) two areas of higher risk – filling carousels; and
- b) fully enclosed volumes with 100% confinement; there is a risk of gas accumulating in remote areas (the main shaded areas on [Figure D.1](#)).

Figure D.1 — Example of facility requirements grade map



Annex E (informative)

Graphic symbols for fire and gas detection location drawings

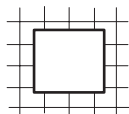
COMMENTARY ON ANNEX E

BS 1635 has been produced with the agreement of the Fire Standards Policy Committee. During preparation ISO 6790 (withdrawn) was considered and its symbols have been used wherever possible, with consideration of the content of BS 1635.

If an organization has established symbols, this annex does not preclude continued application.

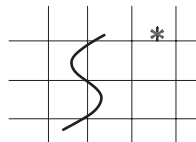
The initiating equipment (detectors) has been organized into elements and symbols. Any detector is symbolized as a square.

4.201 detector, any type

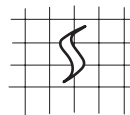


The detector requires an element, including:

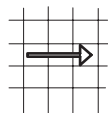
Smoke - the letter S



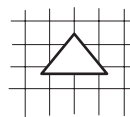
Flame - a tiny flame



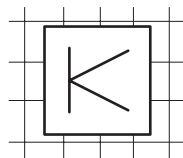
Beam - an arrow



Gas - a triangle



Heat - K (for kelvin)



Detail required in the latest F&G detection location designs is not included. The amount of information (detail) provided by the graphic symbol provides:

- a) location;
- b) tag reference;
- c) device type;

- d) orientation (for flame detection and beam detection); and
- e) detection points (for aspirated smoke or gas detection).

Greater detail is to be referred to in a separate schedule.

Each detector (smoke, gas or flame) continues to be as a square with the elements in [Table E.1](#) added.

Table E.1 — *Graphical symbols*

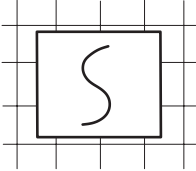
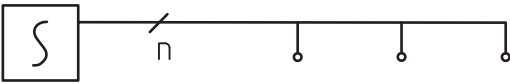
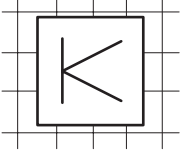
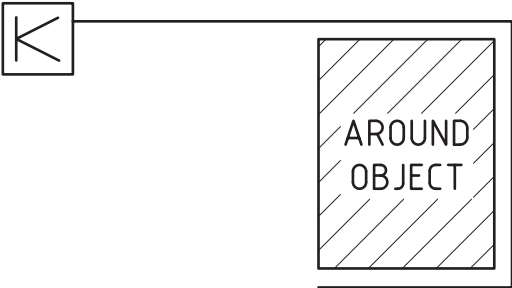
Detector	Symbol	Notes
Point smoke detector		No change – ionization or optical to be specified in separate schedule.
Beam (Open-path) smoke detector		Height (S) to be specified in separate schedule. Tx = transmitter, Rx = receiver.
Aspirating/sampling smoke detector – single pipe		The S symbol indicates control unit/interface. The end of the sample pipe indicates location of sample points.
Aspirating/sampling smoke detector – multi-pipe		The end of the sample pipe indicates location of sample points. n = the number of detection pipes.
Point heat detector		No change – temperature or rate of rise to be specified in separate schedule.
Linear heat detection		The K symbol indicates control unit/interface. The line indicates the run of the detection line. Details of heat detection to be specified in separate schedule.

Table E.1 (continued)

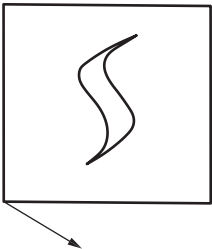
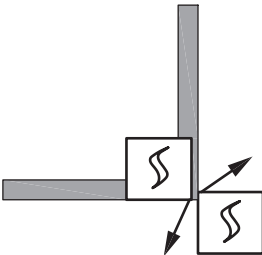
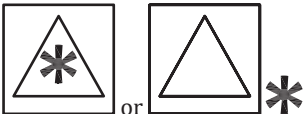
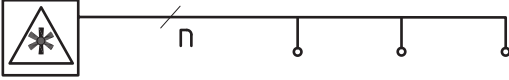
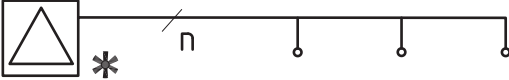
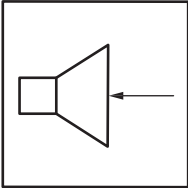
Detector	Symbol	Notes
Flame detection	 A sample of how these can be displayed 	The base of the arrow indicates where the flame detector is installed. The direction of the arrow indicates direction of field of view. Specific details required in a separate schedule need to include: • type of detection; • X, Y and Z location of detector; • pan in degrees (horizontal orientation in relation to building north); and tilt in degrees (vertical orientation to the horizon).
Flammable/toxic point gas detector	 Sample for chlorine point gas detector 	* = chemical notation/formula of gas being detected [i.e. methane detector would be "CH₄" or chlorine would be "Cl"].

Table E.1 (continued)

Detector	Symbol	Notes
Flammable/toxic open-path (bbeam) gas detector	 Or 	Tx = transmitter, Rx = receiver
Aspirating/sampling toxic gas – single pipe	 or 	The triangle symbol indicates control unit/interface. The end of the sample pipe indicates location of sample points.
Aspirating/sampling toxic gas – multi-pipe	 Or 	The end of the sample pipe indicates location of sample points. n = the number of detection pipes.
Acoustic gas detector		Inside the detector are the standard symbols for sound detection (microphone).

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BS EN 45544-3, *Workplace atmospheres – Electrical apparatus used for the direct detection and direct concentration measurement of toxic gases and vapours – Performance requirements for apparatus used for general gas detection*

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